STUDIES IN HEALTH TECHNOLOGY AND INFORMATICS 253

German Medical Data Sciences: A Learning Healthcare System

Proceedings of the 63rd Annual Meeting of the German Association of Medical Informatics, Biometry and Epidemiology (gmds e.V.) 2018 in Osnabrück, Germany – GMDS 2018

GMDS 2018

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IOS Press

Advances in digital and information technology have meant that medical informatics and its associated fields are of ever-increasing importance in the modern healthcare environment.

This book presents selected papers from the 63rd annual conference of the German Society of Medical Information Sciences, Biometry, and Epidemiology, GMDS 2018, held in Osnabrück, Germany, in September 2018. The society encompasses not only medical informatics, biometry and epidemiology, but also medical bioinformatics, systems biology and health data management. The title of this year's conference is "The Learning Health System: Research Based, Innovative, Connecting", and 38 full papers of the 164 oral presentations and 65 posters delivered at the conference are included here. A wide variety of scientific topics are covered, including standards to enable the interoperable interchange of information; metadata management; record linkage; IT issues for health care networks; interprofessional teaching and training; eHealth legislation; analysis of miRNAs and RNA-Seq data, among others.

The contributors are all specialists in their field, and this book disseminates some of the innovative ideas which are urgently needed to meet the challenges facing a constantly developing digital healthcare environment.

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GERMAN MEDICAL DATA SCIENCES: A LEARNING HEALTHCARE SYSTEM

Studies in Health Technology and Informatics

International health informatics is driven by developments in biomedical technologies and medical informatics research that are advancing in parallel and form one integrated world of information and communication media and result in massive amounts of health data. These components include genomics and precision medicine, machine learning, translational informatics, intelligent systems for clinicians and patients, mobile health applications, data-driven telecommunication and rehabilitative technology, sensors, intelligent home technology, EHR and patient-controlled data, and Internet of Things.

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The Virtuous Circle of Learning

Preface by the Editors

The 63rd Annual Conference of the German Association for Medical Informatics, Biometry and Epidemiology (GMDS) builds upon the motto "The Learning Health System: research based, innovative, connecting" and herewith highlights the unique combination of disciplines under the roof of GMDS and their contribution to a Learning Health System (LHS): medical informatics, biometry, epidemiology, medical bioinformatics, systems biology and health data management. This volume witnesses the breadth of these contributions and demonstrates their input to the learning circle of an LHS from *data to knowledge*, *knowledge to performance* and finally *performance to data*[1]. The German national medical informatics initiative fuels research on how integrated large amounts of data can be processed to become knowledge and how performance related data are fed back into the circle. Biometry, epidemiology, bioinformatics and medical informatics provide the research methods and innovative algorithms to turn data into knowledge. Although these thoughts have accompanied biomedical and health research for many decades, the new vision about an LHS is the systematic and large scale feedback loop between science and practice: the virtuous circle of learning. It gives research a new momentum towards better patient care.

In light of an LHS, it is clear that digitised healthcare is not an aim in itself but a must and an instrument to ultimately serve the patient.

Yet, there is still so much to learn about a Learning Health System. Exchange with colleagues from within Europe and from abroad is essential to obtain new insights. This idea guided our motivation to invite dedicated international guests to the conference to give keynote speeches and introductions to workshops. It also guided our intention to carry on the undertaking of an open call for full papers in English, a two stage peer review process and the publication of accepted papers in the Series *Studies in Health Technology and Informatics*, which was initially established for GMDS2017 in Oldenburg. It allows the authors to present their achievements to an international audience, also beyond the GMDS Annual Conference. In addition, full papers in German are published in the open access journal German Medical Science – Medical Informatics, Biometry and Epidemiology (MIBE) and abstracts in German or English by German Medical Science via www.egms.de.

This volume embraces 38 full papers out of the total of 164 oral presentations and 65 posters of this year's GMDS conference. The papers are sorted alphabetically by the topic of the session they were presented in.

We wish to thank all colleagues who submitted their work to the conference and particularly those who submitted full papers. We are grateful to the many reviewers who dedicated their time and knowledge to make this congress possible.

GMDS2018 takes place in Osnabrück on the site of the HealthCampus, an institution founded by the two local universities that brings together major regional hospitals

[1] Friedman CP, Rubin JC, Sullivan KJ. Toward an Information Infrastructure for Global Health Improvement. Yearb Med Inform. 2017;26:16-23.

and health and social care institutions to form an emerging Learning Health System. It is meant to build up trust and stimulate the dialogue between science and practice.

With this in mind, GMDS2018 adds another building block to establishing LHS inspired virtuous circles of learning.

Ursula Hübner
Ulrich Sax
Hans-Ulrich Prokosch
Bernhard Breil
Harald Binder
Antonia Zapf
Brigitte Strahwald
Tim Beißbarth
Niels Grabe
Anke Schöler

Preamble

Andreas Stang
President of GMDS

It is a pleasure to congratulate the editorial board and contributors for the present volume of scientific contributions to the annual meeting of the German Society of Medical Information Sciences, Biometry, and Epidemiology (GMDS). It reflects a wide spectrum of research activities. 2018 is the first year in which the newly founded special field (Fachbereich) entitled "Medical Bioinformatics and Systems Biology", contributes explicitly to the annual meeting. The GMDS is obviously a professional scientific society that unites diverse disciplines under one roof and hereby contributes to the required interdisciplinarity in science.

This year, hot topics are addressed including the consequences of the General Data Protection Regulation (GDPR) of the European Commission, new developments in digitalization of biomedical data, artificial intelligence, big data, just to name a few. The volume contains work of specialists of their fields and the conference provides an important platform for exchange of innovative ideas which are urgently needed to meet the challenges in a constantly developing digital environment. Some products related to the Medical Informatics Initiative (MI-I) can be studied in this volume. Furthermore, a wide variety of scientific topics are covered including the provenance standards to enable the interoperable interchange of provenance information, metadata repository, record linkage, patient-reported outcome, IT issues of health care networks, eHealth legislation, analysis of miRNAs and RNA-Seq data among others.

Besides high-end research results, the conference offers several opportunities for continuous education including workshops, tutorials, and seminars.

I thank the organizers for their patience, perseverance and enthusiasm during the preparation phase of the conference and hope that the conference becomes a huge success.

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1. Clinical Trial Information Systems

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U. Hübner et al. (Eds.)

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Investigating the Capabilities of FHIR Search for Clinical Trial Phenotyping

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Abstract. Clinical trials are the foundation of evidence-based medicine and their computerized support has been a recurring theme in medical informatics. One challenging aspect is the representation of eligibility criteria in a machine-readable format to automate the identification of suitable participants. In this study, we investigate the capabilities for expressing trial eligibility criteria via the search functionality specified in HL7 FHIR, an emerging standard for exchanging healthcare information electronically which also defines a set of operations for searching for health record data. Using a randomly sampled subset of 303 eligibility criteria from ClinicalTrials.gov yielded a 34 % success rate in representing them using the FHIR search semantics. While limitations are present, the FHIR search semantics are a viable tool for supporting preliminary trial eligibility assessment.

Keywords. FHIR, phenotyping, clinical trials

1. Introduction

Clinical trials are the foundation of evidence-based medicine; they are used to evaluate new treatment options such as drugs or surgical approaches in a well-defined and controlled environment. Whether a given patient can be considered a suitable participant for a trial depends on the extent by which his demographic and clinical characteristics match a set of eligibility criteria defined by the designers of the trial. In order to search for eligible patients for a study, a variety of clinical trial recruitment support systems (CTRSS) have been developed, which, based on implicitly or explicitly formalized in- and exclusion criteria, can query for potential trial patients from data available in the electronic health record [1].

The Health Level 7 Fast Healthcare Interoperability Resources (HL7 FHIR) standard is being developed to address the challenges of an increasingly digitized healthcare ecosystem. The specification leverages emerging and well-established industry standards for exchanging data between healthcare applications based on the experiences gained and lessons learned from developing previous standards such as HL7 v2, v3, and Clinical Document Architecture (CDA) [2]. FHIR defines a set of resources representing common clinical objects and concepts as well as operations for creating, modifying, and searching for them based on defined attributes. In FHIR, resources are considered the basic building blocks of integration. They generally represent clinical concepts such as patient, medication, and practitioner. Every resource has a set of attributes: a *patient*

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resource, for example, encompasses demographic data such as the date of birth and gender of a patient and an *observation* resource contains information about a lab test or measurements compiled by a clinical device. Such resources may reference – or may be referenced by – other resources.

The RESTful create, read, update, and delete (CRUD) operations of the FHIR API are complemented by a search functionality which allows for filtering the set of resources by parameters supplied to the HTTP URL query [3]. In this study, we investigate the capabilities of these search semantics and demonstrate how this constraint-based search can be used to formulate clinical trial inclusion and exclusion criteria to find eligible patients. We limit the scope of this work by exclusively focusing on the features provided by a standard-conformant FHIR server and resources out-of-the-box and not on the role FHIR may play as part of a larger CTRSS.

2. Methods

As input for our investigation and experimental validation, we downloaded the HTML representation of the 100 latest (as of January 1st 2018) trials that were open for recruitment from ClinicalTrials.gov and randomly selected a subset of 25. We created a tool to automatically parse and extract the eligibility criteria from this data. The tool considers each bullet point in the 'Eligibility Criteria' section of the HTML page describing the trial as a single criterion.

We investigated in how far this subset of criteria can be encoded using the search functionality provided by the FHIR DSTU3 specification as implemented by a HAPI-FHIR server[2]. In order to retrieve the final set of patients from the inclusion and exclusion criteria, we created an additional tool which receives the FHIR query representations of both sets of criteria as input and returns a final set of eligible patients by excepting the set of patients fulfilling the exclusion criteria from those fulfilling the inclusion ones.

3. Results

From the 25 trials, we extracted a total of 303 eligibility criteria; 119 describe inclusion conditions, 184 exclusions. The mean number of inclusion criteria per study was 4.76 (IQR=4, Range=1-12) and for exclusion criteria it was 7.36 (IQR=9, Range=1-20).

Table 1. Numerical overview of the analyzed criteria and their representability.

	Representable	Not Representable	Total
Inclusion Criteria:	48	71	119
Exclusion Criteria:	56	128	184
Total:	104 (34 %)	199 (66 %)	303

Table 1 shows the statistical results of the encoding process. A criterion was considered representable if it described a requirement that could be expressed by the syntax and semantics of the FHIR search. In the following, we will give some examples of these representations and the limitations we encountered.

[2] http://hapifhir.io/

Table 2. List of trial inclusion criteria and their query representation of one clinical trial[3].

No.	Criterion	URL Query Representation
1	Diagnosis of a major depressive disorder	/Condition?code=http://snomed.info/sct|370143000
2	Age greater or equal to 18 and less or equal to 34	/Patient?birthdate=le2010&birthdate=ge1984
3	Ability to fluently read, write, and speak Dutch	/Patient?language=nl
4	Sexes Eligible for Study: Female	/Patient?gender=female
5	Logical conjunction of all criteria	/Patient?gender=female &birthdate=le2010&birthdate=ge1984 &language=nl &has:Condition:patient:code =http://snomed.info/sct|370143000

The degree by which the search methodology defined in the FHIR standard can be used for representing clinical trial eligibility criteria strongly depends on the kind of data requested in the criteria, the kind of data available in the queried resource, as well as the semantic complexity of the criteria. The query API only supports simple numerical or categorical comparisons when filtering resources. Table 2 shows the inclusion criteria for one of the studies we analyzed and their representation as a search query. In order to now find all patients that match these inclusion criteria, we combine them using the logical AND for a composite search of patient resources resulting in the final query (5).

The example trial in Table 2 demonstrates the general category of criteria which can easily be represented using the FHIR RESTful API search syntax: categorical comparisons (1, 3, 4) and simple numerical range evaluations (2) using quantitative prefixes (`le` = less or equal, `ge` = greater or equal, etc.). Typical examples of these criteria relate to demographic characteristics, such as patient age, gender and basic diagnosis. They are commonly present in the trial criteria as well as the underlying electronic health record (EHR) data source [4].

4. Discussion

The FHIR RESTful search provides an easy way to search for patients based on common inclusion and exclusion criteria. Using FHIR for phenotyping leverages the capabilities of an open standard, thereby easing the reuse and exchange of these queries.

In Table 2 criterion (2) shows that by default the patient resource only exposes the birthdate attribute as a valid search parameter. This makes it necessary to explicitly define the age as a period relative to the current year. However, resources provide the ability to be extended by custom attributes and search parameters. For example, the Patient HL7 profile adds an extension with additional search parameters such as age to the patient resource [5], thus simplifying the criteria query to `/Patient?age=ge18&age=le34`. This allows for the query to be reused over time without modifications. This problem not only applies to the patient age, but also more generally to any situation in which absolute duration information is required from timestamps stored in FHIR resources. For example, exclusion criteria often require eligible candidates to not have undergone a specific form of therapy, or have taken some

[3] https://clinicaltrials.gov/show/NCT03388177

medication, within the past N days prior to the study begin. Extensions of resources can thus simplify constraint-based searches. However, which extensions are required to better support the search semantics for phenotyping in the context of clinical trials remains an open question.

When attempting to represent more complex eligibility criteria, one quickly runs into the limitations of the FHIR constraint-based search approach. For example, one study[4] aims to include children whose BMI is less than the 95th percentile for their age, with children between 2 and 5 years old considered for the study. In this case, one would first have to determine the age of a child, potentially calculating the BMI based on weight and height, or retrieving it from extension of observation resources, and finally determining whether the BMI is less than the 95th percentile for that particular age. Such queries, with inter-data dependencies and necessary computations, cannot be expressed.

Table 3. Example of a complex temporal inclusion criterion and the required explicit exclusion criterion necessary for correctly identifying eligible patient cohorts.

No.	Criterion	Query Representation
1	Patients who are 1 year post liver transplant, but within 5 years of transplant	/Patient?_has:Procedure:patient:code =http://snomed.info/sct|274025005 &_has:Procedure:patient:date=ge2013 &_has:Procedure:patient:date=le2017
2	Explicit exclusion criteria	/Patient?_has:Procedure:patient:code =http://snomed.info/sct|274025005 &_has:Procedure:patient:date=ge2017

Furthermore, complex temporal constraints are in general difficult to represent. Instead, only basic time ranges can be expressed. One complex inclusion criterion is shown in Table 2. Here, we are searching for patients who had a liver transplant performed on them with the temporal constraint of it happening at least one year ago but not more than five. While the query may seem correct at first, this approach shows one limitation of the constraint-based search: a patient who had a liver transplant in 2015 and an additional transplant mid-2017 would be included by criteria (1) even though the patient is not fully eligible. In order to avoid such a scenario, an additional explicit exclusion criterion (2) has to be defined, executed and the intersecting elements discarded.

On the topic of exclusion, the FHIR search standard only provides semantics for declaring that resources must *not* fulfill a given constraint for search parameters of type number, date, and quantity (e.g. using a 'not equals'-prefix) and not, for example, for disease coding. This makes it difficult to generally represent both inclusion and exclusion criteria in a single query. Instead, intersecting patients between the inclusion and exclusion criteria need to be explicitly removed from the final result set. To address this, we used the second tool described above to build the set of patient resources by excepting the set of excluded patients from the set of included ones.

The HL7 Clinical Quality Language (CQL) [6] offers a much richer set of features for formalizing, among others, trial eligibility criteria. This richness requires stronger familiarity with its constructs as opposed to the RESTful paradigm which is in wide use even beyond the healthcare sector. Included in CQL and specified in FHIR is also FHIRPath, a path-based navigation and extraction language. It defines mathematical

[4] https://clinicaltrials.gov/show/NCT03394326

operations, subsetting, filtering, and additional functions for data processing. Evaluating the potential of FHIRPath for clinical trial phenotyping is subject for future work.

5. Conclusion

In this paper, we have investigated how the standardized FHIR RESTful search operations and capabilities can be used for clinical trial phenotyping by representing eligibility criteria as a set of constraint-based filters on FHIR resources.

While the FHIR standard does not explicitly provide the ability to query patients based on trial eligibility criteria, the available operations are helpful in supporting prescreening efforts in order to reduce the set of candidates based on common, expressible criteria prior to manual verification by medical professionals.

6. Conflict of Interest

The authors declare that there are no conflicts of interest in this work.

7. Acknowledgments

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2. Consumer Health Informatics and Public Health

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An Online Differential Cost Estimator Tool to Support Cost Planning of Demographic and Health Surveys Using Mobile Electronic Device or Paper and Pen Methods: A Function of Sample Size

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Abstract. The use of mobile devices for a house to house interview-administered survey data collection is becoming a practice along with the paper-based data collection tools. Though the electronic data capture mechanism is supposed to improve the efficiency of the data collection mechanism and the quality of the data, there is limited evidence on the cost-effectiveness of the technologies. This project aims to develop an online pre-implementation survey cost estimator to support the planning and decision of implementing agency. Scalable costs with sample size were estimated using parametric cost estimating technique. In this article, we used Introduction, State of the art, Concept, Implementation, Lessons Learned (ISCIL) format to present the overall development process of this online cost estimator.

Keywords. Costs estimator, mHealth, economic evaluation, observational studies

1. Introduction

Household surveys conducted in Health and Demographic Surveillance Systems (HDSS) or at large-scale surveys such as Demographic and Health Surveys are the common types of survey for collecting data on population and health [1; 2]. To get the right information at the right time, to the right person and through the right means, we need to give emphasis to the data collection and management processes. Though all are important, the data collection mechanism is usually the source of significant blessing and curse for data quality [3; 4]. To improve the data quality, efficiency and cost-effectiveness of the data collection process, the use of a mobile device such as tablet computers for the data collection is becoming popular [5; 6]. The success history of those mobile devices based data collection lay on the proper implementation of the system in a cost-efficient manner. Recent systematic reviews are showing inconsistent evidence to concretely decide about the cost-effectiveness of those mobile devices in

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balancing the data quality and work efficiency with the incurred costs in the process [5; 7]. A lack of well-estimated cost information for healthcare information technology hampers the effective decisions in planning, budgeting, and funding of implementations. This may lead to miscalculating led failure of resource allocation and uncertainty in the decision makers.

This project aims to describe an online cost estimator tool using parametric cost estimating technique. The tool can accept input parameters from users and pass it to the models and formula and produce cost estimates of electronic mobile device and paper and pen data collection tools.

2. State of the Art

Several cost estimations tools existed in various categories which are enterprise spreadsheet-based tools [8; 9]. The tools are setting and scenario specific project management tools other than our thematics area. Cost simulation for clinical trial cost [6] and a web based survey cost calculator which only considered cost of data [10] can be taken as similar state of arts. According to the author's literature search, there is not any online cost estimator tool which estimate survey research cost using paper and electronic data collection tools. The added values of our cost estimator are the use of cost parameters that are scalable according to the survey sample size. Also, the scalable comparative graph using initial user entry parameters is possible.

3. Concept

To give the user a refined differentiation between the cost of a survey for pen and paper and a mobile device, it is mandatory first to identify the actual differences in expenditure for both tools.

House-to-house surveys consist of several time expenditures: traveling to the next destination, conducting the interview, transferring the data back into a system and analysis of the data. Additionally, there are expenditures in material, namely: traveling, printing and tablet costs. The only meaningful difference in time expenditure is the data transfer time. Using mobile device takes virtually no time with an appropriate system in place while the data needs to be entered into a computer in case of paper and pen. Apart from the traveling cost, which can be excluded, the material cost is vastly different for both approaches. There is a high startup cost per mobile device, but no need to pay additionally per questionnaire. Compared to a pen and paper survey, the electronic approach forces the implementing agency to make a sizeable purchase of a tablet computer per data collector.

3.1. Asking the Right Questions

The net cost for a pen and paper survey is the transfer cost of the data into a system plus the printing cost per questionnaire. The net cost for mobile device approach equals the number of devices acquired. All other costs are roughly even for both variants and are not needed in this estimation. The data collector cost is included considering the estimator might be adapted in the future in a situation where different data collection

time per tool is needed. A cost estimating tool which asks for information where the user does not know about it will be unusable, so it is imperative to ask questions that can be easily answered by the users. For example, it is unrealistic to ask a user for the total transfer cost of his study. Instead, we ask how many samples he wants, what the average duration is to fill out the questionnaire and how much a data collector is paid per hour. The complete list of parameters where the user needs to input is listed below in figure 1.

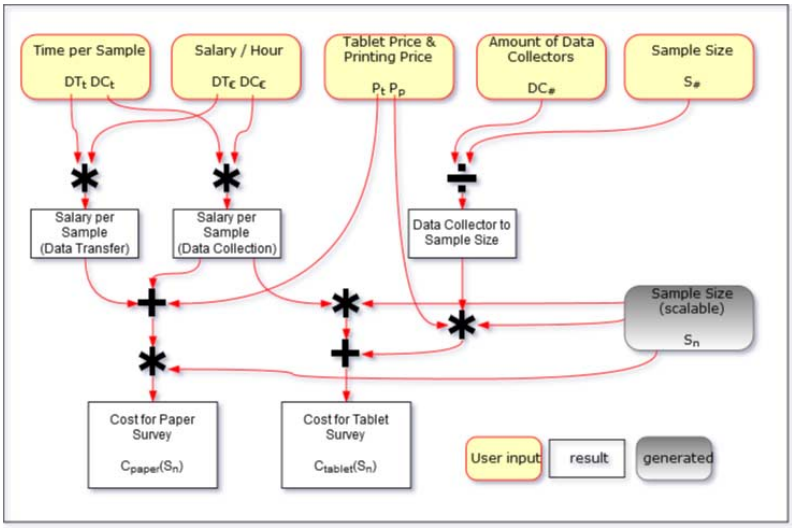

Figure 1. A graphic representation of both cost differentiation.

3.2. Formula / Equation

A graphic representation of both formulas and sub-results can be seen in figure 1. Following is equation 1 for the differentiating cost for the pen and paper approach.

$$C_{paper}(S_n) = \left(DC_€ DC_t + DT_€ DT_t + P_p\right) S_n \tag{1}$$

Given any sample size S_n greater than 0, $DC_€$ the given price per hour for data collectors, $DT_€$ for clerks, DC_t and DT_t is the average time per one questionnaire for collection and transfer and P_p paper printing cost. Multiplied with S_n, equation 1 will result in a linear graph. Equation 2 describes the differentiating costs for the tablet computer approach.

$$C_t(S_n) = DC_€ DC_t S_n + \left\lceil \frac{DC_\#}{S_\#} S_n \right\rceil P_t \tag{2}$$

Given any sample size S_n greater 0, it will calculate the data collection costs analogue to equation 1. A native solution would be to set the tablet cost equal to the amount of data collectors taken times the tablet price. This was not done, because this formula is supposed to work for any given sample size. Scaling up only the sample size

while leaving the amount of data collectors constant, would result in a longer collection period, which is not a fair comparison to make, considering data collectors for a pen and paper approach can be scaled up without additional cost. However, our formula does not display time, so we cannot use it as a means to show a higher time expense. Instead, we decided to take the number of data collector $DC_{\#}$ to sample size $S_{\#}$ (the initial fixed value given by the user) *ratio* multiplied by the scaling sample size, rounded up, to keep the collection period roughly equal. The output will be a nonlinear stair-like curve. Each stair marks a new purchased tablet for the corresponding increase of sample size.

4. Implementation

The system is implemented as a responsive designed web application to ensure a barrier-less user experience on any device with a browser and internet access. It was programmed using HTML5 for content, JavaScript with the extension JQuery to express dynamic behavior and CSS for design. We used chart.js to display the graph. For data input, simple forms are divided into tabs. The tabs are separated in a trial setting/data entry section and comparative result section. The result presented is the differential cost estimate (Figure 2).

Figure 2. Examples screen shoots input, comparative result, and simulated graph.

5. Lesson Learned / Discussion

Although estimating the actual economic cost of each data collection method is quite complicated and study specific, this estimator has attempted to highlight and compare the major differential costs for both the electronic and paper methods. The costs from this estimator considered provider perspective where the provider will use this information for planning and deciding from the available data collection tools. Cost estimation is not a full economic evaluation as it does not make explicit comparisons in the costs and consequences or benefits of alternatives. Full economic evaluation includes the comparison of costs and their outcomes for the available alternatives [11]. However, cost estimation can provide useful insights for stakeholders by identifying potential amounts of costs, understanding key drivers of costs, and preparing to estimate the resources required to sustain or expand a solution or to develop more

comprehensive economic evaluations [12]. This estimator does not calculate the full cost of conducting a survey in both tools instead; it estimates the exclusive independent costs for each tool which can be scalable over the sample size. Including all parameters of costs to our cost calculation model was a challenge. For example, if we need to ask the users to enter a majority of the input variables, the users may get challenged to know the exact amount of costs for the specific variables. On the other hand, derivating some of the cost variables might compromise the accuracy of the derived variables. To balance this, at the current version of this work we have estimated for potentially cost incurring parameters in both methods as described in the literature [11].

6. Conclusion

Incorporating cost accounting model/economic model is a challenge at this level of the work. Identification of the appropriate model which can accommodate other potential cost parameters is working in progress. This pre-implementation cost estimator believed to help the decision of implementer agency on the selection and implementation data capturing tools based on the cost.

7. Conflict of Interest

The authors state that they have no conflict of interests.

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Analyzing the Readability of Health Information Booklets on Cardiovascular Diseases

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Abstract. A low level of patient health literacy represents a major reason for worse prognosis or reduced therapy adherence. Health information booklets are a major tool for improving patient's health literacy. This paper presents a computer-based readability analysis of patient information booklets from the cardiovascular domain. The study relies on 34 English booklets mostly on heart disease, prevention and procedures. It compares five different, well-established readability instruments. On average, readers of the assessed booklets have to visit school at least until the 9th U.S. school grade when applying the Flesch-Kincaid formula. According to the Gunning-Fog metric, readers would have to attend school until the 11th grade. The presented study demonstrates the feasibility of a fully automated text processing tool-chain for patient information booklets. The results reveal that readability metrics should be carefully interpreted and only be interchanged with caution.

Keywords. Readability, health literacy, patient education, heart diseases

1. Introduction

One of the main problems of written patient information material is the gap between the language of experts and lay people. Even with a higher level of health literacy, vocabulary and concepts of diagnosis and treatment may not be easy to understand [1]. In addition, even for patients with higher levels of education understanding medical concepts might be difficult [2]. A low level of patient health literacy represents a major reason for worse prognosis, associated with less preventive actions (e.g., influenza vaccination) or reduced therapy adherence [3]. This can result in a poorer general health status for patients, especially for life-threatening incidents such as cardiovascular diseases. According to [4], this group of diseases is one of "the five leading causes of total years of life lost in 2016 [...]" from a global perspective.

Therefore, the improvement of the average health literacy in the general population and the provision of suitable information for individual patients is a major aim in most western health-care systems. In this context, low readability levels help to improve the average health literacy of the general population. Readability can be checked for with different instruments [5]. However, for extensive patient brochures, the manual computation of the related formulas is difficult, time-consuming and demands human

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and/or financial resources. This paper presents a computer-based readability analysis of patient information booklets from the cardiovascular domain. For the comparison of different readability instruments, the study relies on 34 English booklets mostly on heart disease, prevention and procedures.

2. Methods & Material

Readability is a term to describe the properties of written text with respect to the readers' understanding of a document. It depends on (a) complexity of the text structure, (b) the sentence structure and (c) the vocabulary used. For the English language, well-accepted readability instruments to rate text difficulty in U.S. school grades (G) are the:

i) SMOG readability formula G_S – [6],
ii) Flesch-Kincaid readability formula G_{FK} – [7],
iii) Automated Readability Index (ARI) G_A – [8],
iv) Gunning-Fog metric G_{GF} – [9] and,
v) Coleman-Liau metric G_{CL} – [10].

2.1. Selection of Booklets

Patient information booklets on cardiovascular diseases and prevention were collected. All booklets had to be published by a credible national organization and freely available on the Internet. In our selection we chose (a) the NIH National Heart, Lung and Blood Institute (U.S.), (b) British Heart Foundation (UK), and (c) the National Heart Foundation of Australia. Print-only booklets or multimedia health content, were not taken into account. In total, eight different topics related to the cardiovascular domain were taken into consideration, including: Heart attack and related diseases, stroke, vascular complications, effects of high cholesterol levels, prevention and procedures, and rehabilitation. We identified 38 booklets provided by the aforementioned institutions. As four patient information brochures were considered too short in text length and thus excluded, only n = 34 booklets were selected for further assessment.

2.2. Computational Processing Steps

All booklets included in the analysis were originally downloaded as PDF documents. Therefore, PDF parsers from the Apache Tika framework were applied to extract the actual text content. As a second step, the extracted text was cleaned from disturbance artifacts (e.g., different hyphen encoding schemes). Next, the aforementioned readability metrics were computed for every booklet by a self-implemented analysis software written in Java. For sentence detection, we relied on the Apache OpenNLP library and their widely accepted sentence models for the English language. Liang's hyphenation algorithm [11] was used to estimate syllable counts.

2.3. Statistical Methods

Readability grade levels of G_{FK}, G_A, G_{GF} and G_{CL} were tested pairwise against SMOG values with four 2-sided, paired *t*-Tests (multiple level of significance α=0.05, adjustment for multiple testing acc. to Holm-Bonferroni method [12]). The alternative

hypothesis (H_a) is thus defined as the difference in means for G_{FK}, G_A, G_{GF} and G_{CL} respectively, is not equal to zero. For the analyses conducted, the statistical software suite R was used in version 3.3.3 (2017-03-06) on a Windows 10 Enterprise 64bit computer.

3. Results

3.1. Characteristics

The acquisition of booklets was carried out on October 22nd, 2017. A total of 38 different patient booklets was downloaded. Four booklets could not be investigated since the lengths of the related text sections were not sufficient for the analysis. Finally, 34 booklets could be assessed within this study (see Table 1).

Table 1. Grad levels for selected readability instruments for *n=34* patient information booklets on cardiovascular diseases, with grade levels as: G_S = SMOG, G_{FK} = Flesch-Kincaid, G_A = ARI, G_{GF} = Gunning-Fog, G_{CL} = Coleman-Liau. Frequencies: Σ_w (words) – Σ_{cw} (complex words) – Σ_s (sentences).

†: As $\Sigma s < 30$, G_S adjusted according to SMOG Conversion Table II developed by Harold C. McGraw, Office of Educational Research, Baltimore County School, Towson, MD.

Booklet	Origin	G_S	G_{FK}	G_A	G_{GF}	G_{CL}	Σ_W	Σ_{CW}	Σ_S
aterial_01	UK	10.361	8.001	8.069	10.217	9.972	1128	125	78
aterial_02	UK	10.973	8.510	9.391	11.087	10.996	1511	181	96
aterial_03	UK	13.884	11.525	12.019	15.065	12.893	12679	2442	689
cholesterol_01	UK	11.637	9.211	9.547	11.944	11.065	1247	173	78
cholesterol_02	US	11.992	9.730	10.571	12.475	11.738	2403	337	140
heart_attack_01	AU	11.257	9.168	9.670	11.388	11.860	1220	168	83
heart_attack_02	UK	10.466	8.282	8.685	10.352	10.717	1408	160	97
heart_disease_01	AU	12.698	10.850	11.428	13.406	13.272	1676	289	103
heart_disease_02	AU	11.053	9.663	9.958	11.476	10.530	2610	279	145
heart_disease_03	AU	10.977	9.953	11.497	11.506	12.110	2144	217	115
heart_disease_04	AU	13.821	11.660	11.993	14.982	13.029	2912	564	161
heart_disease_05	UK	13.861	11.484	12.339	15.049	13.598	12374	2449	694
heart_disease_06	UK	13.720	11.466	12.121	14.848	13.403	13156	2557	744
heart_disease_07	UK	13.180	11.257	11.914	14.146	12.188	14692	2337	755
heart_disease_08†	US	10.013	7.747	8.679	9.365	11.480	360	38	28
heart_disease_09	US	10.799	8.669	9.886	10.766	12.290	2112	265	147
heart_disease_10	US	11.912	11.073	12.986	12.978	12.456	2354	260	110
prevention_01	AU	11.655	9.745	10.510	12.090	11.489	5388	686	308
prevention_02	AU	11.643	10.213	11.642	12.086	12.858	2145	271	122
prevention_03	US	11.208	8.980	10.359	11.345	12.458	2910	382	191
prevention_04	US	11.350	9.105	10.515	11.536	12.606	3020	408	197
prevention_05	US	11.796	9.758	10.881	12.178	12.487	4150	580	252
prevention_06	US	11.113	8.810	11.114	11.207	13.533	958	125	64
prevention_07	US	11.464	9.369	10.693	11.734	12.399	21634	2846	1337
prevention_08	US	10.139	7.567	8.527	9.870	10.940	1240	137	91
prevention_09	US	12.399	11.282	13.555	13.525	13.025	9441	1148	436
procedures_01	AU	12.275	9.752	10.902	12.826	13.162	1564	264	103
procedures_02	UK	10.512	7.859	7.819	10.355	10.035	1285	157	94
procedures_03	UK	9.946	7.542	7.154	9.664	9.064	824	84	59
procedures_04	UK	10.652	8.278	8.225	10.659	9.730	1213	137	79
procedures_05	UK	10.977	8.246	7.859	11.001	9.762	1019	134	71
rehabilitation_01	AU	11.802	10.136	9.870	12.198	11.125	1366	189	82
rehabilitation_02	UK	11.551	9.984	9.554	11.794	11.724	1101	163	75
stroke_01	UK	10.718	8.080	8.394	10.631	11.086	1100	150	85
Mean ($\bar{G}$)		11.576	9.499	10.245	11.934	11.796	4011	609	233

In sum, 14 booklets (41.17%) were available from the UK, 11 (32.36%) from the U.S. and 9 (26.47%) from Australia. Most booklets referred to heart diseases in general (10; 29.4%) followed by brochures on prevention (9; 26.5%) and text material describing cardiovascular procedures (5; 14.7%).

3.2. Readability

The results in Table 1 list the computed grade levels G of the readability metrics selected. On average, readers of the booklets have to visit school at least until the 9th grade in the U.S., when assessed with the Flesch-Kincaid metric: G_{FK} = 9.499, SD = 1.275. By contrast, according to the Gunning-Fog metric, the same reader would have to attend the same school until approx. the 11th grade (G_{GF} = 11.934, SD = 1.569). For all 34 booklets, the absolute numbers of words (Σ_w), complex words (Σ_{cw}) and sentences (Σ_s) vary greatly, ranging from: $min(\Sigma_w)$=360; $max(\Sigma_w)$=21, 634; $min(\Sigma_{cw})$=38; $max(\Sigma_{cw})$ =2, 846; $min(\Sigma_s)$=28; $max(\Sigma_s)$=1, 337. Mean values: $\bar{\Sigma}_w$=4, 011, $\bar{\Sigma}_{cw}$ =609, $\bar{\Sigma}_s$=233.

3.3. Comparison

The 2-sided, paired t-Test shows a significant difference between the means of G_{FK}, G_A and G_{GF} when compared against G_S (each $P < .001$), as listed in Table 2.

Table 2. Pairwise readability grade level comparison of mean values ($\delta_{\bar{G}}$) against *SMOG*; *n = 34*.

Readability Instrument	$\bar{G}$	SD	$\delta_{\bar{G}}$	P	95% CI of $\delta_{\bar{G}}$
Flesh-Kincaid	9.499	± 1.275	2.084	< 0.001	[1.918; 2.250]
ARI	10.245	± 1.614	1.338	< 0.001	[0.989; 1.686]
Gunning-Fog	11.934	± 1.569	-0.351	< 0.001	[-0.520; -0.183]
Coleman-Liau	11.796	± 1.222	-0.214	0.172	[-0.526; 0.098]

For the comparison between *SMOG* and *Coleman-Liau*, H_0 cannot be rejected, i.e., there is no significant difference between both metrics in our study setting.

4. Discussion

The length of some of the assessed material exceeded more than 500 sentences, e.g., *prevention_07, heart_disease_05 | 06 | 07* and *aterial_03* (see Table 1). These book- lets would be hard to assess when conducted manually, demanding time- and human resources. In this context, an automatic processing tool, as outlined and applied in this study, can be used for several readability instruments at once to overcome these obstacles.

However, several limitations apply for the study setting as we selected merely book- lets from highly credible organizations. Therefore, our results cannot be applied to every patient information brochure found on the Internet. Yet, the authors value trustworthiness of such patient information material over its heterogeneity.

Moreover, disturbance artifacts – e.g., different kinds of hyphens, markup language (i.e., XML or HTML meta-tags), mis-encoded characters from different encoding schemes, etc. – may still be included in the extracted raw text material.

Furthermore, for all computed instruments, an exclusive computation of readability does not reflect the individual knowledge or the motivation of patients or family

members. In addition, aspects such as illustration and type-setting were not assessed in this study. Consequently, health information should not merely be rated on the basis of readability grade levels.

5. Conclusion

The findings of our study mostly confirm the results of previous work published in this field [13–15]. All assessed booklets exceeded the 7th-grade reading level as found in a study by Walsh et al. which analyzed consumer health information offered by the American Heart Association [5]. Wang et al. stressed that different readability instruments should not simply be interchanged when assessing the same text material [16]. Future work includes a detailed technical description of the underlying tool-chain as well as a free service for the research community.

6. Conflict of Interest

None declared.

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3. Data Quality

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German Medical Data Sciences: A Learning Healthcare System
U. Hübner et al. (Eds.)

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Interactive Feedback of Data Quality in Clinical Research. A Case Study from an Infectious Diseases Cohort

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Abstract. To improve data quality, clinical cohort studies require ongoing centralized data monitoring. Results of data quality monitoring are often reported in a general format, without individual feedback for different types of users within the research consortium. Several R packages provide the possibility for an easy to use approach to analyze and communicate information in interactive web-based forms. This article describes a pilot, which is applied in a German infectious disease cohort using interactive feedback and gamification to report data quality, to ultimately increase the quality of research outcomes.

Keywords. Data quality, interactive feedback, dashboards, infectious disease cohort, gamification

1. Introduction

Evaluating data quality (DQ) during a clinical study needs to be an ongoing procedure [1,2]. Unfortunately, DQ assurance is not applied to every clinical study. Clinical trials that apply ongoing data monitoring, often report their results in pdf-Files sent to the principal or local investigators to plan further actions which increase DQ. This approach alone, however, is generally not effective in reducing data inconsistencies.

Evaluating DQ needs to be accompanied by a targeted communication of its results to the different user groups: study nurses, (principal-) investigators and the research community. General forms of communications are text or tables. Figures are an important instrument for research communication, but figures are the results of stories told by its underlying data [3]. All used elements need to have one goal: to motivate the users to increase the value of their data.

In this article, we present an approach to implement targeted individual and interactive web-based feedback using different packages in R.

1.1. Infectious Diseases Cohort Study

The Transplantation Cohort (TxC) of the German Center for Infection Research (DZIF) collects information about infectious complications from patients with different types of organ transplantations (Tx) from five large German university hospitals. The Helmholtz

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Centre for Infection Research (HZI) performs the epidemiological evaluation of the collected information. For this, data needs to be aggregated to provide an overview of study performance and DQ.

1.2. Evaluation and Visualization of Data Quality

DQ is multidimensional, but still not a standardized term [4]. The Technology, Methods, and Infrastructure for Networked Medical Research (TMF) published a guideline for the evaluation of DQ in clinical cohort studies and disease registries [5]. Here, several DQ indicators (e.g. completeness or value distribution) are defined to support data managers, who need to adapt important indicators to quantify trial performance [6] and to generate feedback for its users.

Web-based feedback about DQ is not a novel approach. It is already applied in private companies to measure metrics that show trends of key performance indicators. However, DQ feedback is rarely implemented in publicly funded clinical cohort studies, and its impact on trial performance is not researched well enough [7–9].

2. Methods

Received data exports consist of multiple csv-files of different records accompanied by metadata (e.g. type of data field, allowed data range) and data dictionaries. Data is twice pseudonymized, still allowing linkage of data between the records.

We developed for our case study a framework for DQ feedback using flexdashboard as visualization approach [10]. For data analysis and visualization, the programming language R was used [11]. For each analysis, a dataframe containing aggregated information and graphical objects were separately stored in an RData-file, allowing the import into the flexdashboard.

2.1. Evaluation of Study Performance and Data Quality with R

For automated DQ assessment we used R tidyverse packages [12]. We used the DQ dimension completeness, which we defined as the ratio of not entered fields divided by all available data fields of a measured variable. For selected continuous variables (e.g. length of ICU stay, surgery time), data distribution was assessed, providing case-based warnings for study nurses, if the data points could be identified as outliers.

Study performance was assessed by descriptive analysis of study population and biosamples. Comparing actual and target count of follow up (FU) visits was conducted to assess FU performance. If patients have an Tx date in the cohort study, regularly FU visits need to be performed, after 3, 6, 9, 12 months and annually thereafter.

The package lubridate [13] was used to evaluate the FU. Lubridate uses interval class objects. The analysis was performed by measuring, whether the expected FU is within the time interval (+/- 28 days) of the required FU. Fig. 1 provides an example, where the first two FU visits match, but the third visit was performed after the allowed time interval, and was subsequently reported as a warning.

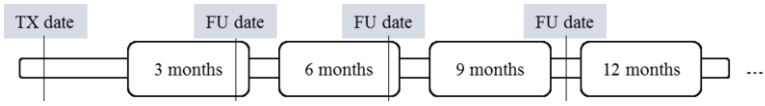

Figure 1. Follow up (FU) evaluation in accordance to entry date.

2.2. Interactive Visualization of Information with R

We used knitr and datatables (DT) packages [14] for rendering tables. For the DT package, we used an extension, providing the users an option to directly export the table into standard data formats (CSV/XLS/PDF). Graphs were rendered using ggplot2 and plotly [15]. For colorblind purposes, we applied the color palette viridis [16] on graphical elements.

3. Results

The general frame of the dashboard contained the overview of study population, biosamples, FU visits and DQ. Fig 2 shows an example of the dashboard DQ frame.

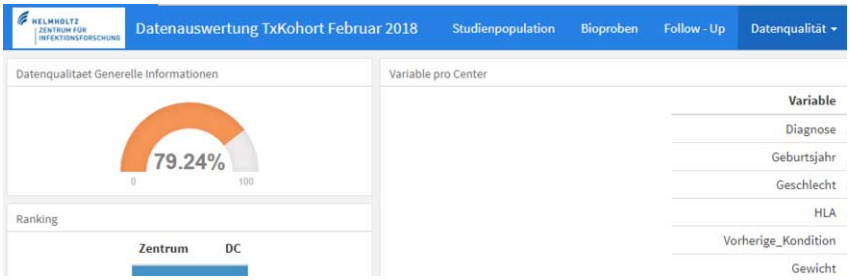

Figure 2. Example of the upper DQ screen.

3.1. Demographic and Population Information

First page of visualized information is the overview of study performances. Here, the number of overall enrolled participants is shown as a bar graph, distinguished into every type of organ, each bar graph stratified by center. The graph is scripted using the syntax of plotly. The interactive element grouped by center, which provides the possibility to show a bar graph of all types of organs by a single or multiple selected centers.

Second element is a DT element, providing an additional overview that shows total numbers of patients by center and by type of organ.

Another important element is the recruitment chart, which highlights positive or negative trends of patient enrollment per center. We used a plotly line graph, to provide the possibility of an interactive graph to show lines of selected centers.

Forth element is a set of important values of all other forms, including total number of biosamples, FU performance values and overall DQ rates. This is helpful to show the user all important information on the landing page without working through forms.

3.2. Feedback on Biosamples

The head of the page provides three value boxes, summarizing total number of blood samples, total number of other samples and the total number of unique patients that provide these samples.

Below the value boxes two tables are provided to the user, showing different types of blood samples (EDTA, PBMC, RNA, or Serum blood samples) per center. The second table has the same structure, showing types of other samples (e.g. stool or urine samples), by center. Both tables have tab stops that subset the data by type of organ. When moving to the biosample page, first tab stop shown, is the total number of samples, which is the sum of all subsets (types of organs).

During data modelling it was discovered, that some patients provide samples for two organs. These patients are provided as a case-based list, allowing study nurses to check their plausibility. It is often the case that kidney transplanted patients could also have a pancreas transplantation. However, all combinations should be checked, therefor a list of warnings is exported.

3.3. Feedback on Follow-Up

Two main elements are reported; graphs, which show the actual and target number of visits of all patients with entry date, and tables with total numbers and percentages by center. Graphs and tables have tab-stops, which allow showing results per type of organ.

The graph is a line graph with colored area to y=0. We are using green to visualize the correctly performed FU and red for the not correctly performed FU (not performed or not matching the required interval). The larger the red area, the lower is the FU performance. Each center has an individual graph.

3.4. Feedback on Data Quality

Elements used for feedback about DQ are gauges, tables with center rankings (gamification element), line graphs with data quality trends and tables showing item completeness for each data item per center (see Fig. 2).

Thirteen different records are analyzed (e.g. general information, immunosuppressive drugs, bacterial infections, laboratory results, etc.). Every form has the same structure to provide the user a simplified overview of its DQ. Primary assessment is the evaluation of data completeness, which is accompanied by single plausibility checks (e.g. scheme of immunosuppressive drug). The mean of data completeness and data plausibility is defined as DQ.

4. Discussion

This regularly created report is provided via encrypted email as an html file to the research consortium. Other options should be discussed with regards to methods (e.g. providing a server with login to the user) and data security.

Flexdashboards have also limited flexibility (e.g. the usage of tab stops in only one element instead of all), other more code intensive feedback options like shiny

applications or dashboards options using JavaScript solutions (e.g angular.js in combination with d3.js) need to be considered.

One mandatory feature that needs to be added is that case-based lists need to be provided as check lists, simplifying work flows of study nurses.

5. Conclusion

Further research needs to assess the impact of interactive (web-based) communication on DQ. For the TxC, a quasi-experimental study design in accordance to [17] needs to be developed, e.g. by using a stepped wedge design and providing interactive feedback for a period of time to one center, after that adding other centers and evaluating the change of DQ.

6. Conflict of Interest

The authors declare no conflict of interest.

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Practical Extension of Provenance to Healthcare Data Based on the W3C PROV Standard

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Abstract. Secondary use of healthcare data is dependent on the availability of provenance data for assessing its quality, reliability or trustworthiness. Usually, instance-level data that might be communicated by HL7 interfaces entail limited metadata about involved software systems, persons or organizations bearing responsibility for those systems. This paper proposes a strategy for capturing interoperable provenance data needed by data stewards for assessing healthcare data that are reused in a research context. Aimed at a realistic level of granularity even system-level metadata will support a data steward trying to trace the origins or provenance of healthcare data that have been transferred to the research context. Those metadata are extracted from the 3LGM^2-system, used for modelling hospital information systems. Based on the W3C provenance specification interrelated activities, entities and agents can be integrated and stored in RDF triple stores and therefore queried and visualized.

Keywords. Secondary use of EHR data, interoperable provenance data

1. Introduction

In medical research, genotype and phenotype data from multiple sources as well as workflows and log records of corresponding data processing steps are distributed among heterogeneous information systems that are used under the authority of different healthcare actors [1]. Provenance is regarded as the equivalent of a logbook: Capturing all the steps that were involved in the actual derivation of a result, and which could be used to replay the execution that led to that result [2]. In the comparatively young field of bioinformatics the demand of reproducible science is accomplished by provenance-aware workflow systems e.g. Galaxy [3]. These workflow systems take advantage of an ecosystem of almost harmonized services, e.g. from EMBL or NCBI which can be integrated in compact workflows for generating experiment pipelines. Such an approach with executing prospectively defined process models are not suitable for capturing the provenance of healthcare data. The recording of relevant context data for data capturing and processing steps of interest needs to be treated differently.

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In the following a twofold approach for capturing provenance data is proposed: The data processing activities and entities at instance-level are enriched by linkages to responsible agents at system-level. In research context data stewardship would gain a significant added value, reviewing the quality of medical data, if links to responsible applications, persons or organizations are available, see chapter 5 "Conclusion".

2. Methods

The de-facto standard W3C PROV for interoperable provenance data [2, 4] is based on a conceptual model with basically three classes *activity*, *entity*, *agent* and nine binary relationships like "*wasGeneratedBy*" or "*startedAtTime*", see Fig 1. Data processing steps like "*laboratory_data wasGeneratedBy laboratory_ analysis*" are represented as RDF-triples of corresponding instances. Using semantic web formats like OWL for ontologies, RDF for instance data and several data interchange formats like RDF/XML, N3 or Turtle, facilitates the utilization of powerful tools. For processing, querying or visualizing provenance graphs consisting of interconnected triples; e.g. the ontology editor PROTÉGÉ [5], SPARQL-query tools [6] and visualization tools [7] can be used.

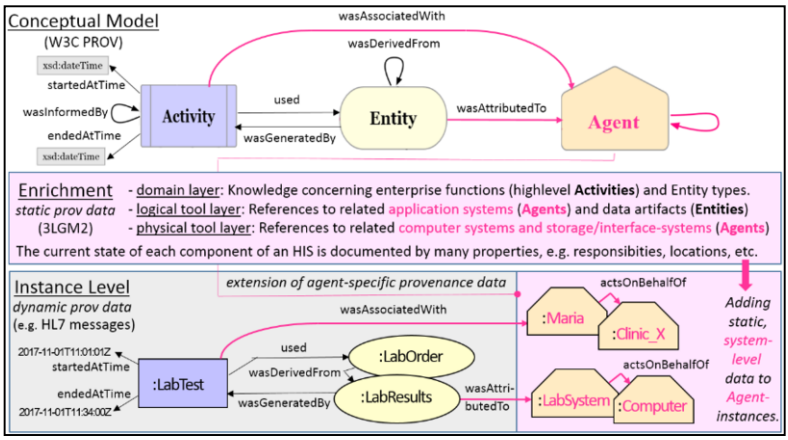

Figure 1. General approach for capturing interoperable provenance data.

In order to capture provenance data of processing steps, analogously to the bioinformatics pipelines, the workflow tool TAVERNA [8] has been evaluated. By launching a defined workflow with suitable service connectors instance-level data from different sources can be accessed. TAVERNA inherently records service invocation, intermediate and final workflow results and exports provenance data as RDF triples formatted in Turtle syntax conformant to the W3C PROV standard. However, the lack of suitable service connectors in routine clinical application systems like a laboratory software, leads to the conclusion that TAVERNA is not suitable for accessing the provenance of healthcare data. For that reason, available clinical data is used directly for deriving a W3C conformant model of provenance data, e.g. by starting from HL7 V2 messages. The main motivation for this paper is the observation that provenance data is not available or is not communicated, e.g. attributes for almost all data elements in databases like "created_by", created_date". Therefore, missing information about responsibilities are supplemented by linkages to suitable metadata at the system level, e.g. about agents like administrating persons related to application software systems.

3LGM2 is a tool for modelling hospital information systems at three interrelated layers presented in Fig 1 [9]. The enterprise functions at the domain layer correspond to W3C PROV *activities,* which are supported by software systems at the logical tool layer, known among W3C as *agents*. These systems use data storage or message artefacts correlated to W3C *entities,* that are realizations of Entity types in 3LGM2 models. References to the physical tool layer allow accessing information about responsibilities for hosting application systems opposed to context data concerning responsibilities to operate software systems at the logical tool layer.

The 3LGM^2 tool provides a set of predefined queries, like the availability of enterprise features because of network component failures or generic assessments like functional redundancies within a Hospital Information System (HIS). However, it does not support simulations of enterprise functions by processing real data and adding provenance at all three layers [10]. With respect to missing instance-level data, context data describing W3C agent instances, the static system-level data, structured documentation on the state of a distributed information system is very valuable.

3. Results

A simplified laboratory use case has been chosen to demonstrate the proposed strategy for integrating instance- and system-level provenance data by using the mentioned tools. Aiming at an improved secondary use of healthcare data in the medical research context, consumers like data stewards, members of use-access-committees or researchers should be supported in assessing data´s context and history.

3.1. Phase 1: 3LGM^2 Model Providing System-Level Data to the PROV Agent Node Type

At all three layers the consecutive data processing steps “capturing of laboratory data”, “mapping to LOINC” and “re-used in a research IT platform” are modelled. All components are described in detail via dialog windows, as shown in Fig 2. Of special interest are involved application and computer systems providing provenance data for agents.

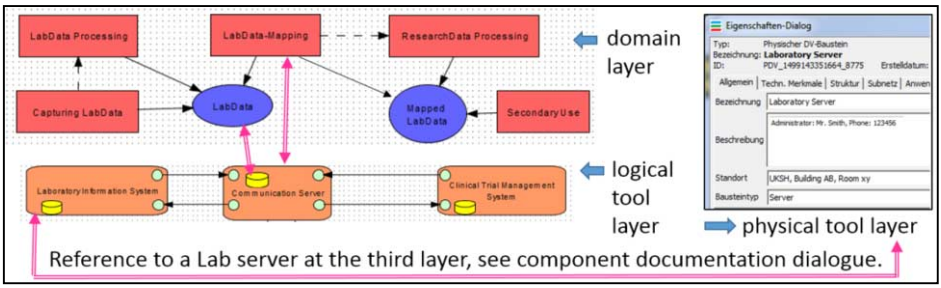

Figure 2. Simplified 3LGM2 model with exemplary inter-layer relationships based on a laboratory use case.

3.2. Phase 2: Instance-Level Provenance Data Extracted from HL7 Messages

Instance-level provenance data can be obtained for example by analyzing and annotating HL7 messages with respect to suitable PROV classes “ACT” (e.g. LabTest “UMIC^ Urinalysis” in the OBR segment) and “ENT” (e.g. LabResult like “UGLB^Glucose” in

OBX segments). The header segment provides the corresponding instance of the PROV class “AGNT”. The laboratory information system presented in the 3LGM^2 model at the logical tool layer in Fig. 2 are provided by HL7 V2 messages. The LabData-Mapping where proprietary names are mapped to LOINC (e.g. UGLB^Glucose to 25428-4) is not explicitly shown.

3.3. Phase 3: Integrating Instance- and System-Level Data from Phase 1 and 2

RDF (Resource Description Format) is used for representing instance data from HL7 messages [11] as well as system level data from the 3LGM^2 model complemented by general data about responsible persons and organizations. To avoid name collisions and to refer to defined ontologies namespaces like “f3LGM” [12] and “FOAF” [13] are necessary. For space reasons, the following examples illustrate just the rough idea.

```
<LabTest_2018_9876543...>
  rdfs:label "UMIC^Urinalysis" ;
  rdf:type prov:Activity ;
  hl7:segment “OBR_1 (LabResult)” ;
  hl7:message <http://hl7.org/message/2018/a63d332-17ea-4fa3-.....> ;
  prov:startedAtTime  "2009-05-04T12:13"^xsd:dateTime ;
  prov:wasAttributedTo  <LabSystem#> . “Taken from HL7 message header”
```

```
<LabSystem#>
  rdfs:label "Laboratoryinformation System" ;
  rdf:type prov:SoftwareAgent, f3LGM:Appl.System ;
  f3LGM:basesOn “OPUS::L” ;
  prov:actedOnBehalfOf <Computer_12> .

<Computer_12>
  rdf:type prov:SoftwareAgent , f3LGM:Comp.System ;
  f3LGM:belongsTo  <subnet_123> ;
  f3LGM:isLocated <UKSH_room_456> ;
  f3LGM:ContactPerson  <Person_789> .
```

```
<Person_789>
  rdf:type prov:Agent, foaf:Person ;
  foaf:givenName “Mr. Smith” ;
  foaf:mbox <mailto:smith@example.org> ;
  prov:actedOnBehalfOf <Organization_159> .

<Organization_159>
  rdf:type prov:Agent, foaf:Organization ;
  foaf:name “University Hospital ” .
```

Note(f3LGM): Namespaces must start with letters.

3.4. Phase 4: Analyzing the Complete Provenance Graph

Due to the mainstream technology many opportunities of analyzing the integrated RDF dataset occur. Straightforward, the turtle file with provenance data can be visualized [7].

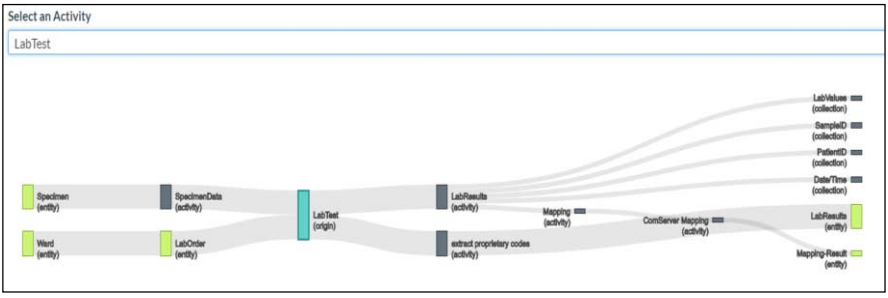

Figure 3. Web-based PROV-O-VIZ tool for visualizing W3C PROV conformant provenance data.

For a deeper analysis, semantic web tools like the ontology editor PROTÉGÉ [5] or triple stores like LUPOSDATE [6] are used for evaluating SPARQL queries. For enabling inferences based on the PROV ontology with suitable axioms like inverse properties, e.g.

"Activity prov:generated Entity" or subclass relationships, e.g. *agent* with subclasses like Person, the exported turtle file and PROV OWL file are merged.

4. Discussion

Extracted healthcare data from HL7 messages and linked metadata provided by a separated HIS modelling tool like 3LGM^2 can be integrated suitably by using RDF triples conformant to the PROV model. This was prototypically implemented to demonstrate the feasibility of this approach for making provenance data available und usable. However, the 3LGM^2-tool needs improved interfaces for accessing agent-related data automatically and a suitable mechanism for uniquely identifying agents.

5. Conclusion

Adding metadata about responsible persons or other relevant context item alone is certainly not enough in the end. Regarding the potential granularity of provenance data targeted to application systems, data elements or values, the proposed approach can be extended by using more ambitioned methods presented by Curcin et al. [14].

6. Conflict of Interest

All authors declare no conflict of interest.

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4. Data Sharing and Secondary Use

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German Medical Data Sciences: A Learning Healthcare System
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A Web Service to Suggest Semantic Codes Based on the MDM-Portal

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Abstract. Annotation with semantic codes helps to overcome interoperability issues for electronic documentation in medicine. However, the manual annotation process is laborious and semantic codes are ambiguous. We developed a publicly accessible web service to alleviate these drawbacks with a sophisticated and fast search mechanism based on more than 330,000 semantic code suggestions. These suggestions are derived from semantically annotated data elements contained in the Portal of Medical Data Models manually curated by medical professionals. Integrating this suggestion service can support the manual annotation process and strengthen uniform coding. Integration is demonstrated for two separate data model editors. Usage statistics show over 5,500 suggestion requests per month for semantic annotation of items. The web service may also prove helpful for automatic semantic coding.

Keywords. Metadata, semantics, data curation, unified medical language system

1. Introduction

Electronic documentation is becoming ubiquitous in modern medicine and provides key benefits regarding data quality [1] and cost-efficient documentation [2]. It might even lead to improved caregivers' decisions [3]. However, the adaption of electronic documentation goes along with a growing demand for semantic interoperability [4]. Many documentation processes are still developed from scratch and lack harmonization, impeding automatic data integration [5]. Annotation of items with unique semantic codes is one possibility to diminish this interoperability issue. Furthermore, such codes enable semantic analysis of electronic documentations [6, 7]. The Unified Medical Language System (UMLS) is a common choice for semantic coding in the medical field combining several terminologies such as ICD-10, LOINC and SNOMED CT [8].

However, semantic coding suffers two severe drawbacks. Firstly, semantic annotations are ambiguous. Inter-coder reliability has been proven to be low or unsatisfactory in SNOMED-CT [9] or UMLS [10]. Secondly, the manual annotation process is laborious. In our experience, creating a coded item takes approximately two minutes by a medical professional. Dugas et al. [11] already demonstrated the feasibility of a UMLS code suggestion mechanism based on codes from the Portal of Medical Data Models (MDM-Portal) for uniform and faster semantic annotation.

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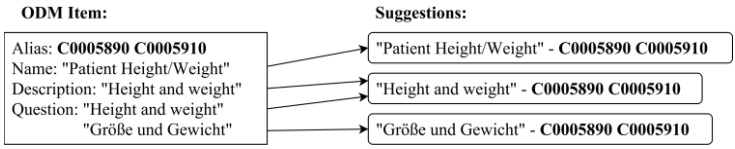

Figure 1. Pairs of text and semantic codes generated from an ODM *Item* element.

The objective of this work is to leverage this approach concerning search quality and performance. Moreover, our suggestion service should be publicly accessible through a standardized web interface, supporting integration into other systems. This work can also be considered as a first step towards a reuse repository for semantic codes and their feature sets to improve automatic semantic coding as proposed by Christen et al. [12].

2. Methods

The web service suggesting semantic codes of data elements is based on pairs of text and associated UMLS codes from semantically annotated data models in the MDM-Portal. This approach is identical to [11]. Data models in the MDM-Portal are stored in the CDISC Operational Data Model (ODM) format [13]. ODM defines a hierarchical XML structure for metadata of clinical studies as *Protocol-StudyEvent-Form-ItemGroup-Item-CodeList-CodeListItem*. Since models from the MDM-Portal contain semantic annotations mainly on the level of *ItemGroup*, *Item* and *CodeListItem*, we only take into account those three. UMLS codes are assigned based on the text content (name, description, question and label) of the ODM elements. As illustrated in Figure 1, identical pairs of text and semantic codes are taken into account only once. We decided to use such simplistic pairs to keep the search mechanism simple and to enrich it with many text variations.

Apache Solr is used to store the pairs of text and codes. It offers full-text search and a sophisticated query language. Similar to a database table, a schema defines entries in the so-called Solr index. In our case, it consists of a unique identifier, the pair along with the definitions of the UMLS codes and the URLs of its occurrences. We implemented the application with the Java web framework Spring Boot to make the service publicly accessible. Moreover, we use the libraries SolrJ to simplify the communication with Solr and Java Architecture for XML Binding (JAXB) to handle the XML-format of ODM.

Five tests were verified by manual review of the Solr index: (1) import of data models, (2) reimport of same files without adding new pairs, (3) deletion of overlapping data models, (4) import of overlapping models and (5) deletion of all models. We selected a test set of 20 ODM files, ten by special characteristics, another ten randomly.

3. Results

Our intention was to develop a web service to suggest semantic codes based on semantically annotated data models in the MDM-Portal. Figure 2 shows a conceptual overview of the resulting application. The MDM-Portal (right-hand side) automatically synchronizes with the suggestion service by sending or requesting the deletion of data models through a web interface. The service translates incoming queries (left-hand side) for Apache Solr. The search results are encoded as JSON.

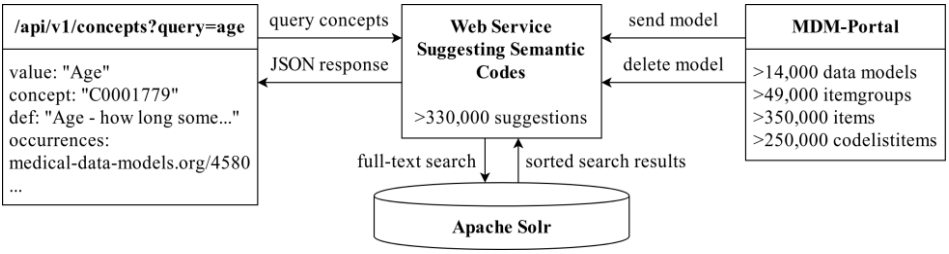

Figure 2. A conceptual overview of the developed web service suggesting semantic codes.

Table 1. Statistics for ODM elements and annotated texts based on data models from the MDM-Portal.

Element	Occurrences	Unique occurrences	Annotated Texts	Unique Annotated Texts
ItemGroup	49,106	36,896	30,982	10,163
Item	354,732	217,627	630,425	298,118
CodeListItem	264,270	98,589	126,520	34,090
Total	668,108	353,112	787,927	336,424 (342,371)

The web service is publicly available through the MDM-Portal domain with the path *MDR*. This landing page contains a documentation of the available web interfaces. The suggestion service, specifically, is accessible via */api/v1/concepts*. It supports the optional parameters *query* for searching, *sort* for sorting criteria and direction, *page* and *size* for pagination of the results, and the mandatory *apiKey* for authentication. For instance, retrieving semantic codes for *age* can be done by specifying the *query* parameter accordingly[2]. The access was secured with API keys to prevent web scraping and to avoid overload situations. The necessary keys will be issued upon request.

Apache Solr takes into account the complete annotated text and UMLS code due to its full-text search capabilities. For each search request, Solr uses an internal score ranking how well an annotation pair fits the query. In order to boost frequent pairs, we defined the sum of this internal matching score and the logarithm of the number of occurrences weighted with one tenth as the default sorting criteria. Moreover, the query parameter supports Boolean connectives, exact matches via quotes, and fuzzy search among others. The average time of ten requests measured with Firefox's internal network tool for *age* (total results: 3863, response size: 163.81 kB) were 18.5 ms (max: 23 ms, min: 15 ms) and for *cardiovascular* (1123, 5.49 kB) 7.1 ms (max: 12 ms, min: 6 ms). The response time seems to be proportional to the number and size of the results.

Table 1 summarizes statistics about the content. The ODM elements *ItemGroup*, *Item* and *CodeListItem* are distinguished. The first two columns show the number of occurrences with and without duplicates contained in the MDM-Portal. These elements do not have to contain semantic coding. The third column shows the number of text and codes pairs generated from semantically annotated elements. Identical texts in a single element are only taken into account once. In the last column, counts of unique pairs for each element are given. The total count also disregards identical texts across different elements. Hence, it is lower than the sum given in brackets. The suggestion service is based on 787,927 UMLS annotated texts and contains 336,424 unique annotation pairs.

We integrated the semantic coding web service into two ODM model editors. The ODMEdit tool implemented in R [11] and an internal editor of the MDM-Portal written

[2] https://medical-data-models.org/MDR/api/v1/concepts?query=age

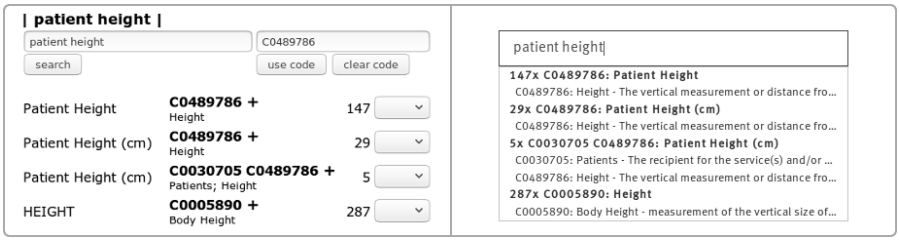

Figure 3. Integration of the suggestion service into ODMEdit implemented in R [8] and into the MDM-Portal editor written in Java. Results are truncated to top four suggestions determined by our sorting strategy.

in Java. Both use standard methods for web communication to query the suggestion service. The left part of Figure 3 shows the integration into ODMEdit with the example input *patient height*. It separates the text of the annotation pair, on the left side, and the associated UMLS codes along with their short description in the middle. On the right-hand side, the count combined with URLs to the according data models in the MDM-Portal is given. The right part of Figure 3 shows the internal editor that offers suggestions while typing. It includes the full description for UMLS codes, but does not include URLs to the original data models. Both model editors can readily adopt the suggested codes allowing a direct re-use of semantic annotations. Usage statistics of the suggestion service in ODMEdit show 5,575 requests per month with 2,860 unique queries. The three most frequent search terms are *Age* (112 occurrences), *age* (104), and *pregnancy* (72).

4. Discussion

Dugas et al. [11] stored suggestions in a database and performed searches with exact and partial matching. We could significantly improve the response time and offer a more sophisticated search mechanism by applying the platform Apache Solr. Moreover, the suggestion service is now available through a standardized web interface simplifying the integration. This was demonstrated by embedding it into two separate data model editors.

Suggestions are derived from annotated data models in the MDM-Portal. Hence, the quality of the semantic annotations depends on the portal's content, which is curated by medical professionals. However, a review of this quality does not lay in the scope of this work. Furthermore, the suggestion service only contains a subset of the UMLS vocabulary. There is also a web interface to explore the whole UMLS maintained by the U.S. National Library of Medicine [8]. However, solely relying on this service still suffers the problem of ambiguous codes and the lack of reusing a rich expert-coded text corpus.

Automatic semantic annotation of medical documents is still a difficult problem. As suggested by Christen et al. [12], our approach could serve as a repository of semantic annotations that provides an additional knowledge source for (semi-)automatic systems.

The developed suggestion service has limitations. First, the search mechanism is based on simplistic pairs of text and semantic codes. In future work, these might be enriched with further information such as the context in which a semantic code occurs. Moreover, our approach is limited to CDISC ODM as input format and tightly connected with the MDM-Portal. Therefore, it is only possible to add annotation pairs by uploading new models to the portal. However, since the search results are based on pairs of text and codes this could easily be extended to further formats that offer the generation of such pairs. In addition to that, entries solely exist for the UMLS vocabulary simply due to the

vast collection of such codes offered by the MDM-Portal. The same approach would also work with a different medical vocabulary or terminology.

5. Conclusion

In this work, we presented a web service for suggesting semantic codes that aims to alleviate two significant drawbacks of semantic coding, the laborious manual annotation process and ambiguities of codes. The resulting application provides search results based on over 780,000 UMLS annotated texts of which more than 330,000 are unique. Integrating this web service into external applications is simple and we encourage requests for scientific collaborations. Reusing same suggestions in different settings would strengthen uniform coding outside ODMEdit and the MDM-Portal and ought to deliver valuable feedback for improvements. Future work includes enriching the coding mechanism with additional information such as the context in which an annotation occurs.

6. Conflict of Interest

The authors declare that they have no conflict of interest.

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Metadata Import from RDF to i2b2

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Abstract. Metadata management is an important task in medical informatics and highly affects the gain out of existing health information data. Data Warehouse solutions like Informatics for Integrating Biology and the Bedside (i2b2) are common tools for identifying patient cohorts and analyzing collected clinical data while respecting patient privacy. The Resource Description Framework (RDF) is designed for highly interoperable ontology representation in various formats, facilitating ontology and metadata management. Our approach is to combine i2b2's and RDF's benefits by importing the easy-to-edit RDF ontology into the extensive-research-enabling i2b2 software. We do so by using a SPARQL Protocol and RDF Query Language (SPARQL) interface, that enables RDF data queries, and developing a java program, which then generates i2b2-specific SQL insert statements. To demonstrate our solution's feasibility, we transcribe our lung disease specific ontology to RDF and import it into our i2b2 data warehouse.

Keywords. Biological ontologies, metadata, organization and administration, automatic data processing, information systems

1. Introduction

In medical informatics, metadata management is an important and demanding task. It is indispensable for data harmonization, data integration, data quality management and data comparison. Applied on clinical data, these processes result in large pools of consolidated, corrected and annotated data, enabling large-scale clinical data analysis and trial patient recruitment [1].

The clinical data warehouse software i2b2 serves as storage system for clinical data as well as metadata and offers a reliant and effective tool to support clinical trials by either prospectively finding cohorts of patients fulfilling specific constraints or retrospectively making further use of already collected routine health care data. Established in 2004, 7 years later already 60 academic health care centers have adopted this software [2]. It is funded by the National Institutes of Health (NIH). A large community continuously enhances its features. For example, by developing solutions for challenges created by i2b2's founders at Harvard MIT Division of Health Sciences and Technology. From 2006 to 2012 there are 124 publications listed, enabled by these challenges [3]. From 2016/01 to 2017/10 PubMed lists 46 articles related to i2b2, which shows that i2b2 and its applications are of ongoing interest and usefulness to the research community.

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RDF is a W3C standard and designed for ontology representation and described resources may be annotated with literals (e.g. labels or codes) and connected to other resources for large knowledge-graphs. It is commonly used in many fields like content management, content discovery, data integration, semantic annotation and schema mapping. Several use cases show Health Care applications' clear dependency on these fields [4]. RDF statements consist of triples < Subject, Predicate, Object >, which describe the subject in more detail, either through literal attributes or through relations to other resources. RDF has a variety of representations, e.g. N3, Turtle, JSON or XML [5–7].

For research (meta)data, FAIR Data Principles offer a measurable set of principles to improve data quality in terms of data being Findable, Accessible, Interoperable and Reusable. Many implementations show how RDF serves as a reliant component for fulfilling these principles [8]. One may enrich RDF resources by referencing concepts from clinical ontologies like SNOMED-CT or LOINC through their globally unique identifier / code [9,10]. For this purpose, systems like Dublin Core (DC) or Simple Knowledge Organization System (SKOS) provide standardized relations and annotations [11,12]. By referencing common resources in a standardized way, RDF data becomes interoperable, findable and reusable.

Although several solutions exist for importing routine health care data [13], there are still very few tools for i2b2 ontology administration [14]. Particularly, there is no published solution for importing RDF metadata into i2b2.

This paper aims on developing a generic solution for metadata import of RDF metadata into an i2b2 database and prove its feasibility.

2. Methods

There are two effective ways for importing data into the i2b2 metadata structure: Either by working directly on the database or by using the provided web service API. The latter becomes less efficient for large-scale data import and is less flexible because of API restrictions. Therefore, we decided to query the database directly through SQL statements. To achieve this, we have to transform RDF resources into appropriate SQL statements, taking into account their relations (especially their hierarchy) as well as some annotations like labels and codes of other ontologies.

Four database tables are used for the i2b2 metadata representation: "i2b2", "table_access", "concept_dimension" and "modifier_dimension" [15]. The tables "i2b2" and "table_access" contain information about the visual representation of concepts in the user interface. This is where labels and descriptions are stored. The "concept_dimension" and "modifier_dimension" tables contain coding information. They are used internally for execution of user queries. Modifiers are implemented in i2b2 since release 1.6.00 to enrich existing concepts (and when needed its sub-concepts) with additional specification options. I2b2 can only display concept trees, so we can only import RDF graphs without loops.

The W3 Consortium published SPARQL [16] as a standard for querying RDF. By using an engine with a SPARQL-interface, we can provide accessibility (see FAIR principles) of RDF resources. Jena Fuseki Server is a popular open source software designed for querying loaded RDF data via SPARQL interface. It supports a wide range of input formats like Turtle, XML or JSON.

For each concept or modifier, we have to query the RDF data for the concept's primary label, codes, as well as sub-concepts and applied modifiers. Then we add a row to the i2b2 table. In case of annotated codes, which are necessary for concepts and modifiers to be found and considered in the results, we also add a row to the respective "*_dimension" table. In case of a root concept, we also add a row to the "table_access" table.

Knowing how to query the RDF data and what the i2b2 database insert statements have to look like, we are able to formulate an algorithm that will generate all insert statements by successively gathering all required concept and modifier information from the RDF graph (Figure 1). This algorithm recursively runs through a method that, for each concept, queries all its sub-concepts, modifiers, notations and its preferred label, generates the proper SQL-statement and then continues with all sub-concepts. Visual attributes for the i2b2 web client (e.g. folder, leaf, hidden sub-concepts) are deduced by looking at the number of sub-concepts. For modifiers, the same procedure is applied.

The German Center for Lung Research (DZL) uses i2b2 as central data warehouse software, collecting data from various sources and merging them for large-scale retrospective data analysis, sample ordering and patient recruitment. The advantages of properly setup metadata has already been shown in the introduction. To evaluate our metadata import algorithm, we transcribe our DZL ontology to turtle N3 syntax and load it into a Jena Fuseki server, offering the SPARQL interface for querying the data. The DZL ontology was composed by lung researchers, medical documentalists and data managers using our Collaborative Metadata Repository (CoMetaR) editing framework [17].

For hierarchical ordering of concepts and modifiers we use the standard RDF predicates skos:topConceptOf/skos:hasTopConcept and skos:broader/skos:narrower. For modifier indication, we use rdf:hasPart/rdf:partOf. For labeling and description, we use skos:prefLabel and dc:description. Concept's and modifier's codes are indicated through skos:notation. Table 1 shows an example of entries in the postgres database used by i2b2. All software components (RDF files, Jena Fuseki Server, our java program and i2b2 software) run on the same Linux system.

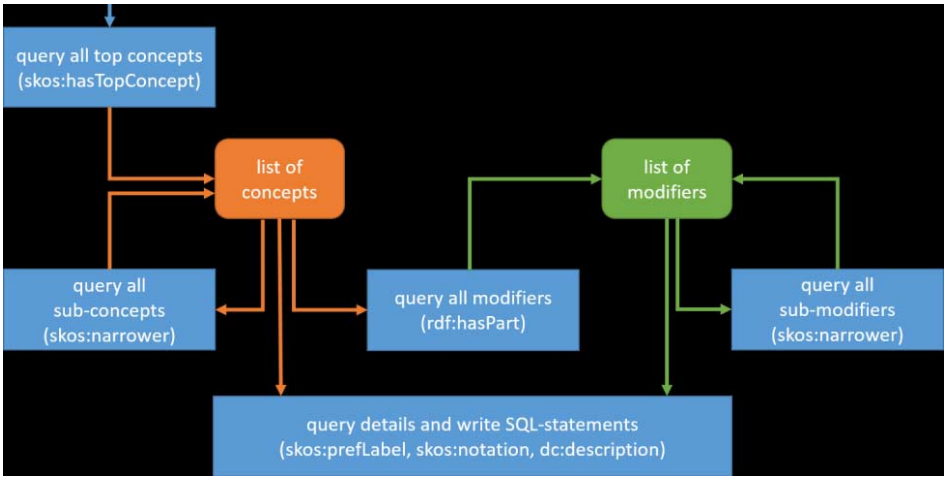

Figure 1. Visualization of the RDF-to-i2b2 transformation algorithm.

Table 1. Example entries in respective database tables for i2b2 (postgres).

Database table i2b2

c_fullname	c_name	m_applied_path
\i2b2\dzl:Dataset\dzl:Follow-up\	Follow-up	NULL
\i2b2\dzl:Dataset\dzl:Follow-up\dzl:Vitalstatus\	Vital sign	NULL
\L:8462-4\	Diastolic blood pressure	\i2b2\dzl:Dataset\dzl:Follow-up\dzl:Vitalstatus\S:75367002\%
\i2b2\dzl:Dataset\dzl:Follow-up\dzl:Vitalstatus\S:75367002\	Blood pressure	NULL

Database table concept_dimension

concept_path	name_char	concept_cd
\i2b2\dzl:Dataset\dzl:Follow-up\dzl:Vitalstatus\S:75367002\	Blood pressure	S:75367002

Database table modifier_dimension

modifier_path	name_char	modifier_cd
\L:8462-4\	Diastolic blood pressure	L:8462-4

3. Results

We developed a solution for integrating RDF ontologies into i2b2 databases by loading the data into an application that provides a SPARQL interface processing it through an algorithm, which queries the interface for RDF concepts recursively and generates SQL insert statements, before executing these SQL-statements. The ontology is accessible via SPARQL through https://data.dzl.de/fuseki/cometar_live/query. A visualization is available at https://data.dzl.de/cometar/.

3.1. Range and Performance of the Implementation

Our entire DZL ontology was successfully imported into i2b2. Currently it includes 653 concepts and 39 modifiers organized in four trees with 518 leafs. Out of all, 601 concepts and 32 modifiers are associated with a code. In total, the ontology is spread across 40 text files in turtle syntax (ttl). During transformation by our java algorithm, all insert statements are split up into two SQL files, since some have to be executed on i2b2metadata schema and some on i2b2demodata schema. Loading the ttl-files into Jena Fuseki Server takes less than one second. The java program for generating all i2b2 insert statements runs between 9 and 21 seconds. Executing the SQL files on PostgreSQL takes about two seconds.

4. Discussion

We developed a generic algorithm to import metadata from RDF to i2b2 and proved its feasibility. Although i2b2 can represent some basic elements of standard RDF vocabulary (e.g. SKOS), its capabilities are rather limited. For example, relations like skos:closeMatch and graphs with loops are not representable in i2b2. On the other hand, data type restrictions are supported, but our algorithm does not yet consider them. Editing

RDF ontologies in turtle N3 syntax by hand may prove difficult. In this case, RDF editing tools like Protégé may facilitate this task. Since execution time of our implementation lies under one minute, it is possible to perform live i2b2 ontology updates after editing the RDF ontology.

5. Conclusion

In order to combine the benefits of i2b2 and RDF, we investigated possibilities to transfer RDF metadata to i2b2 data warehouse systems. We found that import of hierarchical information, labels and codes is feasible and developed an algorithm for this task.

6. Conflict of Interest

The authors state that they have no conflict of interests.

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A Method to Use Metadata in Legacy Web Applications: The Samply.MDR.Injector

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Abstract. Whenever medical data is integrated from multiple sources, it is regarded good practice to separate data from information about its meaning, such as designations, definitions or permissible values (in short: metadata). However, the ways in which applications work with metadata are imperfect: Many applications do not support fetching metadata from externalized sources such as metadata repositories. In order to display human-readable metadata in any application, we propose not to change the application, but to provide a library that makes a change to the user interface. The goal of this work is to provide a way to "inject" the meaning of metadata keys into the web-based frontend of an application to make it "metadata aware".

Keywords. Metadata repository, semantic interoperability, JavaScript, MDR client

1. Introduction

Metadata is instrumental for semantic interoperability in the field of medical informatics [1]. The ability to understand and interpret exchanged information among heterogeneous systems and healthcare actors increases the benefit and value of the gained information. While the separation of instance data and its describing metadata is established in various standards and technologies [1–4], metadata are not always well-integrated into applications. In an ideal world, all applications integrating data from several data sources would be "metadata aware": They would store data (actual content) according to metadata (information about its meaning) stored in a separate component, a metadata repository (MDR). They would extract metadata such as designations (e.g. "smoking status"), definitions (e.g. "An indication of a person's current tobacco and nicotine consumption as well as some indication of smoking history.") or even rules for syntactic validation (e.g. permissible values "current smoker", "former smoker", "never smoker", "non-smoker") [5] as needed via well-defined web interfaces. In reality, however, few applications support the externalization and resolving of metadata.

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While it is possible to define data elements in an MDR and use their identifiers as variable names, users would be presented with cryptic keys rather than human-readable variable names and would be forced to look up their meaning in the MDR. As of today, there are two ways to resolve metadata keys to variables names:

1. Modifying the application's backend to resolve the keys into their metadata, which is then displayed in the frontend. However, modifying an application is not always desirable or even possible, e.g. when using proprietary software.
2. If the data undergoes a data integration process, another option is to resolve the keys during the transformation stage. However, this would lead to a loss of the MDR keys in the application, as only the resolved values would be available.

We present a third approach, addressing it through modifications in the frontend alone while leaving the backend untouched: A generic method and a reference software implementation (Samply.MDR.Injector, in the following: the "Injector") to provide metadata in web applications by resolving cryptic data element identifiers to their corresponding human-readable designations and definitions. The goal of the project is the responsiveness of the function even if the MDR is unavailable, as well as the avoidance of over-fetching.

2. Methods

2.1. Metadata Repositories

An MDR is an application that stores and provides so-called "data elements", each of which is denoted with a unique key, which includes the version of the data element. MDR usually provide a web-based interface to resolve a given key to its metadata, e.g. its designation. The Injector requires that MDR keys can be resolved via a RESTful request. In this work, we use Samply.MDR, an open-source ISO 11179-based implementation developed in the German Cancer Consortium [6]. Listing 1 shows an exemplary REST call and its result.

```
GET /api/mdr/dataelements/{key}/labels HTTP/1.1
[...]

[
    {
"language": "en",
"designation": "smoking status",
"definition": "An indication of a person's current tobacco and nicotine
consumption as well as some indication of smoking history."
    }
]
```

Listing 1. Fetching data from a metadata repository using an HTTP web interface (example).

2.2. Cross-Origin Resource Sharing

JavaScript is a programming language that can be integrated into web browsers. The most common variant of JavaScript is the client-side JavaScript, which combines the

scripting ability of its interpreter with the Document Object Model (DOM) of a web browser. With JavaScript, the HTML document can be changed dynamically. The changes can affect the content or the design of a document [7]. To prevent a potential malicious script from one source from gaining access to sensitive data of another source, browsers implement the so-called “same-origin policy”, a security mechanism to prevent the subsequent loading of scripts from third-party servers [8]; in this case, the MDR. In order to overcome this barrier, “Cross-origin resource sharing” (CORS) [9] allows the MDR to be accessible via JavaScript, by using a new cross-origin request header "Access-Control-Allow-Origin: https://application.researchnetwork.org" in the MDR application.

3. Results

A JavaScript library was implemented as a reusable tool to inject metadata into a web application. In order to use it in a given webpage, the library has to be configured with parameters describing the MDR instances’ interface. While loading the web page, the library seeks for keys to be resolved and collects them internally. This method allows batch retrieval, thus reducing the cross-origin requests to a minimal amount and improving the performance. Upon successful connection, the key is resolved and substituted with the desired information. As the library works asynchronously, the document is loaded before metadata is retrieved. If the MDR query fails, missing keys will be displayed in red, as seen in Figure 1. However, the application itself will still be responsive.

Figure 1. The Samply.MDR uses URNs (Uniform Resource Name) as keys. URN before injection (left), failure to resolve metadata (middle), injected metadata (right).

Integrating the Injector into an existing web application involves two steps:

1. The script needs to be included into the head part of the HTML document
2. The script needs to be activated by a single method named *search*

As seen in Listing 2, for instance, the library is constructed with specified arguments: the HTML tag to be searched for keys (1), the URL of the MDR’s REST interface (2), a regular expression to identify the MDR keys (3), the interface-dependent attributes (4-5), which contain the requested information, and the ID of the popup text (6).

```
    search('output','http://mdr.ccp-
it.dktk.dkfz.de/v3/api/mdr/dataelements/%s',
    'urn:dktk:dataelement:\\d+:\\d+','designations','designation',’definition’)
```

Listing 2. Search method with exemplary parameters (1-6).

4. Discussion

The proposed library suits the characteristics of modern web applications [10]: Moving the program logic from the server backend to the (more and better-equipped) clients, thus distributing the workload effectively on the communicating participants, which has negligible effect on the performance even of dated web browsers. In addition, the use of JavaScript and the RESTful information retrieval within the web browser profits from its well-evolved HTTP implementation: the caching mechanism prevents over-fetching information, reduces the number of connections by storing previously requested metadata and can even, to some extent, deal with outages of the remote server. User settings, such as language preferences, are defined in the request headers and can improve the usability of the rendered webpage, by automatically displaying the description in the preferred language – assuming the MDR provides this information.

While the current implementation has been developed for use with Samply.MDR, it can easily be adapted to any metadata repository as long as it provides a RESTful communication interface.

The greatest benefit, however, is a novel way for integration into existing applications: Often the backend systems are closed-source or, if the source code is published, difficult to adapt. In contrast, the proposed approach requires simple modifications to the frontend layer only. These modifications are identical for any web application, rendering this approach generic. For example, well-established data warehouses and data analysis tools such as i2b2 [11], TranSMART [12] or even proprietary closed-source solutions can be made “MDR aware” with the same change to their HTML output. Since this change occurs in the respective HTML template rather than the program’s source code, it is easily preserved across software upgrades. In the clinical context, however, the approach faces two limitations: Firstly, the manufacturer or the medical devices law may prohibit even minor modifications on the frontend. Secondly, an institution might restrict the use of JavaScript for security reasons. However, such restrictions will not only disable the Injector but also reduce functionality of the application itself as well as of many other public web sites.

The trade-off between security and improved functionality will have to be evaluated by the user, but the authors would argue in favor of the latter. As a conceivable compromise, the CORS requests can be limited to certain applications using the Injector. This procedure includes a registration of each application at the MDR. This approach is not able to overcome the start-up problem of an offline MDR system but provides an effective alternative to keeping the application responsive. This simple integration is a first step in the direction of improving any web application to be "metadata aware".

5. Conclusion

The described method provides a flexible and reliable way to inject Metadata instead of cryptic keys in a given webapplication. A prototypic implementation has been published as free software and can be downloaded at https://bitbucket.org/medicalinformatics/samply.mdr.injector. This solution is currently being implemented in components of the German Biobank Alliance and the German Cancer Consortium.

6. Conflict of Interest

The authors declare that they have no conflict of interest.

7. Acknowledgements

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Samply.MDR – A Metadata Repository and Its Application in Various Research Networks

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Abstract. Collaboration in medical research is becoming common, especially for collecting relevant cases across institutional boundaries. If the data, which is usually very heterogeneously formalized and structured, can be integrated, such a collaboration can facilitate research. An absolute prerequisite for this is an extensive description about the formalization and exact meaning of every data element contained in a dataset. This information is commonly known as metadata. Various research networking projects tackle this challenge with the development of concepts and IT tools. The Samply Metadata Repository (Samply.MDR) is a solution for managing and publishing such metadata in a standardized and reusable way. In this article we present the structure and features of the Samply.MDR as well as its flexible usability by giving an overview about its application in various projects.

Keywords. Metadata, data integration, interoperability, research networks

1. Introduction

Collaboration in the analysis of medical data is becoming common, as collecting data through the recruitment of patients at a single institution may prove insufficient for, inter alia, large cohort studies or research projects on rare diseases. Collaborators must

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therefore share their data with others. This, however, requires the integration of data from all participating sites and their various heterogeneous sources. Generally, this is achieved by consenting on a common dataset. Based on its specification, every participating site has to map all of their relevant data elements. Traditional approaches such as defining and exchanging data element definitions with Excel or Word documents have many limitations, especially in larger research networks. Furthermore, this information would have to be manually translated into mapping or transformation rules. One way to address this is to formally describe all data elements in a metadata repository (MDR) according to internationally accepted standards that provide clear guidelines as to the content and structure of the metadata. Finally, open and standardized access to metadata for data collected in healthcare and medical research would ease data sharing and integration [1]. In order to tackle this challenge, the Samply.MDR was designed and stepwise enhanced in functionality throughout its application in various research networking projects.

2. Methods

One such common metadata standard is ISO 11179, whose six parts generically describe how metadata can be used to describe data elements to preferably cover all possible use cases and domains, e.g. within the manufacturing industry or the healthcare system. Our work focused on ISO 11179 part 3 edition 3 [2], which specifies a metamodel divided into two layers of metadata. A representational layer comprises information about data storage in a database, e.g. the datatype and restrictions such as a range or a list of permitted values of a specific data element. The conceptual layer outlines how to formalize the meaning of data elements, e.g. the relation between a value and its corresponding code item from a classification. To ensure that the metadata content can be adapted to different use cases, any further necessary metadata attributes can be specified as slots (key value pairs), e.g. additional implementation specific information which is needed by the respective software solution relying on the metadata.

Samply [3], a collection of IT tools and components to support collaboration and data sharing in research networks implements, inter alia, parts of ISO 11179. Its development began in 2011 with the idea of supporting smaller biobanks in sharing their biomaterial collections including their annotation data [4]. The intent in using metadata as a natural requirement was (1) to foster the reuse of existing data elements and (2) to support the formulation of inquiries to networked biobanks based on their respective data elements. Samply is published as open source [https://bitbucket.org/medicalinformatics/] to facilitate vendor and implementation independent data and system integration.

3. Results

Samply.MDR [5] is developed as an extensible and integrable MDR implementation to support the specification of data elements in a structured, formalized and standardized way. Its development began in 2012 as part of the IT for the Clinical Communication Platform (CCP-IT) of the German Cancer Consortium (DKTK) [6]. As another project, the development of an Open Source Registry System for Rare Diseases (OSSE) [7], posed additional requirements, the decision was made to reimplement the PHP-based prototype in JSF - in accordance with the other components of the Samply toolset – and to publish it as open source software in November 2015. It comprises a relational

database and tailored implementations for the data access layer, data transfer objects and data access objects in the backend. A browser-based graphical user interface as well as a REST-API can be used to manage and access the metadata. All data elements are organized in namespaces and can be identified by a unique Uniform Resource Name (URN). A user can edit data elements according to his or her access rights for the respective namespace. Reading access to data elements can also be restricted to authorized users by hiding namespaces.

Reusing existing (not hidden) data elements is explicitly desired. A user can issue a full text search for matching data elements and simply import a match into his or her own namespace to reuse it. If necessary, parts of the metadata such as the designation and definition can be modified, e.g. by adding a translation of both in another language. To preserve comparability, any other change such as adding a new value to the value domain is not possible, as this would break comparability of the data element in its old and new revision. In this case the user can use the existing data element as a template for a new one. Table 1 provides a simplified overview of possible metadata attributes.

Table 1. Simplified overview about metadata attributes in the Samply.MDR to describe a data element.

Metadata Attribute	Description	Example Value
Identifier	The identifier as URN [namespace: element type:identifier:revision]	urn:miracum:dataelement:17:1
Designation	The name or label (1-n languages)	Heart Rate
Definition	The extensive human-readable description (1-n languages)	Heart Rate of a patient, measured by palpation on the wrist (radial pulse)
Validation Type	The data type, the list of permissible values or the reference to a catalog	Integer
Validation Rules	Restrictions on the validation type	Range between 0 and 250
Unit of Measure	The unit of measure	bpm
Slot	Key value pairs to describe further aspects (use case specific)	‘sharing_allowed’:‘true’

As a first part of an implementation of the conceptual layer, classifications and terminologies can be imported as coding systems or catalogs. All or any subset of codes contained in a catalog can then be defined as permitted values of a data element.

In the following the authors representing the projects give a brief overview about how these are using and developing Samply.MDR.

CCP-IT: Samply.MDR holds the harmonized common dataset upon which all DKTK partner sites have consented. The dataset and its metadata are used for the formulation of inquiries at the central search broker. An inquiry is then downloaded to the respective bridgehead, the local part of the CCP-IT at every partner site, and is executed against the data stored in the local data warehouse, which involves a locally defined and stored tailored mapping to its used data schema.

OSSE [8]: In addition to CCP-IT, OSSE relies on the metadata stored in the same instance of Samply.MDR to design, render and validate electronic case report forms. The user can drag and drop the desired data elements into a form which can then be published and enabled in the registry. Based on the metadata of the enabled forms, the OSSE registry software creates the database at runtime and validates the user’s input.

OnkoWiki [9]: The project aims to establish a resource for medical documentation staff to look up data elements and usage guidelines for the documentation of tumor cases. The user interface of Samply.MDR has been adapted to include additionally required metadata attributes (i.a. Pubmed links). Slots have been used in the backend to minimize necessary modifications. To support the process of maintaining the Common ADT/GEKID record format for tumor cases [10], collaboration features, such as commenting on data elements, have been added.

BBMRI-ERIC CS-IT/ADOPT [11,12]: In the research network Biobanking and BioMolecular resources Research Infrastructure - European Research Infrastructure Consortium (BBMRI-ERIC) and its subprojects all the required IT tools are developed to interconnect biobanks across Europe, enabling them to share their resources and facilities. Currently, every biobank stores its information about samples in different ways and data integration is therefore necessary to reach the expected level of interoperability. The BBMRI search tool Sample Locator relies on the metadata stored in Samply.MDR.

GBA: The German Biobank Alliance (GBA) [http://bbmri.de/ueber-gbn/german-biobank-alliance/] networks eleven German biobanks with a harmonized IT platform for the efficient exchange and usage of biomaterial and its associated clinical data as well as for participation in BBMRI-ERIC. As one of the first steps, an installation of Samply.MDR has been set up and the biobanks have started to specify their data elements within their individual namespaces. The collected metadata is being analyzed and the results will be used to identify common data elements which all or at least most of the biobanks could take into account in inquiries.

MIRACUM: MIRACUM (Medical Informatics in Research and Care at University Medicine) [http://www.miracum.de/], one of the four consortia funded by the BMBF within the Medical Informatics Funding Scheme [http://www.medizininformatik-initiative.de], is a network of eight German university hospitals, two technical universities and one industrial partner. Three out of four funded consortia consider ISO 11179 for the specification of metadata. MIRACUM has designed hospital-based data integration centers with the Samply.MDR as local installations at every partner site. These hold the metadata of locally-used data elements and a central MDR will be used to publish and communicate all agreed-upon dataset specifications. Local ETL processes will be implemented based on the metadata to ensure that all relevant data can be queried and used as intended by the distributed search concept of MIRACUM.

4. Discussion

Although one of the ideas is to foster harmonization and reuse of data elements, the current implementation of Samply.MDR lacks functionality to actively support the user in avoiding redundancies. As the current implementation does not cover the entire ISO 11179 standard and mainly provides features to register, specify and manage data elements according to the representational layer of the metamodel, the user can only perform a full-text search, which should be replaced, or at least complemented, by a semantics backed feature. This would improve usability, as the result set contains fewer candidate data elements the user can reuse instead of defining a new one. Also, such enhancement would ease the curation of all the metadata. The incorporation of relations and mappings between data elements, which is currently worked on, will further improve the harmonization and deduplication.

Comparing the concept and implementation of Samply.MDR with other existing approaches in detail, i.a. the tools ART-DECOR, caDSR or semanticMDR, as well as comparing ISO 11179 with other standards, i.a. HL7 FHIR or ISO 13606, is out of scope of this article. However, we will further investigate approaches to ensure interoperability between Samply.MDR and other standards and implementations.

At the moment we are in the process of collecting, evaluating and pushing the various project specific modifications to the public repository. Either way, the project welcomes pull requests or other contributions from the community at any time.

5. Conclusion

The broad use of Samply.MDR as well as the existence of many other similar solutions show that a metadata-based approach can help in developing and establishing IT tools to support collaborative research and data integration. Furthermore, the number of research networks relying on Samply.MDR indicates, that using and developing existing open source solutions can be feasible to allow them to focus on other aspects of their goals.

6. Conflict of Interest

The authors declare that there is no conflict of interest.

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Using Graph Tools on Metadata Repositories

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Abstract. To exchange data across several sites or to interpret it at a later point in time, it is necessary to create a general understanding of the data. As a standard practice, this understanding is achieved through metadata. These metadata are usually stored in relational databases, so-called metadata repositories (MDR). Typical functions of such an MDR include pure storage, administration and other specific metadata functionalities such as finding relations among data elements. This results in a multitude of connections between the data elements, which can be described as highly interconnected graphs. To use alternative databases such as graph databases for modelling and visualisation it has already been proven to be beneficial in previous studies. The objective of this work is to evaluate how on-board techniques rely on matching and mapping using a graph database. Different datasets relating to cancer were entered, and algorithms for metadata management were applied.

Keywords. Metadata repository, graph database, Neo4j

1. Introduction

Secondary use of electronic health record data depends among others not only on data quality but also on the understanding of the data elements. To gain a general valid understanding of the data, metadata have become widely accepted, which are usually stored in a metadata repository (MDR). The tasks of such a metadata repository include pure storage, management and administration as well as matching and mapping of metadata. Matching refers to the discovery of related or equivalent data elements; mapping refers to the relationship between data elements such as conversion rules. This results in a large number of connections between the data elements. Summarised metadata itself is a highly connected object which gains its value from meaningful semantic connection to other data objects [1]. The modelling of metadata in graph databases like Neo4j [2] seemed promising, and its feasibility was successfully proved in previous research [3]. In this paper, the matching and mapping of corresponding metadata represented as graphs are investigated: For evaluation purposes, three datasets regarding the topic of cancer were inserted into the graph database, and algorithms for matching and mapping were applied.

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2. Methods

The modelling of data elements could be performed based on the information from the source system: data element information should be defined as nodes with a set of properties, where *DataType*, *Origin* and *SelectOption* are modelled as labels and interactions are described by the relationships: "is_option", "is_datatype" and "is_source" [3]. For complex data types like enumerations, the values of the selection list are stored separately in value domains assigned to the data element by "is_choice", see Figure 1.

2.1. Matching of Data Elements

Metadata matching is the process to discover data elements which are similar or at least related to each other in some way. For all data elements, classical string matching algorithms are provided resulting in new edges termed "is_similar" with a value range from zero to one, where one is an adequate match. This approach proposes a similarity algorithm that combines two string-based measurements: the *five-gram* algorithm and the *metric Longest Common Subsequence* (mLCS) [4]. The first one splits the given terms into subsequences of five characters and measures their similarity based on correspondence. The mLCS is based on the metric longest subsequence that both terms share. The subsequence differs therein from the substring - the subsequence it is not mandatory to have the same position in the terms. To minimise false positives, a quality threshold rejects matches with a low score. This approach achieved good results in earlier designs [5]. In addition to the mLCS and nGram, a part-of-speech tagger was used. Properties with more than 50 characters are processed to identify the significant words, e.g. nouns, of the property and use them for a more precise mapping.

For the complex datatypes like enumerations, this procedure had to be changed; the value domains and their choices must, first of all, be standardised. However, a similarity measure is not sufficient; the choices can only be matched when equal, which means with an equivalence measure of one. If choices from different data elements, and therefore different value domains are matched with each other, the similarity measure for the value domains can be calculated. The value domains can then be used to conclude the complex datatypes themselves.

2.2. Mapping of Data Elements and Merging of Choice

Metadata mapping is the process to discover the semantics of a data element in datasets. Metadata merging is the process of actually reducing the complexity of the graph by reducing the number of nodes. This process ends up with a set of mappings between the data elements and a reduced number of choice nodes.

The choices of the value domains (Fig1 a) are compared with the described methods in Section 2.1 (Fig1 b) to detect if the corresponding value domains may share choices in their characteristics. If this is the case, the number of vertices can be reduced (Fig1 c). The simple and complex data elements are compared with one another according to their origin. The results of the matching are displayed as edges with the similarity rate of each applied algorithm. Using the similarity score, a similarity path can be calculated between one single data element from dataset A and all other data elements from dataset B (Fig1 d).

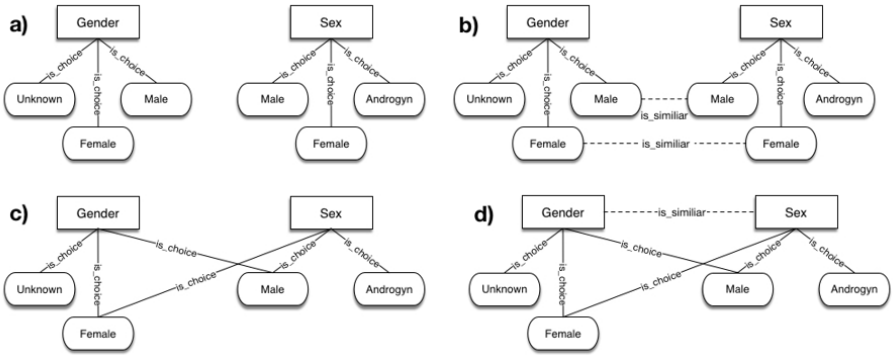

Figure 1. The process of mapping and merging choices and data elements based on similarity measures.

2.3. Exploring the gained Information

The newly gained information is stored in edges and the corresponding attributes. Neo4j comes with a built-in visualisation for the stored data, see Figure 2. The discovered links between metadata can be seen easily and enable an intuitive way to understand and discover the dataset. The shown set of vertices can be refined by manual selection or with Cypher, a declarative SQL-inspired query language. In addition to the fast adaption of the graphical style, an entire Neo4j database can be exported as a single Cypher script, so the exchange of information is simplified.

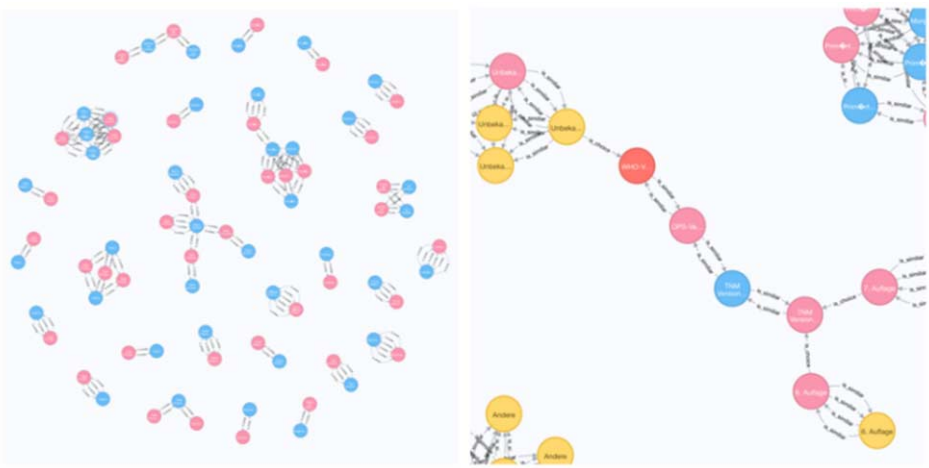

Figure 2. Results of the Neo4j graph database: Left side showing a mapping of the Finnish Cancer Registry [7] dataset according to the Einheitlicher onkologischer Basisdatensatz von Arbeitsgemeinschaft Deutscher Tumorzentren e.V. (ADT) und Gesellschaft der epidemiologischen Krebsregister in Deutschland e.V. (GEKID) [6] dataset. The right side is showing a chaining of data elements across several datasets arisen from a structural mapping of the choices and value domains.

Besides Cypher, Neo4j also supports a new dynamic query approach by using GraphQL, a graph-based query language developed by Facebook. In contrast to the popular Restful pattern this language provides one single route, which accepts queries written in JSON. The unique feature of GraphQL is providing the response format as part of the actual query. So, the requesting client can decide which parameter and attributes are essential and necessary for its processing. Therefore, the problems of over- and underfetching can be overcome.

3. Results

Three datasets, with an overall of 666 data elements, were used for the evaluation, all data collections refer to the topic of cancer: The Einheitlicher onkologischer Basisdatensatz von Arbeitsgemeinschaft Deutscher Tumorzentren e.V. (ADT) und Gesellschaft der epidemiologischen Krebsregister in Deutschland e.V. (GEKID) [6] (*ADT/GKID dataset*); the dataset of the Finnish Cancer Registry [7] and the MIABIS 2.0 [8] (*MIABIS dataset*). All three datasets have been matched and mapped beforehand to the locally documented data elements so that a gold standard is available. All datasets were imported into the graph database and either of them compared to one another using the matching and mapping algorithms described earlier. The experimental setup was able to find all equivalent choices and therefore able to reduce the number of choice nodes from 392 to 371 nodes. For the complex data elements, the gained information from the mapping of value domains was considered in the evaluation and resulted in three of four matchings of complex data elements. Simple data elements of datatype string, integer or boolean could just be matched using the string-based measurements. The algorithms have been applied to both German and English translations of both data element names as well as the description and the results are promising. The setting could identify 44 out of 54 possible connections, which results in 81% accuracy.

4. Discussion

Previous studies proved that a metadata repository based on a graph approach has great potential. The extension with matching and merging algorithms could confirm the result, but the proposed matching algorithm needs to be enhanced. However, the interactive support assists users to deal with thousands of data elements, especially in the context of clinical trials management. The approach has promising results and is the first step to enable a better understanding of different datasets and their reuse. The Neo4j shows to be well suited, due to the excellent visualisation and the GraphQL support. The built-in graph representation of the given datasets allows a compelling overview of data quality and data accuracy, see Figure 2. The adaption to different data sources is easily possible, and all information can be exported as a Cypher file for exchange purposes.

5. Conclusion

This study raises general concerns about the technical architecture of metadata repositories, like the question if graph databases with their characteristics might be the better

choice for storing metadata [9]. The capability of efficiently interconnecting the datasets and the flexible information provision using the graph-based approaches like GraphQL promotes these database systems as superior candidates. However, moving away from relational databases also comes with the loss of the ACID guarantees. A feasible compromise could be a hybrid approach: a relational database with the transaction safety and a graph-based interface for querying and mutation of the information.

6. Conflict of Interest

The authors state that they have no conflict of interests.

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User Expectations of Metadata Repositories for Clinical Research

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Abstract. Metadata Repositories (MDR) are databases for data elements that can be utilized in research as well as in medical care. These data elements are not the actual patient data (facts), but a complete definition of the variables or characteristics used, including coding, unit of measurement, data type and other aspects. The aim of the project described here was to evaluate possible application scenarios for MDRs by a larger group of experts. The focus was not on specific software, but on the community's basic expectation of such a database of data elements. To achieve this goal, a questionnaire was designed that contained questions on general aspects of setting up a registry for data elements in biomedical research as well as more specific points with regard to necessary functionalities, desired contents, tools for community work and the quality of data elements. One of the main results was that the users attach more importance to the quality of the content than to the efficiency in implementing their documentation concepts. At the same time, they consider the effort involved in using existing software systems to be too much compared with the benefits and have concerns about the use of their designs by third parties.

Keywords. Metadata repository, user acceptance, clinical research informatics

1. Introduction

The process of preparing a documentation concept for clinical and epidemiological studies includes the specification of the variables to be collected in form of *data elements* [1]. This is a complex and time-consuming task in terms of completeness, correctness and comparability. On the one hand, the documentation concept should be simple and comprehensible and contain only the data elements relevant to the research question. On the other hand, the unambiguous, detailed definition of the acquisition characteristics has a considerable influence on the significance and quality of a study. At present, the scope and level of detail of the documentation concept depends to a large extent on the expertise of the biometrician in charge. Compulsory guidelines or standardized sets for describing data elements, such as CDISC's SDTM format for submitting clinical trial data to the FDA[2] or the German regulated ADT/GEKID basic data set in the domain of oncology[3], exist only to a very small extent. For the purpose of a faster study specification and comparable features, a central database of frequently used, community-tested data elements would be desirable.

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[2] https://www.cdisc.org/standards/foundational/sdtm
[3] http://www.gekid.de/adt-gekid-basisdatensatz.html

In order to build up a pool of data elements, they have to be described by means of a flexible and detailed information model, which is oriented to the requirements of clinical researchers in terms of generic methods for naming, identification, classification and representation, but also to the structures of actual documentation concepts. Preferably this is not a proprietary development, but an established standard such as HL7 Clinical Document Architecture (CDA) [2], ISO 13606/openEHR Archetypes [3], CDISC ODM [4, 5] or ISO/IEC 11179-3 [6].

In addition to a solid technological basis, it is important to ensure early on that a product or service is not developed without understanding the user [7]. In the present project, user requirements for a MDR and its operation were determined by a structured inquiry.

2. Methods

An anonymous survey with subsequent statistical evaluation was chosen as a method for the evaluation of possible application scenarios for MDRs. With the help of a structured survey, a large community can be involved as the questions can be answered electronically without a human interviewer. At the same time, a current prospective database is available. A further advantage is that all participants can answer the same questions and, in contrast to free text comments, fixed answer options are predefined. We mainly used statistical variables with ordinal characteristic values, often represented by a five-level Likert scale. Absolute/relative frequencies were determined for categorial characteristics, mean value, standard deviation, minimum, maximum, value distribution and 5%, 10%, 25%, 50%, 75%, 90% and 95% percentiles were calculated for the few quantitative characteristics and missings were counted.

In the narrower sense, the survey was aimed at people who regularly deal with information technology support for clinical and epidemiological research projects. This includes the following target groups:

- *Data managers* who design clinical trials based on the study protocol
- *Computer scientists* who develop databases for clinical studies, e. g. with the help of CDMS
- *Statisticians* and *epidemiologists* evaluating data from research projects

The actual clinical researchers were not considered to be the primary target group, since they generally use the services of representatives of the above groups to carry out scientific hypotheses.

The survey was divided into two sections. Section 1 contained 19 questions on general aspects of building a central library of biomedical research data elements and scenarios for use in an organization. It had been developed with a focus on rapid responsiveness and included such questions that were considered particularly important. This part should also be able to be processed by fillers with a low time tolerance. Section 1 contained 3 subsections: Subsection 1 was concerned with the assessment of the basic benefit of a MDR for the research community, especially with regard to efficiency, quality, trust, comparability and long-term documentation of data elements or the creation of documentation concepts. Subsection 2 covered the demographics of the participants, their role in the study process and the characterization of their organization. Section 3 dealt with the integration of an MDR into local workflows, the connection to

existing systems, SOPs and user groups, but also with potential concerns or obstacles to deployment. The preferred operating model for an MDR was also part of the questions.

Section 2 was directed more towards experts or interested potential users. It contained 22 questions in 4 subsections and referred to more specific points with regard to necessary functionalities, import/export formats, desired content, tools for community work and the quality of data elements.

The implementation and execution of the survey was carried out with an internet-based survey tool. REDCap [8] proved to be very suitable for this purpose. The survey was conducted in April/May 2014. The target audience was the members of various working groups of the TMF - Technology, Methods, and Infrastructure for Networked Medical Research, an umbrella organization supporting clinical research as well as the German-speaking CDISC user group. They were asked to take part once by e-mail.

3. Results

During the survey period, 34 people took an active part in the survey, filling out at least one field. 24 respondents answered the entire questionnaire, i. e. sections 1 and 2. Additionally, 54 respondents have opened the survey without entering any data. Due to the sample size, selection and anonymity of respondents, the survey is not representative in an empirical sense. The most important results are presented and interpreted below.

Participants' demographics: The overwhelming number of participants came from academic research institutions or clinical study centers. On average, their employer supervises 15 new clinical trials per year. 83% of organizations use an electronic data management system. The use of e-CRFs has become standard (53%), but paper-based data capture is still common (30%), mixed studies from e-CRF and p-CRF occur in a quarter of the studies (24%). Fillers use a local institutional standard with conventions for naming/encoding data elements (31%).

Efficiency when using an MDR: One of the basic ideas of an MDR is that documentation forms can be created faster by reusing finished data elements, certain section or entire forms from a central database. Only a minority sees the possibility of reusing entire forms (1.3%). The majority of respondents believed that they could reuse a multitude of data elements or groups, however modifications will be necessary (57.6%).

Quality: Furthermore, forms created with the help of an MDR promise a higher quality because they consist of practical, harmonized data elements. A number of questions were related to the participants' expectations regarding different quality criteria: routine suitability, completeness, consistency, degree of coverage of a domain and durability. The highest priorities were high coverage and routine suitability (see Figure 1).

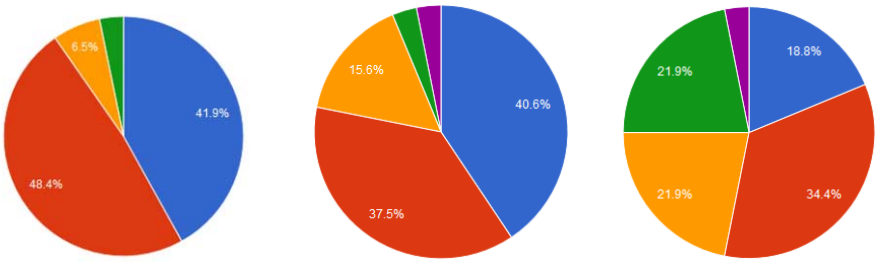
Trust: In a community-based MDR, the range of content producers is expected to be heterogeneous, and often several variants of data elements will exist. The question therefore arises as to which producers enjoy the trust of the community, so that their content can be reused. The greatest trust is placed in a formal quality assurance process (87.5%), Standard Development Organizations (68.8%) and the community itself (64.6%).

Utilization: Documentation forms created using an MDR contain structured, reusable data elements. By reusing the same data elements in different research projects, data sets become semantically comparable. From this, various post-use scenarios can be

derived. The greatest potential is attributed to its use in meta-analyses (90.3%), the least to the reduction of duplicate data acquisition within a research network (48.4%).

Long-term documentation: Another advantage of an MDR is the permanent provision and archiving of a documentation concept. Different levels of complexity can be distinguished here: simple list of all data elements, case report form (CRF), annotated CRF, annotated CRF with validation rules, possibly further levels such as creation history, comments, notes on completion, etc. The expectations of the respondents are inconsistent in this respect, with no clear preference.

Integration into local workflows: An important point is the optimal integration of an MDR into routine operation, i. e. the work processes defined within the SOP system. The participants see an MDR primarily as a tool for the electronic implementation of survey forms and CRFs, so the MDR would be used at a rather late stage and not so much in the preparation of the study protocol or the definition of the variables to be collected in the documentation concept. Accordingly, it would be used foremost by data managers (77.8%) and database programmers (81.4%), with biometricians cutting back on it (62.9%).



Routine suitability: data elements are clearly formulated, can be filled in quickly and generate few queries (e. g. through options for typical units of measurement, unknown fields, ...).

Completeness: Data elements contain all necessary substructures, e. g. for the calculation of certain scores (BMI, SOFA, GCS,...).

Consistency/redundancy: Conflicting, missing or redundant data elements are detected within a documentation concept.

Figure 1. Example from subsection "Quality". Legend: options from a five-level Likert scale: full voice to (blue), partial voice to (red), draw (orange), majority voice not too (green), no voice at all to (violet).

When asked about obstacles about the use of MDR software on site, the most common concern was that the effort required to operate the MDR might be greater than the benefit. Issues of copyright law, for instance on translated assessments (e. g. German version of SF-36) and the intellectual property of one's own organization were also raised. Many users want a centrally managed MDR but want to keep the option of a local installation available. Accordingly, a recognized open source license is desired. A majority of respondents would post their own content if naming and disclosure were guaranteed on equal terms.

4. Discussion

There are a number of limitations in this study. Differences in the ratings between the different user groups were not evaluated. The number of respondents was moderate and not representative of the community. Clinical researchers with a non-technical

background were not part of the target group addressed; their expectations must be met by future surveys. A more detailed analysis and the underlying data set can be downloaded from the Leipzig Health Atlas[4].

5. Conclusion

The current status of medical research and care information systems is characterized by a lack of semantic foundation. This affects the efficiency, quality and performance of research projects and can potentially have a serious impact on the medical care of individual patients. The evaluation of possible application scenarios for MDR carried out here showed what researchers expect from a MDR. It also showed that a community-operated system in the sense of a voluntary group of experts could be accepted if the strict regulations customary in clinical research regarding the traceability of changes and the citation of previous work were observed. It is particularly important, however, that an MDR must not be a stand-alone product in addition to routine work processes, but that it must be possible to integrate them.

6. Conflict of Interest

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Data Discovery for Integration of Heterogeneous Medical Datasets in the German Center for Lung Research (DZL)

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Abstract. The German Centre for Lung Research (DZL) is an association of Germany's leading university and non-university institutions dedicated to lung research. Institutes and disease areas within the DZL manage their own data in several databases and registers using different software tools. Aim of our data integration effort is to provide a single central data warehouse frontend, where all patient related data is combined and made accessible. A two-stage survey was used to determine the data collections suitable for data integration. Integration was performed via extract-transform-load (ETL) steps using custom software. Original software (e.g. eCRF) used by the data collections did not need any modifications. The survey yielded 68 data collections. Until Jan 2018, 20 collections were successfully integrated. 10 collections were withdrawn by their owners while the integration of 38 was delayed. Data discovery, the process of finding existing data collections in a large research network, proved to be the step most underestimated. From technical point of view, data integration proved to be of minor complexity in comparison to the effort required for harmonization/mapping of data elements and management of common terminology.

Keywords. Database, data integration, lung research, data warehouse

1. Introduction

The German Centre for Lung Research (DZL) is a research network funded by the German Federal Ministry of Education and Research (BMBF) consisting of leading universitary and non-universitary institutions. Using translational research methods, the DZL seeks to jointly develop new approaches for the prevention, diagnosis and therapy of serious lung diseases. Geographically, the DZL is organized into five partner sites consisting of a total of 28 institutions in Germany. Research is performed and supervised by ~ 250 principal investigators (PIs) – Physicians and scientists with noteworthy reputation in their field of research.

Institutes and disease areas within the DZL manage their own data in several databases and registers using different software tools. Aim of our data integration effort is to provide a single central data warehouse frontend, where all patient related data are combined and can be queried and accessed by any researcher within the DZL.

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Data Integration

We refer to data integration as "the computational solution allowing users, from end user (GUI) to power users (API), to fetch data from different sources, combine, manipulate and re-analyse them as well as being able to create new datasets and share these again with the scientific community" [1]. Our approach to data integration is "minimal invasive" in the sense that no modifications are required to the original data collections or corresponding software. Data entry, administration and primary storage are handled by the respective software used by the researchers, while a central data warehouse (DWH) is used for secondary storage and analysis. Data transfers occur only for patients who signed an appropriate informed consent.

For a central data warehouse to be useful, all contained patient related data must be linked both vertically (record linkage of patients) and horizontally (mapping to common/core data elements). Metadata in the form of common data elements and terminology can be managed using different approaches. To respect the FAIR guidelines for data sharing [2], we chose a metadata registry approach based on standard technology [3] publicly available at https://data.dzl.de/cometar/. A common terminology for patient phenotypes and bio specimen was developed and agreed upon earlier.

2. Methods

The first step to data integration is "data discovery", the issue of finding the data relevant to a project [4]. The subsequent processes for data integration are applied between "operational source systems and the data presentation area" and "are commonly known as extract-transform-load (ETL)" [5]. Planned to use a breadth-first approach in favor of a depth-first approach, meaning that we aimed at integrating most of the data collections by importing few common data elements first. Later, the full depth of each collections data elements would be assessed, mapped and imported.

2.1. Data Discovery

Although information about biobanks and some large registries and cohorts were on hand by the management, there was no previous complete record on existing data collections available. Thus, we devised a two-stage survey to be distributed throughout the DZL to gather this information.

The first stage of the survey was meant to reach all PIs and collect the following basic information: *Name of the data collection, start and end date of collection, software used and contact person for IT questions*.

The second-stage survey would then be distributed only to the respondents of the first stage, asking for more detailed information: use of a broad informed consent allowing data sharing, current number of patients, inclusion of bio material as well as a technical information: software version, export formats, primary table for patient identification, tables with visit identification/time, example data export and a mapping table with 290 common data elements: 100 common data elements like height, weight, smoking status, lung function, lab values etc. and 190 phenotypes or diagnoses, like pneumonia, adenocarcinoma or pediatric asthma.

2.2. Extraction of Data from the Operational Source Systems

For data extraction from source systems, we planned to use existing export functionality of the respective software systems. Previous evaluation of a small sample set of available software systems showed that most vendors implement either table-based data exports (e.g. comma separated values, CSV) or XML files (e.g. CDISC-ODM).

2.3. Transformation and Data Harmonization

"One of the most important factors for the biological field to thrive is to standardize the data" [1]. Due to the historic nature of many data collections, we expected most data sources to use non-standardized custom notations. While the second-stage survey asked for a mapping of source fields to common data elements to be performed by the data collection's owners, we anticipated an additional manual step for review and quality control of the mapping. This manual harmonization step is performed by the central data manager in collaboration with the respective local data manager and the data collection's owner. To this extend, the data manager reviewed the individual data mapping documentation in conjunction with our online metadata repository [3] containing common data elements in standard terminology.

Structural transformation as well as terminology mapping is performed via custom software. A configuration file contains information about structural relations in the source data as well as the mapping to common terminology. The configuration needs to be changed only when structural modifications are made to the data collections (e.g. new tables, forms added, removed or modified).

2.4. Loading the Data into the Data Warehouse

The third step of loading the data into the data warehouse is performed via encrypted transfer over the internet. Our custom transfer software is executed at the site or institution owning the data together with a source system specific configuration as described in the previous section. Pseudonymization is performed on-the-fly with a separate pseudonymization service based upon generic data protection concepts for Germany [6]. Data transfers will always replace the data previously submitted with the same configuration. Since the data owner initiates and performs all transfers, he keeps full control over the transferred data and can remove his data at any time.

3. Results

3.1. Data Discovery

The first-stage survey for data discovery was distributed in Q3 2013 via Email (CC) to all site coordinators. A reminder was sent in Q1 2014. Until Q3 2014, we received a total of seven responses for data collections. A second attempt for the first-stage survey was distributed in Nov 2016 with a different approach: The site coordinators were addressed individually and received letter template to be sent to the PIs. Additionally, an Excel sheet was provided to each site coordinator with the names of the site's PIs where contact and response dates had to be filled out individually per PI. A reminder was sent at

18.01.2017. Final responses to the first-stage survey were received from the five site coordinators between Jan and Apr 2017 – resulting in a total number of 68 original data collections. Of these, 56 use a single software system, whereas 12 make use of multiple software systems (up to three) due to change in software (old vs new) or separation of data (e.g. biomaterial and clinical). A breakdown is shown in table 1.

Table 1. Software systems used by the 68 original data collections. Software mentioned two or less was grouped into "Other" and "Other LIMS" respectively for laboratory information management systems. Some collections make use of multiple software, resulting in a total sum higher than the number of collections.

Software	Frequency
MS Access	12
CentraXX	3
MS Excel	18
FileMaker	8
ISHmed	5
Other	11
Other LIMS	6
SQL	4
WebSpirit	5

The second-stage surveys were sent in Mar/Apr. 2017 via e-mail personally addressed to each of the PIs who responded during the first stage. The majority of responses to the second-stage survey were received between March and July 2017, although some sporadic responses came in as late as Nov 2017. Of the 68 original data collections, 10 were withdrawn at this stage due to incompatible informed consents or sponsoring.

3.2. Data Export and Data Integration

Data integration steps were performed for each data collection, as described in sections 2.3, 2.4 and 2.5, as soon as sufficient information was retrieved from the respective second-stage survey. Until Jan 2018, we were able to integrate 20 data collections into the central data warehouse. The 38 remaining collections are not yet integrated for different reasons: 6 local preparations, 5 pending ethical vote, 12 change in software, 7 internal discussion, 9 waiting for information. Our progress over the course of 2017 is shown in Figure 1. By Jan 2018, a total number of 17916 patients are available in the data warehouse. On average, 26 distinct details are available per patient.

All data collections supported delimited text exports (e.g. CSV). For nine collections (FileMaker, SecuTrial) we chose XML over CSV, because it contained more details. The XML was processed with XML Stylesheet Transformations (XSLT) to generate CSV – which in turn could be processed analogous to other CSV exports.

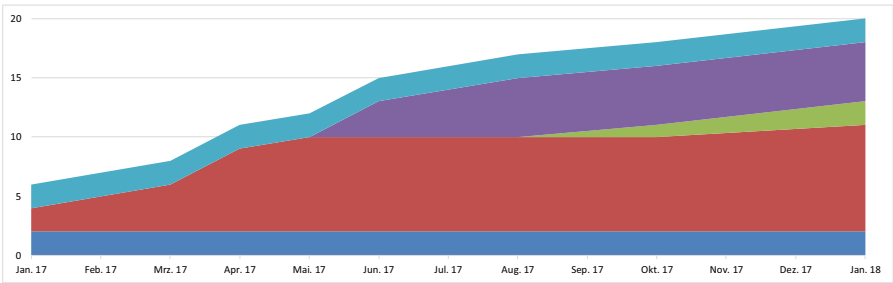

Figure 1. Progress of integrating data collections throughout 2017. Colors correspond to the five geog. sites.

4. Discussion

"Data integration should not just rely on software engineers and computational scientists, but needs to be driven by the actual users whose communities need to define, adopt and use standards, ontologies and annotation best practice" [1]. From technical point of view, data integration proved to be of minor difficulty. Most underestimated was the first step of "data discovery". The long period of two years between the first and second attempt for the survey was used for development of the IT infrastructure and data privacy concept. Hierarchical structures common to research networks in the medical field impeded initial communication with researchers collecting data. E.g., information and surveys to be distributed to all PIs had to be reviewed, modified and approved by several organizational structures and we were unable to distribute the data discovery survey directly to the researchers as we planned originally.

Integration of a single data collection took on average three days of technical effort for the data manager, depending on the structure and quality of the data. Technical difficulties included combination or separation of values (e.g., if a single common information derived from multiple values of the data collection or if a single source value contains multiple information). The majority of time was used for understanding the relational model behind the exports and mapping of data elements.

The small number of 17916 patients for 20 data collections is due to the fact that small collections are commonly less hesitating with data sharing. Large collections usually seek multiple consultations with data protection officers and ethical committees agreeing to data transfers. Thus, we expect a considerable increase in patient count for our data warehouse in the following months, when the next collections are integrated.

5. Conclusion

Data integration is an important but difficult task for large research networks. The data discovery aspect is easily underestimated and should be undertaken with direct involvement of all hierarchical levels of personnel.

6. Conflict of Interest

The authors declare that there is no conflict of interest.

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Predicting Matching Quality of Record Linkage Algorithms on Growing Data Sets

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Abstract. A record linkage algorithm tries to identify records which belong to the same individual. We analyze the matching behavior of an approach used in the E-PIX matching tool on the very limited attribute set of first name, last name, date of birth and sex. Our benchmark set contains almost 37,000 records from the Popgen biobank. We develop a model which allows us to predict the workload on clerical review for data sets growing up to a factor of 10 or even more, without the need for a data set of this size. Based on this model we show two parameter sets with comparable detection rate of true duplicates, but where only one of them scales well on growing data sets. Our model provides realistic example records for each predicted matching of an upscaled data set. Thus, it enables to identify the parameters which need to be adjusted in order to improve the quality of the matching candidates. We also show that unreviewed merging of records is prone to homonym errors on data sets with 200,000 records and the limited attribute set above, while the merged record pairs are obviously different in clerical review.

Keywords. Record linkage, data integration, parameterization

1. Introduction

Record linkage is the task to decide whether two records in a person database belong to the same person. Also referred to as duplicate record detection problem, an application can be found in quality checking of a single database, where we want to identify and eliminate duplicates. Another application arises, when information from different data sources has to be integrated into one common database. Such applications can be found, e.g., in health care, where medical treatment data from a hospital shall be combined with research data from study centers, or in the context of cancer registries [1].

Algorithms that address the record linkage problem cover rule based approaches, probabilistic methods, and machine learning techniques like decision trees, support vector machines or more advanced approaches [2]. A comparison and evaluation of these methods can be found in [3]. The linkage algorithms often base on similarity measures obtained by single field comparison, e.g. done by character-based similarity metrics like simple edit distances or more elaborated distance measures like the affine gap distance that puts different weight on longer insertions. Phonetic similarity measures like Soundex or the Kölner phonetic can be used to account for phonetically similar names with large edit distance in their written form. For a survey see e.g. [4].

The empirical results from [3] suggest that a binary field comparison would not decrease the record linkage quality, given a sufficient number of attributes. In cases of

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combining different databases, the linkage algorithms are limited to those attributes which are available in all sources. For our data integration scenario in the health care embedded biobanking at the university hospital in Kiel [5], this means we are limited to the attribute set of first name, last name, date of birth and sex and are thus required to use string comparison in the record linkage. The tool which is planned to be used for record linkage in Kiel is the E-PIX (Enterprise Patient Identifier Cross-referencing) [6], which already is in active use for data integration at the University Medicine Greifswald. The tool performs a probabilistic record linkage utilizing the Levenshtein edit distance [7] and the Fellegi-Sunter algorithm [8].

The Fellegi-Sunter approach is a probabilistic method and is usually based on an expectation maximization. Two records are compared field-wisely and transformed into a real-valued pattern vector, e.g. using the Levenshtein distance. The key is to estimate the conditional probabilities that this pattern vector is observed either assuming a match or a no-match. The decision process itself is then based on a global weight value r derived from the probability estimations. Via two threshold values, the record pair gets categorized in "good match", "possible match" and "no match". Good matches are intended to be linked unreviewed, and no-matches to be rejected unreviewed. Possible matches are presented for clerical review. The computation of the global weight value r is tuned by further attribute weights and attribute-specific thresholds that indicate to which extent a single attribute would be considered to match. We evaluate different sets of parameters on the study data sets from the biobank Popgen [9] and predict the behavior when the number of records grows up to the 10-fold, as motivated by a planned integration of the Popgen data with treatment data from the University Hospital in Kiel.

Changing the matching parameters in a running setup shall be avoided to retain reproducibility and reliability of the matching process. Therefore, it is crucial to estimate the long term workload for clerical review when the data sets are growing. While having a good quality of review candidates is one key to keep the workload low, our aim goes further and might imply dropping some positive matching candidates, when otherwise the workload would just be beyond personal capacity. We also focus on the unreviewed linking of good matches, which can result in false-positives (homonym errors) that remain undetected in a productive environment.

2. Methods

We implemented a simple and flexible version of the record linkage algorithm similar to the one used in the E-PIX tool. Special characters are replaced, e. g. "ä" by "ae" and "-" by a blank, and for attribute comparison the Damerau-Levenshtein distance (allowing transpositions) is used to compute $p_a = dlev(u_a, v_a)/max(|u_a|, |v_a|)$. For attribute specific thresholds t_a and weights w_a, we compute the global weight $r = (\sum w_a(1 - p_a))/(max(1, \sum w_a p_a))$. To predict the matching behavior for growing data sets, we develop a model that relates the number of records with the number of possible matches on a typo-free data set. This model is meant to capture matches of records which correspond to different persons and thus are unwanted noise. We restrict our analysis to possible matches of records with same date of birth and thus analyze a subset of the matches found by the algorithm. Assuming a data set S with n records sampled at random from a population with probability distribution P, the number of records in S with date d is binomially distributed with $p_d = P(\text{date}=d)$. We can show that, when the data set grows to kn records,

the number of expected (false-positive) possible matches grows approximately by k^2 for $n>>k$. This enables us to predict the number of possible matches in a k-fold data set, however, the number of possible matches in the original data set might be too low for a good prediction. Also, we need to eliminate all correct possible matches beforehand. To overcome these issues, we modify the sampled records in S and manipulate the date of birth. We merge c subsequent days as follows: Each date is converted into the number of days since a fixed date in the past divided by c. The integer portion of the ratio plays the role of the date, i.e. we consider pairs of records with same integer value. Merging c days $d_1,...,d_c$ to a bin b_1 and assuming the same date probability p_d for all of them, we expect $(c^2n^2-cn)p_d^2/2$ pairs of records in bin b_1. On the other hand, there have been $c(n^2-n)p_d^2/2$ expected pairs within the c days before merging. Thus, the growing factor of the expected number of pairs due to the merging of c days is $(c^2n^2-cn)/c(n^2-n)=c+(c-1)/(n-1)$. Assuming the same probability $p_{d,m}$ for each such pair to be a possible match, this factor holds for the expected number of possible matches as well – the number of possible matches grows linearly in c. We now relate c and k and obtain $c\approx k^2$. Thus, merging k^2 days gives the same expected number of possible matches as using a k-fold data set.

Note that duplicate records with typos in the name, but not in the date, contribute to the total number of possible matches as a constant offset – their number does not change when merging dates. Thus, linear regression allows us to quantify the offset. The effect of duplicates with typos in the date should be negligible due to the dominating linear growth of the false matches and due to their distribution on the merge bins.

3. Results

We perform the matching algorithm on the data set of the studies from the Popgen biobank [9] with a total number of 36,894 records. As first parameter set for the attributes first name, last name, date of birth, and sex we use the threshold values (0.8, 0.8, 1.0, 1.0) and the weights (8, 7, 15, 2). These parameters have been motivated from a first E-PIX configuration and allow first and last name to differ in up to 20% with a decent penalty, while date errors receive a large penalty. Result values $32\geq r\geq 29$ are defined as good match (unreviewed merging), values $29>r\geq 5$ are possible matches and subject to clerical review. Besides 996 identical records (removed in preprocessing), the data set contains 34 good matches. Reviewing them via considering additional address data confirms all 34 matches to be correct. All these matches contain typos like “Margret” and “Magret” or “Günter” instead of “Günther”, while last name, date of birth and sex are equal. Further, the algorithm identifies 80 possible matches. They contain examples for errors in every attribute. We could identify 9 correct matches with

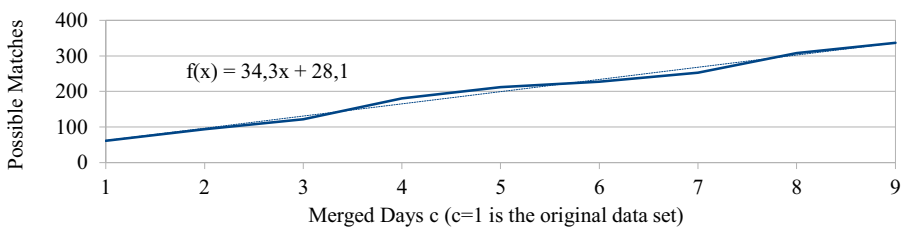

Figure 1. Possible matches with same date on the Popgen data set (n=36,894) when up to 9 subsequent days are merged. The linear regression is given as formula and shown as dashed line.

date errors and another 15 with typos, additional first names and changes of the last name, probably due to marriage. The true matches range over the full range we allow for r. From these results, the parameterization might seem to be a reasonable choice. However, it is not: We now restrict ourselves to matches with equal date of birth, which is 62 on this set, and apply our prediction model by merging subsequent dates. In Figure 1, we observe the growth of the number of possible matches linearly in the number of merged days c. A linear regression estimates the offset to 28 records which should be correct matches in our data set and is close to the 34 matches identified in our review. Next, we simulate the possible matches in a k-fold sized data set via merging k^2 subsequent dates from our Popgen data set. We observe 1206 possible matches already in the 6-fold data set size and 3274 possible matches on the 10-fold size, imposing a high workload in clerical review. Since our model also provides example names which would occur as good matches, we are able to point out a more startling effect on homonym errors: When reaching the 6-fold record number of about 200,000 records, we observe 5 good matches, like e. g. "Adrian Sauer" and "Adrian Baier" (names changed), which belong to different individuals and hence are false-positives. We analyze this effect further: Assume two persons have the same first name and the same date of birth. Which last names would then lead to a good match? If we consider the 3,000 most common last names in Germany, and consider the last name Bauer, then these names would result in a good match: Sauer, Bayer, Baier, Bader, Baur, Lauer, Bär, Brauer, Auer, Hauer, Mauer, Baer, Rauer. Thus, unreviewed merging of good matches is prone to homonym errors on our limited set of attributes.

We propose another set of parameters for the matching algorithm and use the threshold values (0.7, 0.7, 1.0, 1.0) and the weights (7, 7, 8, 1). Result values $23 \geq r \geq 20.5$ are considered as possible match for review. There are no good matches by definition. In total 60 possible matches are found. Among them are the 34 good matches of the first parameter set and the 18 possible matches with differing date of birth, as well as 8 matches from the other 80 possible matches we had before. The second parameter set leads to 58 of 60 confirmed correct matches, while 9 further true matches known to be in the Popgen data set are missed. Figure 2 shows the predicted number of possible matches with same date matchings for the second parameter set. Besides the results shown here, we performed our analysis on a second available study data set with n≈45,000 and were able to confirm our qualitative results and conclusions.

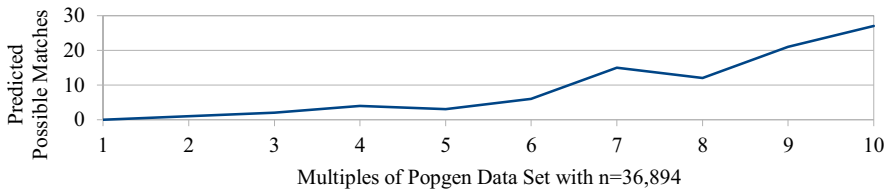

Figure 2. Predicted possible matches with same date for a sample set of kn persons (up to $10n$=368,940), based on the Popgen data set (n=36,894) and using the model merging k^2 subsequent days.

4. Discussion

The first parameter set seems to be a reasonable choice for our Popgen data set with 36,894 records. All 34 good matches are correct and we have at most 56 incorrect possi-

ble matches. However, when the data set grows to 200,000 records, we get 5 false-positive good matches and over 1,000 incorrect possible matches even when restricting to equal date matches. Hence, the quadratically increasing workload renders this choice of parameter infeasible for larger data sets. Our second parameter set in comparison looses 9 correct matches, but the number of incorrect possible matches for 200,000 records is estimated below 10 for equal date matches. This is an interesting observation since we allow an error rate of up to 30% of the characters in the name. This parameter set would be a reasonable choice for record numbers of 500,000 or more. When considering unreviewed merging of good matches, we predicted a growing number of homonym errors for data sets with 200,000 records and more. Thus, for linking medical records with our attributes we highly recommend to review all good matches as well.

5. Conclusion

When parameterizing record linkage algorithms for growing data sets, it is not sufficient to observe good test results on some reference data set. We have shown one parameter set that worked well on our reference data, but revealed an infeasible scaling behavior and also resulted in false-positive record linkage. We developed a model to predict the matching behavior for large multiples of such reference data sets. The model provides a general framework for benchmarking of record linkage parameterizations and provides example matches that allow to tune the parameters according to the expected data.

6. Conflict of Interest

The authors state that they have no conflict of interests.

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Enabling Pedigree Visualization and Analysis in tranSMART

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Abstract. This paper examines the relevance of genetic pedigree data in the context of medical research platforms. By surveying currently available tools for visualization and analysis of this data type and by considering possible use cases that could make usage of the combination of singular patient data and pedigree data, the advantages of integrating the data type into a medical research platform were shown. In a practical work step, an integration procedure of pedigree data into tranSMART was created. Furthermore, a tool to analyze and visualize pedigree data in combination with other patient data was implemented into SmartR, a dynamic analysis tool inside of tranSMART. Finally, we address limitations and future development strategies of the tool.

Keywords. Pedigree, visualization, tranSMART, medical research platform

1. Introduction

Modern medical research platforms are able to greatly simplify the integration, combination and querying of the diverse structured and semi-structured medical data created in medical research. Therefore, they greatly simplify making use of large and diverse data sets [1].

Currently, genetic epidemiologic research is mostly excluded from the advantages of these databases, as most workflows rely on observations of subject groups instead of single patients. Unfortunately, neither workflows nor the integration of pedigree data are supported by currently available medical research databases [2].

This integration however could be highly valuable, as multiple publications show the advantages of combining pedigree data with genetic data of single patients [3,4]. Other publications encourage the visualization and analysis of patient data on a familial level [5].

Therefore, the major goal of this work was to offer integration support of pedigree data into a major medical research platform (tranSMART [6]). Furthermore, a prototypic tool was added into the platform to allow the visualization of the pedigree data as an actual pedigree diagram and to execute ad-hoc pedigree analyses.

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2. Methods

To decide, if the integration, visualization and analysis of pedigree data into a medical research database generates benefit, a literature research was conducted using PubMed and Google Scholar. Search terms included: pedigree, tool, ped-data, family tree, visualization, analysis, epidemiology, and family-based. Only papers including a technical description of the used tools and solutions were selected. In the same literature research, feasible use cases that benefit from the utilization of pedigree data and also established applications to analyze and visualize pedigree data were sought. The latter was conducted to benefit from an amplified requirement creation for our similar tool, to enhance usability and to provide useful functions from the perspective of a researcher.

To prepare the implementation of a prototypical tool, a tranSMART v16.2 server was prepared. The data analysis plugin SmartR [7] provides a user interface for displaying visualized analysis in an interactive fashion. Furthermore, it simplifies the integration of R scripts into tranSMART by offering a clear interface to the embedded Rserve process. R is an established programming language for statistical computations in the statistics community. SmartR can be extended with user-created analysis plugins.

The Groovy and Grails Suite (GGTS) was configured using the freely available source code of tranSMART and SmartR to develop the required tool functionalities. As a template for development, source code of a previous developed plugin [8] was chosen as a starting point. This way, similar usability should be ensured.

The development itself was carried out in multiple phases. In a first phase a JavaScript visualization was designed to display a pedigree diagram with additional analysis results inside SmartR. In a second phase data transformations were implemented in R to utilize data forwarded from the tranSMART database in the customized visualizations and pedigree analyses defined by literature. In a third phase the statistical calculations desired in the plugin were implemented in R. To test the plugin, a test set of pedigree data was prepared. The data was stored in the PED and MAP format, which describe familial genotype data [9]. Both data types are needed in combination and are widely used and accepted in genetic epidemiology and in human genetics research [9]. PED files store family data and genetic data per subject in a tab separated value format. MAP files store additional data about the genes referenced in the PED files, again in a tab separated value format. Both file formats possess a detailed format reference defining allowed values and explaining every data column [9].

Finally, the public data set GSE20690 [10] was prepared to be imported into tranSMART using transmart-batch. The data set was chosen to provide additional observations for every patient referenced in the PED file created earlier. This enabled to test how the combination of pedigree data and non-pedigree/non-omics patient data can be performed.

3. Results

The literature research revealed that several tools for analysis and visualization of pedigree data were published. While the main tool for any basic analyses was the command line tool PLINK [11], which is still heavily used, other tools were either web-applications [9,12] or R-packages [13,14]. Most of the R-packages were designed to solve one very specific problem and released for this single project. The web applications were mostly using a more general approach for visualization and analysis. However,

many of the released web-applications were no longer developed or even available, although they [12] provided an easy-to-use solution to analyze pedigree data and to combine it with other data. Nevertheless, these tools required some extra setup, to learn the application handling and to import data and export results due to sparse documentation. Which may have led to poor adoption and abandonment. The presented implementation overcomes those caveats, because it included an already known environment to the user and little effort from the perspective of data integration.

Based on the same literature research, two use cases were defined supporting the assumption that creating a pedigree analysis and visualization tool within a research platform is advantageous: (1) One significant challenge of the current medical research is the investigation of rare diseases triggered by rare variations using high throughput data [4]. The common method to analyze solely the data of one patient generates high numbers of false-positive results for rare diseases, because false results cannot easily be distinguished from rare variations [3]. If high-throughput data of near family members is included into the analysis, the rate of false-positive results can be significantly lowered as close family gene data can be used to reveal paternal and maternal variations [3]. Another advantage will originate from properties of family data: as kindred gene data is interdependent, the probability to find other carriers of the rare disease is much higher than checking non-related subjects [3,4]. This way, the number of cases needed to perform a study can be reached easier [4]. Modern medical research platforms already allow importing high-throughput data [2]. With the integration of pedigree data and accompanying analysis tools, rare disease research could be accelerated. (2) Genetic counseling uses pedigree analysis to calculate risk factors for patients and their unborn offspring. While the mathematical foundations to calculate the correct risk factors are well established, this risk is often calculated incorrectly because only the parents' subjects are included in the calculation instead of all pedigree data available (e.g. grandparents and siblings) [15]. An easy to use tool showing counselors all data available and executing the calculations for them could easily enhance the precision of the calculated risk [15]. Both use cases can be improved with a pedigree visualization, allowing easier insight into complex data. The visualization can enhance understanding and insight of intricate analysis results for researchers and involved patients. As a result of both, the defined use cases and the investigated tools, the implementation of a prototypic analysis and visualization tool inside a medical research platform was chosen. The established open source research database tranSMART was selected providing an extensible data schema as well as an interface for visualization and analysis extensions.

Using the tools presented in the methods section it was possible to create an ETL-process from PED-files into the tranSMART-database. As input a PED and a MAP file had to be used. Additional data had to be created to provide data structures required by tranSMART. For the test data set fictional families were created with patient ids of the GSE20690 study. The converted pedigree data was then combined with the study data and imported into tranSMART. The created prototypic visualization and analysis tool can visualize a pedigree diagram inside of tranSMART using pedigree data, affection data and arbitrary optional data. An example screen of the tool is available in Figure 1. The visualization algorithm was built in compliance with the pedigree visualization conventions released by Benett et al [16]. Furthermore, the tool automatically executes multiple basic pedigree analyses and displays their results below the diagram. These are: (i) the allele frequency; (ii) the list of Hardy-Weinberg-equilibriums; (iii) the largest group of non-related subjects; (iv) the kinship coefficient matrix.

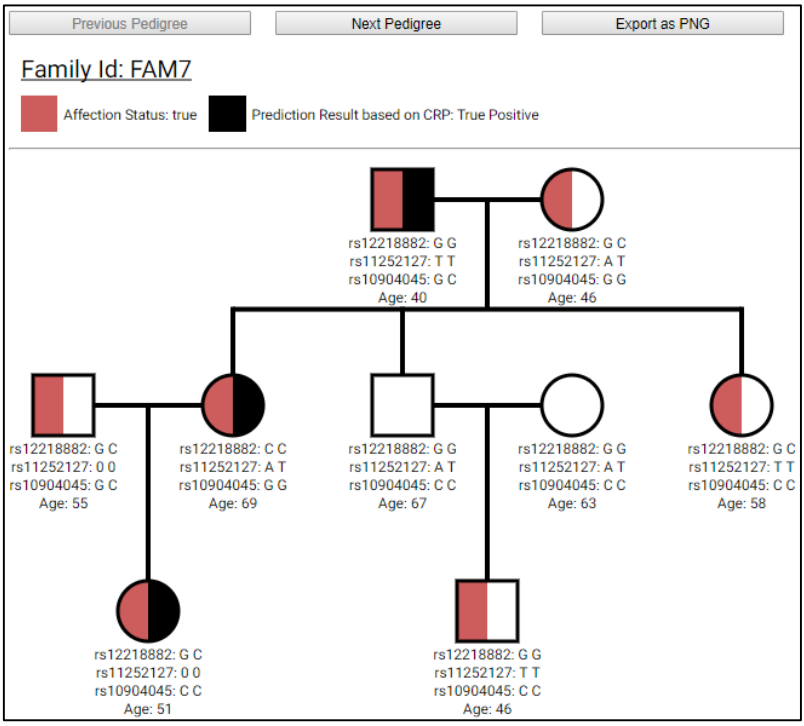

Figure 1. An exemplary output generated by the pedigree analysis prototype. Affected subjects are colored in red. Subjects who have a true positive prediction resulted based on CRP (C-reactive protein) are colored in black. Genetic and age data are shown as text below each subject node.

4. Discussion

The current feature set of the prototype can provide basic but useful insights. Further development with the goal of additional visualizations and analyses has to incorporate in-depth researcher collaboration. This will reveal additional requirements for the tool to be used in active medical research. Also, the support of statisticians is essential to ensure the correct handling of data in the analysis process. A first potential group of researchers to be involved in the development process is suggested by the rare disease use case: researchers working in human genetics using both pedigree data and variation data from high throughput data. Finalizing this use case in a productive environment would be highly beneficial. It is expected to show that there is not only a theoretical but a practical benefit from using the tool and that it is feasible to be integrated into an existing workflow. Additionally, the functionality of the tool should be tested in this environment with novel data showing that it is able to handle diverse settings involving pedigree data. One expected shortcome might be contradictory data from real world cohorts (e.g. loops) which might require additional error handling. However, before using real patient datasets, data privacy concerns must be resolved. As tranSMART already handles roles and rights for every user, only minor changes may be needed to the tool for reaching compliance with data privacy rules.

5. Conclusion

This work revealed that the technological barrier hindering genetic epidemiologists from using medical research platforms can be resolved. Existing applications and use cases showed many possible features to be integrated in a single solution to analyze genetic pedigree data. This solution may propagate the usage of medical research platforms in genetic epidemiology supporting collaboration, data reusability and data sharing.

6. Conflict of Interest

No conflict of interest. The work shown in this paper was conducted as part of a master thesis by Robert Lodahl at the Department of Medical Informatics of the University Medical Center Göttingen (UMG). It was supervised by Ulrich Sax, Professor at the Department of Medical Informatics at the UMG and Michael Krawczak, Chair of the Institute of Medical Informatics and Statistics at the Christian-Albrechts-University Kiel. Partially funded by the BMBF in the project sysINFLAME (01ZX1606C).

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5. Decision Support

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Ontology-Guided Markerless Navigation and Situational Awareness for Endoscopic Surgery

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Abstract. Optical navigation systems help surgeons find their way through the complex anatomy of a patient. However, such systems are accident-sensitive, time-consuming and difficult to use because of their complicated technical requirements such as the setting of optical markers and their intraoperative registration. The BIOPASS project, therefore, provides an innovative localisation system for markerless navigation in endoscopic surgery to support medical decision making. This system comprises several machine learning classifiers to recognise anatomical structures visible in the endoscopic images. To verify the data provided by these classifiers and to alert medical staff about surgical risk situations, we developed a new ontology-based software called OntoSun. Our software improves the precision and the sustainable traceability of the classifiers' results and also provides warning messages that increase situational awareness during surgical interventions.

Keywords. Surgical navigation, situational awareness, endoscopic surgery, biomedical ontologies, knowledge representation, formal ontology

1. Introduction

Although optical navigation systems help surgeons find their way through the complex anatomy of a patient and may reduce their task workload as described in [1], such systems tend to be accident-sensitive, time-consuming and difficult to use because of their complicated technical requirements. These requirements such as the setting of optical markers, preoperative imaging (CT or MRT) and intraoperative registration are described in more detail in [2-3]. The BIOPASS project [4] provides an innovative localisation system for markerless navigation in endoscopic surgery that considerably reduces the intraoperative technical requirements. In order to recognise anatomical structures visible in the endoscopic images and to predict possible subsequent landmarks, the BIOPASS system provides several machine learning classifiers as described in [5]. Nevertheless, statistically based algorithms are not able to reach the precision and sustainable traceability necessary for computer-assisted surgery [6]. To overcome this problem and to improve the surgeon's situational awareness, we developed a new ontology-based software called OntoSun as described in the next section.

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2. Methods

2.1. Design and Ontological Architecture

The BIOPASS System serves as a clinical decision support system (CDSS) for markerless navigation during endoscopic surgery. This system has a client-server architecture with a central communication server as shown in figure 1. The BIOPASS system consists of three basic components: The endoscopic navigation system, the ontology-based software for supporting the surgical navigation and situational awareness (OntoSun), and a set of classifiers to identify anatomical structures visible in the endoscopic images. In order to ensure an effective and real-time communication between the system' components, a service oriented architecture was implemented using the MQTT-Protocol [7]. Consequently, all components run as MQTT clients and can communicate with each other in real time via a message broker. The captured information about the current surgical positions is used to enrich a situational knowledge base created by OntoSun at runtime. This knowledge base has a layered architecture that adopts different types of ontologies at different levels of abstraction. During the ontology development process, we used GFO [6] as a top level ontology and FMA [8] as a domain ontology. To harmonize GFO and FMA entities, we, additionally, developed a core ontology, called BISON [6], and utilized this ontology as a framework for the ontological analysis of functional endoscopic sinus surgery (FESS). As a result, we generated a domain specific ontology, called BISON-FESS, from an excel template using Expert2OWL [9]. This tool enables us to reduce the required development cost and provides a general solution for the implementation of other clinical use cases such as colonoscopy. The resulting ontology provides an OWL representation of domain specific entities that may appear during FESS interventions. Hence, BISON-FESS can also be interpreted as a task ontology [10] because it represents a specific navigation task that has to be accomplished using OntoSun. For this purpose, we implemented the OntoSun navigation algorithm as described in the next section.

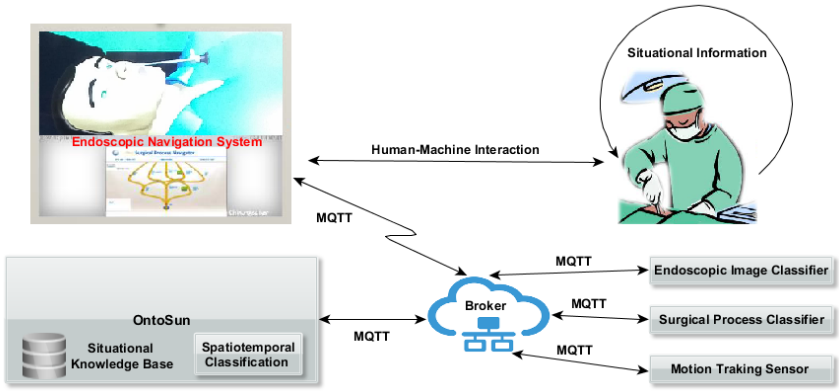

Figure 1. Surgical navigation using OntoSun embedded in BIOPASS CDSS.

2.2. Algorithm and Implementation

The OntoSun algorithm runs across an architecture consisting of three layers. All layers are clearly distinguished by different knowledge representation formats. This distinction

allows a clear separation of responsibilities between all involved developers. The external layer is a service layer that enables the communication with other system components using the MQTT protocol. During surgical procedures, this service interface captures multimodal sensor data as JSON-formatted messages [11] and uses a JSON parser to parse these data. The parsed JSON objects are then passed to the second layer called reasoning layer. The main task of this layer is to provide ontological reasoning of necessary surgical knowledge and to return the results to the service layer. For this purpose, OntoSun integrates the captured surgical information into a situational knowledge base using BISON-FESS. This information includes, inter alia, landmarks which are anatomical structures representing points of orientation. If a new landmark has been detected, the corresponding commands, implemented in the command layer, which is the third layer, are then called to generate a new representation of the current surgical situation and integrate necessary updates into the knowledge base including additional runtime knowledge as described in [12]. To allow such updates and to detect contradictory changes at runtime, we implemented the command layer by using the OWL API [13], and the Openllet DL reasoner [14]. Moreover, this layer provides a set of commands to perform atomic reasoning tasks and queries on the DL knowledge base. Hence, the command layer represents a java interface that provides basic features for bridging the core layer and the DL knowledge base.

During the development process, we adopted the model of test-driven development [15] that allowed us a continuous test, evaluation and refinement of OntoSun. Initially, we implemented a Junit test Suite and chose FESS as first exemplary clinical treatment process. We proceeded with a simple implementation of the basic OntoSun features and iteratively enhanced the developed version until all software requirements were met. We used subversion as a source code repository for version control and iteration planning. At each iteration new methods were added and corresponding tests were made. In addition, we used MQTT messages as input data to test each type of situation with exemplary situational information about FESS procedures.

3. Result

We developed an ontology-based software to support surgical navigation and situational awareness called OntoSun. This software component was embedded into the BIOPASS system as shown in figure 1. OntoSun runs as a web application that continuously receives MQTT messages from other components such as the mentioned classifiers and the movement sensor placed inside the endoscope. OntoSun interacts with these components by exchanging messages about situational information that occur during surgical procedures. Figure 2 depicts an exemplary FESS procedure that consists of three surgical situations. Each surgical situation is defined by its constituent parts, i.e. the anatomical structures visible in the current endoscopic image. The first endoscopic image is a snapshot of the first surgical situation and shows the middle nasal meatus (fma53139). The second endoscopic image visualises the middle nasal meatus and the middle nasal concha (fma57459), which are constituent parts of the second situation. The last endoscopic image is a snapshot of the third situation and shows the middle nasal concha. An anatomical object having the disposition to exhibit the anatomical structures shown in figure 2 is either an instance of the FMA concept fma53139 or fma57459. If OntoSun, for example, receives a message containing the IDs of both FMA concepts, the corresponding commands are then called to generate a new representation of the current

surgical situation. The generated ontological representation, in this case, will describe the anatomical structures visualised in the second endoscopic image i.e. the middle nasal meatus and the middle nasal concha. These anatomical structures will be defined as presentic information i.e. they are instances of the class *Presential* in GFO. In general, OntoSun classifies the occurring anatomical structures of a situation sequentially and after each created presential, the captured information about the current surgical position is used to enrich the situational knowledge as describe in section 2.2. If this enrichment leads to a contradiction, the anatomical structure that has been added is rejected as a constituent part of the new situation. After all FMA IDs in the received MQTT message have been processed as described above, a list of the detected invalid landmarks is published as feedback message to avoid navigation errors.

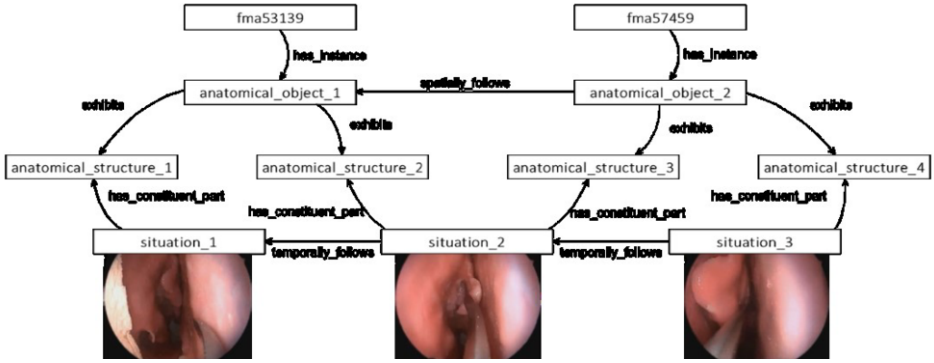

Figure 2. The representation of three consecutively occurring endoscopic images as surgical situations.

Furthermore, OntoSun infers possible subsequent situations and their related anatomical landmarks. If the inference engine has classified the apparent situation as surgical risk situation, OntoSun sends a warning message to the navigation system. This warning message is then displayed on the endoscopy monitor to alert medical staff and to avoid dangerous situations. As a result, OntoSun provides efficient features to increase situational awareness and to support surgical navigation. The inferred surgical knowledge can be used to improve medical decision-making in endoscopic surgery.

During the development process, we adopted an iterative development model that allowed us a continuous test, evaluation and refinement of OntoSun. The resulting software features were successfully tested and evaluated using the Junit test suite as described in section 2.2. The test results revealed the correctness of the implemented methods such as surgical risk detection and navigation support.

4. Discussion

To our knowledge, no papers have been published about markerless navigation and situational awareness. Therefore, we analysed existing ontologies for their potential to represent situational information as well as to support surgical navigation and concluded that none of the ontologies and tools presented in [16-19] could be applied to our project. Neither of these approaches could answer relevant competency questions such as “which anatomical landmarks may occur in the next surgical situation?”, “which surgical situation constitutes a risk situation?” and “which anatomical structures bear this risk?”. In contrast to previous studies, our approach provides not only a framework for

representing the semantics of surgical information but also offers a new tool for the classification and integration of such information in a DL knowledge base during runtime. Using this knowledge base, our tool provides support for surgical navigation and situational awareness. Hence, our approach transcends these previous contributions to surgical knowledge representation. In conclusion, our tool yields promising results, which can be used to support medical decision making and reduce surgical navigation errors. Our future work will include clinical studies to provide an evidence-based evaluation of the whole system outcomes and investigate other clinical use cases such as colonoscopy.

5. Conflict of Interest

The authors state that they have no conflict of interests.
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6. Ethics and Data Protection

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Protecting Record Linkage Identifiers Using a Language Model for Patient Names

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Abstract. Linking information across databases fosters new research in the medical sciences. Recent European privacy regulations recommend encrypting personal identifiers used for linking. Bloom filter based methods are an increasingly popular Record Linkage method. However, basic Bloom filter encodings are prone to cryptographic attacks. Therefore, hardening methods against these attacks are required. In this paper, a new method for such a hardening method for Privacy-preserving Record Linkage (PPRL) technique is presented. By using a Markov chain-based language model of bigrams of identifiers during the encryption, protection against attacks is increased. Based on real-world mortality data, we compare unencrypted and state of the art PPRL methods with the results of the proposed hardening method.

Keywords. Medical data, bloom filter, markov chains, mortality data, privacy

1. Introduction

Linking information from different data sets is an increasingly popular research strategy. Specifically, in Record Linkage, the goal is a dataset containing information from different databases on the same person or organization. Recent EU data privacy regulations [1] recommend encrypting identifiers used for linking. This recommendation will increase the demand for Privacy-preserving Record Linkage (PPRL) solutions, which enable error-tolerant data linkage with encrypted identifiers. For example, PPRL methods have been used for a nation-wide Record Linkage of births in Germany [2], to link health data across states in Australia [3] and to link 114 million records from a cohort-study to health records for epidemiological research in Brazil [4].

2. Methods

Although many different methods for PPRL have been suggested (for an overview, see [5]), only very few techniques are suitable for the standard setting in medical sciences [6]: Usually, many different hospitals and other institutions provide data on the same patients according to a centrally specified encryption and transmission protocol. In general, repeated online computations between the data providers are not permitted by the data protection agencies. Therefore, the actual linking has to be done by a trusted third party using encrypted identifiers. In the technical literature, this kind of setting is

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called a two-party protocol. For this, only two variants of encryption are widely used [7]: Encrypted Statistical Linkage Keys (ESLs) and Bloom filters (BF). ESLs are based on encrypted phonetic codes or subsets of identifiers. An example is the encryption of concatenated identifiers by cryptographic functions such as SHA-3. Bloom filters (as suggested by [8]) split identifiers into bigrams, which are then hashed into a bit vector of a fixed length. In Computer Science, Bloom filters are now considered by some authors [9] as de facto standard in PPRL. The currently recommended technique [10] for mapping bigrams into a Bloom filter is the use of randomly sampled bit positions using the bigram together with a password as a seed for a random number generator, such as Salsa20 [11]. Bloom filters have been subject to cryptographic attacks [12], which is why several hardening techniques have been suggested. After giving an overview of hardening methods, we will propose a new method, based on the idea of inserting additional bigrams to Bloom filters according to the estimated probabilities of a language model. We report on an evaluation using real-world mortality data.

2.1. Hardening Bloom Filters Against Attacks

All known attacks on Bloom filters (BF) for PPRL are either frequency or pattern-based attacks. They require frequent Bloom filters or frequent co-occurrences of bigrams. Depending on the kind of attack, the lower limit on the number of required frequent patterns differs. Therefore, all hardening techniques will try to reduce the number of frequent patterns or co-occurrences. Common to all protection mechanisms is the attempt to use the two elementary cryptographic design principles of confusion and diffusion [13] for PPRL. The confusion of a cipher is large if the relation between the distribution of the encrypted texts and the distribution of the plaintexts is complex. The diffusion of a cipher is large if changing a single bit of the plain text and a single bit of the key changes as many bits of the cipher as possible [14]. Higher diffusion of a specific encoding will require more encoded records for a frequency attack. Niedermeyer et al. [12] suggested the inclusion of random-bits, which would make deterministic filtering as used by Christen et al. [15] much harder. Furthermore, Niedermeyer et al. [12] recommended salting, where a stable plaintext value (such as year of birth) is used as an additional substring for the password used during encoding BFs. [6] suggested the use of rehashing, where a sliding window of bits in a BF is used as an integer, which is hashed again to a second BF. Based on a speed-up for chemometric lookups introduced by [16], XORing of folded BFs was suggested by [17] for PPRL. Since the Hamming weight of BFs is essential for many frequency attacks, [10] proposed the use of Balanced BF, where each BF of length 2*l has a hamming weight of l. Although a comprehensive evaluation of all suggested hardening methods is not available yet, salting combined with balancing of random-hashing BFs is currently considered as hard to break. However, further hardening methods are needed for real-world applications of PPRL to make attacks as difficult as possible. Therefore, we here suggest a new technique for hardening Bloom filters.

2.2. Markov Chain-based Bloom Filter (MCBF) Generation

Current attacks are based on the bit patterns that can be guessed by approximating the underlying frequencies of bigrams. To prevent this, additional bigrams not based on the actual data could be inserted. As long as the additional bigrams are conditionally dependent on the actual bigrams, this should have a small impact on linkage quality.

Therefore, we calculate the conditional probabilities for the follow-up bigrams for all bigram pairs. This computation is essentially the estimation of a Markov chain [18] for a given input data set. Using the name THOMAS as an example, we have the bigrams: TH HO OM MA AS. Starting with the first bigram, TH, the transition probabilities were calculated. Using the transition probabilities as sampling probabilities, we could draw, for example, cl=4 new bigrams (chain length) for each bigram in the name. Using a seed of 42, the following additional bigrams for TH are drawn: HR HI HE HY. These are appended to the list of bigrams. The final obfuscated (deduplicated) bigram list for the name THOMAS is: TH HR HI HE HY HO OB OM OP MM MS MA M_ AU AH AR AE AS SI SA S_ SE. The resulting bigram list is then hashed into the Bloom filter.

2.3. Data and Linkage Quality

To study the performance of MCBFs, we used real-world mortality data. We linked an administrative database containing two years of death certificates (n=14,003) of a large city with external mortality data from a university hospital in the same city (n=909). Both data files were preprocessed - for example, all special characters and symbols were removed. The overlap of both files is n=889. To assess the linkage quality, precision (the number of true positive links divided by the number of all positive links) and recall (number of true positive links divided by the number of true links) were computed. Traditionally, the F-score (the harmonic mean of precision and recall) would have been reported. Given the recent critique of the F-score [19], it will not be reported here.

3. Results

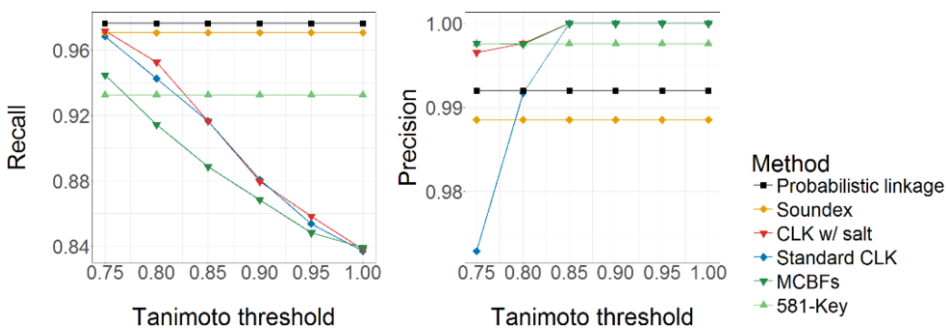

Figure 1. Recall and Precision of Markov Chain-based Bloom filters compared to standard linkage methods.

Using Multibit trees as described in [20, 21], the results for precision and recall depending on the similarity threshold (Tanimoto coefficient) of the Multibit trees are shown in figure 1. We used MCBFs with a chain length cl=3 for each original bigram. Furthermore, we used the date of birth as a password to hash the Markov-chain-based bigrams into CLKs using salting and k=10 hash functions. We tested this variant against standard CLKs (using k=20 hash functions) as described in section 2.1, as well as salted CLKs (with date of birth as a salting variable and k=30 hash functions) and 581-keys [22]. All Bloom filter-based methods used l=1000 bits. Finally, we compared all PPRL methods including Soundex to the widely recognized standard for non-Privacy-

preserving Record Linkage, namely probabilistic Record Linkage on unencrypted identifiers.

Table 1. Recall, precision and corresponding true and false positives. True overlap is n=889.

Method	Recall	Precision	TP	FP
Probabilistic linkage	0.98	0.99	868	7
Soundex	0.97	0.99	863	10
581-Key	0.93	1.00	829	2
CLK w/ salt	0.95	1.00	847	2
Standard CLK	0.94	0.99	838	7
MCBFs	0.91	1.00	813	2

At a Tanimoto threshold of t=0.8, table 1 shows the amount of true and false positives for each method in addition to precision and recall. As could be expected, probabilistic linkage using clear-text identifiers outperforms all other methods. Encrypting identifiers leads to reduced linkage quality. However, salted CLKs and 581-keys perform well. For MCBFs, a slight decrease in recall can be observed while precision remains very high.

4. Discussion

If privacy is a prime concern in research applications of Record Linkage, PPRL methods have to be used. Due to interdisciplinary research, new PPRL techniques have been developed during the last ten years. These new techniques are now increasingly used in real-world applications in health sciences. However, theoretical attacks on these PPRL methods have been published.

5. Conclusion

In this paper, we proposed a new hardening method to prevent all currently known attacks. Since a language-based model is used to increase the extent of cryptographical confusion, currently known attacks would be unable to estimate the actual pattern of q-grams in the identifier represented by the conditional random representation in the Bloom filter. Therefore, MCBFs are less likely to be vulnerable to frequency attacks or pattern mining attacks. This is because the required frequency information is not available and the patterns formed by a given q-gram are not deterministic. Because of the increased resilience against attacks, this may allow the application of PPRL in settings requiring strict privacy. Of course, further research on the cryptographic properties of PPRL methods is needed.

6. Conflict of Interest

The authors declare no conflict of interest.

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7. Health Information Systems: Building Blocks and Workflows

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German Medical Data Sciences: A Learning Healthcare System
U. Hübner et al. (Eds.)

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Measuring the Availability of Electronic Patient Data Across the Hospital and Throughout Selected Clinical Workflows

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Abstract. The workflow-oriented dissemination of electronic patient data is a central goal of IT deployment in hospitals. Against this background, the present study examines two research questions: (1.) Are there differences in the availability of electronic patient data (AEPD) between different clinical workflows and data types and (2.) which structural and organizational factors determine AEPD? Based on a Germany wide hospital survey, AEPD was assessed along six clinical workflows. While AEPD was lowest for ward rounds, discharge showed the highest AEPD with pre- and post-surgery processes ranging in between. With regard to the data types analyzed, patient demographics and observation findings obtained the highest AEPD scores. Electrophysiological results, checklists and warnings were less common electronically and received lower AEPD scores. Multiple linear regression analysis resulted in a significant model that explained 34.4% of the variance of AEPD. Large hospitals and those with a professional information management, a high health IT related innovation culture and a nursing informatics officer possess higher AEPD scores and thus have better clinical information logistics mechanisms at their command.

Keywords. Availability of electronic patient data, clinical information logistics, information management, innovation culture

1. Introduction

Patient-centred information transfer, which is oriented towards clinical workflows and thus transcends the boundaries of specialist departments and professions, has been the declared goal of IT use in hospitals for several decades [1]. On an abstract level, this goal can be described as optimal clinical information logistics [2]. This construct defines the degree to which clinical users are promptly provided with the exact information they need to make the right clinical decision [3]. Transferred to everyday hospital routine, this results in an ideal data flow. In secondary care, this data flow begins with the admission of patients by transferring patient data electronically from previous care levels. During ward rounds, admission and other types of information such as results, kardex, warnings and checklists are made available to the clinicians at the point of care. As soon as a patient is to undergo surgery, all the information collected so far is made available in electronic form to the anesthetists, surgeons and operating-room nurses. If the patients leave the operating-room, their data is also passed on electronically, regardless of whether they are transferred to - the normal or

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intensive care unit. At the end of the treatment, the collected data will eventually leave the hospital along with the patient being made available electronically via the clinical summary. The potentials that result from such a clinical information logistics are manifold and range from a pure increase in efficiency through process automation, to the creation of more transparency in the treatment process, to a more effective and safe treatment [4]. To date, there is no empirical evidence on the extent to which the hospital information systems (HIS) of German hospitals are able to provide a corresponding availability of electronic patient data (AEPD). On the one hand, a large-scale and standardised assessment of AEPD could allow a status quo analysis and thus reveal potential for optimization. In addition, it could be empirically verified which types of hospital were capable of implementing an advanced AEPD. Various studies from international HIS adoption research indicate that structural conditions such as size and ownership determine hospital digitization [e.g. 5]. More recent studies suggest that organizational conditions such as the degree of professionalization of information management (PIM), a pronounced intrapreneurship culture and participatory cooperation between IM, executive board and users can determine the initialization, implementation and institutionalization of HIS [6-8]. Against this background, the present study intends to answer two research questions: (1.) Are there differences in AEPD between different clinical workflows and data types and (2.) which structural and organizational hospital characteristics determine AEPD?

2. Method

A Germany-wide cross-sectional survey was carried out to answer the research questions. The following six clinical workflows were defined to reflect the care process as comprehensively as possible: admissions, ward rounds, pre-surgery, post-surgery to normal station and post-surgery to ICU and discharge. For each of the six processes, a list of patient data was compiled that is ideally provided electronically by the HIS including patient demographics, diagnosis and therapy and observation findings. Dedicated sets of information were assigned for specific processes: medical summary including medication and vital signs (admission, discharge), kardex including medication and vital parameters, warnings and checklists (ward rounds and pre- and post-operative processes)[2]. In addition, several modes of data flows were addressed: mobile availability of data for ward rounds, distinction between post-surgery data flow to ICU and normal ward, and finally data availability for writing electronic medical summaries. In order to check which structural hospital characteristics were associated with a high AEPD, ownership (private/public), size (number of beds) and teaching hospital status (yes/no) were assessed. In order to test the influence of organizational factors, the degree of professionalization of information management (PIM) was assessed by measuring the extent of planning, implementing and evaluating IM activities at the strategic, tactical and operational level with a 15 items scale that had been previously tested for reliability and validity [7]. In addition to PIM, the degree of the HIS related innovation culture (HIC) was gauged by a 25 items scale with high reliability and validity values [6, 8]. HIC consists of five sub-dimensions (CIO[3] and IT

[2] The selection of workflows and data types was based on expert surveys conducted during the questionnaire development. The selection should ensure the highest possible comparability between the hospitals.

[3] Chief Information Officer (CIO) is used as umbrella term to all leaders in charge of IT.

department intrapreneurship, innovation climate, innovation-friendly hospital management and degree of communication between IM and clinical users). Finally, the existence of a medical/nursing informatics officer and the position of the CIO (board member or not) was captured by the survey. The questionnaire was pretested for content and technical plausibility by five CIOs, 11 medical informatics scientists and one clinician. The link to the online questionnaire was sent to 1349 persons in charge of IT in German hospitals (responsible for 1950 hospitals) by e-mail at the beginning of December 2016. Based on the collected data, several scores were calculated to quantify AEPD. First, we calculated the perceived data-type specific AEPD score by the relative number of workflows in which a specific data type (e. g. patient demographics) was electronically available in all workflows consistently. Second, we calculated the perceived workflow-related AEPD by the ratio of the number of data available to the maximum of data possible. If for example the CIO reported that three out of seven patient data to be electronically available in the ward round, the specific workflow-related AEPD score was 42.9%. Both score types were added up across all workflows and data types to an overall AEPD composite score per hospital. In order to identify structural and organizational hospital characteristics that are associated with the AEPD composite score, a multiple regression analysis was calculated with structural and organizational characteristics as predictors. The model requirements were tested for normal distribution, homoscedasticity and multicollinearity.

3. Results

A total of 224 CIOs took part in the survey. Of these, 44 cases had to be ruled out for the analyses because the corresponding institutions did not have an operation room or did not pass the data quality plausibility test. The final sample thus corresponded to a response rate of 14.7% (125 out of 1349 e-mails could not be delivered). The participating hospitals had an average of 543 beds and were, for the most part, public or non-profit institutions (85.6%, n=180). Compared to the basic population, smaller and private hospitals were thus slightly underrepresented [10]. Table 1 shows the means and standard deviations of the data-type related AEPD scores and workflow related AEPD scores. In terms of data types, the results indicate that patient demographics and diagnoses and therapy data, as well as electronic observation reports (text and images) were continuously available to a large extent (means between 65.2% and 73.2%, Tab. 1) in the facilities surveyed. Electrophysiological results, warnings and checklists were continuously available in electronic format to a lower degree ranging from 32.7% to 51,5% (Tab. 1) with the lowest values for kardex. Table 1 furthermore shows that AEPD scores varied with the workflows examined. The highest AEPD score was measured for discharge (74.4%). The lowest (26.0%) was recorded for admission in which only a quarter of the data was made available electronically on a mobile basis. The overall average AEPD composite score was 54.8% ± 19.9. Multiple linear regression analysis resulted in a significant model that explained 34.4% of the variance of the AEPD composite score. Table 2 shows the beta coefficients as well as the p-values and the variance inflation factor (VIF) of the eight predictors. Hospital size was the only structural hospital characteristics that showed a significant influence on the level of the overall AEPD composite score ($p < 0.01$). Organizational factors, in particular the degree of professionalization of the IM (PIM), the development of the innovation culture and the existence of a nursing informatics officer showed a

significant positive influence on the AEPD composite score ($p < 0.05$). Residuals were normally distributed and showed no signs of heteroscedasticity, neither did the VIF indicate multicollinearity.

Table 1. Data type and workflow related AEPD scores (in %; n=180).

Data type related AEPD	x̄	SD	Workflow related AEPD	x̄	SD
Patient demographics	73.2	23.6	Admission	26.0	35.4
Diagnoses and therapies	72.2	21.8	Ward round (mobile availability)	49.0	41.7
Observation reports (text)	69.9	27.4	Pre-surgery	72.8	28.2
Observation reports (images)	65.2	26.6	Post-surgery (to normal station)	68.5	29.4
Electrophysiological findings	49.3	35.0	Post-surgery (to ICU)	64.1	33.9
Kardex	32.7	39.2	Discharge	74.4	21.0
Warnings	51,5	42.5			
Checklists	48.3	42.7			

Table 2. Beta-coefficients, p-values and VIF of the multiple linear regression model (n=160).

Predictors	Beta	p-value	VIF
Structural hospital characteristics			
Size (number of hospital beds)	0.220	0.009	1.713
Teaching hospital	0.108	0.146	1.346
Private ownership	-0.032	0,631	1.082
Organizational hospital characteristics			
Professionalism of information management (PIM)	0.186	0.044	2.053
HIS related innovation culture (HIC)	0.190	0.017	1.506
Nursing informatics officer	0.252	0.002	1.582
Medical informatics officer	-0.052	0.521	1.566
CIO member of the executive board	0.038	0.563	1.042

4. Discussion

This study investigated whether there are differences in the continuous availability of electronic patient data (AEPD) with respect to different clinical workflows and data types. Furthermore, it was examined which structural and organizational hospital characteristics determine AEPD. In four out of the six workflows studied, clinicians are receiving a higher percentage of electronic than of paper-based patient data. Potential for development could be identified for admission and ward rounds, where less than half of the patient data is made available electronically. With regard to data type-related AEPD, it can be seen that on the average patient demographics, diagnoses and therapy data as well as observation findings are continuously available. More complex information types, which in particular can facilitate clinical decision-making (i.e. kardex, warnings, checklists), are less available electronically. The study also identified structural and organizational determinants of AEPD. First of all, AEPD is more pronounced in larger hospitals. These results confirm the findings of previous studies [e.g. 5]. However, the results also provide evidence of controllable conditions for a high AEPD regardless of the size of the hospital. In especially, professional information management (PIM) seems to have a positive effect on AEPD. Thus, a continuous execution of IM activities that is aligned to the hospital strategy can foster the implementation of HIS innovations that ultimately are reflected in a high AEPD scores [7]. An advanced HIS related innovation culture was also found to have a

positive impact on AEPD. However, it can be assumed that the influence identified could be partly due to indirect effects, since a pronounced intrapreneurship at the level of the CIOs and the IT department can evidently support the professionalization of IM [7]. The same applies to a visionary and transformable hospital organization, an IT-related hospital board members and established communication channels between IM, clinical users and hospital board [6-8]. This study is limited with regard to the response rate of 14.7% that might have caused a non-response bias in our sample. The results therefore require further validation. Future research approaches could enrich the operationalization of AEPD by including further factors, in particular the degree of implementation of HIS functions, the depth of integration and the capability to distribute patient data electronically within and outside the hospital.

5. Conclusion

This study provides initial empirical results on the continuous availability of electronic patient data in German hospitals and thus offers a starting point for further insights into the maturity of clinical information logistics in the hospital sector.

6. Acknowledgements

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7. Conflict of Interest

The authors declare no competing interests.

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Learning Health Systems and Laboratory Medicine

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Abstract. Health systems worldwide struggle to provide the optimal treatment amid the rapid pace of scientific progress and frequent structural changes. Learning health systems (LHS) that continuously analyze their data to generate evidence and to enable informed decisions have been proposed as a possible solution. This paper investigates the involvement of laboratory medicine in LHS and evaluates relevant systems according to their scope, enabling technologies, architecture for evidence generation, data-action latency, and social aspects. While laboratory medicine provides a central source of information in many large-scale LHS, there are also systems with a particular focus on laboratories. For example, external quality assessments are performed by sending samples with known quantities to laboratories worldwide to verify their analytical methods. This data are also analyzed for other purposes, such as post-market surveillance of devices. These schemes have operated successfully over several decades. Newer, innovative tools harvest the possibilities of wearable devices and increased connectivity. LHS in laboratory medicine need to be recognized and incorporated into systems of systems to generate better evidence. Further research on the roles of other stakeholders will identify opportunities and obstacles for building and maintaining successful LHS.

Keywords. Learning health system, laboratory medicine, health data reuse, adaptive health systems

1. Introduction

Health systems worldwide struggle to provide optimal care. Increasingly, evidence generation in medicine falls behind the rapid pace of scientific progress and structural changes. Clinical decision-making is therefore often underinformed resulting in suboptimal outcomes. As a remedy, the Institute of Medicine has proposed the development of Learning Health Systems (LHS). These systems are defined as entities in which progress in science, informatics, and care culture align to generate new knowledge as an ongoing, natural by-product of the care experience, and seamlessly refine and deliver best practices for continuous improvement in health and health care [1]. LHS are often associated with large entities such as integrated managed care

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organization [1]. However, some LHS projects like the Connected Health Cities (CHC) program[2] in the North of England are smaller and have a limited regional scope. Other LHS focus on a certain medical sector such as ambulatory care from general practitioners [2].

In an LHS, all relevant stakeholders must trust the system, participate in a shared manner and obtain something valuable in return [3]. Because of its contributions to almost all medical diagnoses, laboratory medicine constitutes a central stakeholder. In this paper, we analyze the role of laboratory medicine, to better understand obstacles and opportunities for creating and maintain a successful LHS.

2. Methods

We searched leading journals of laboratory medicine for papers that describe initiatives resembling a LHS. Additionally, we reviewed publications about LHS in major medical informatics journals and examined the role of laboratory medicine.

The identified systems were analyzed according to five categories [4; 5]: *Scope* encompasses the spatial scope (e.g. regional, national, international), the scope regarding medical specializations (mono-, multidisciplinary), and the scope of stakeholders (patients, industry). The category *enabling technologies* describes innovations that made the success of an LHS possible. The *architecture for evidence generation* deals with the pipelines that connect data generators, information transformers, and knowledge consumers. These pipelines can be unidirectional or contain feedback loops [4]. The *data-action latency* is the time lag between data being available that an action should be taken and the action being taken [4]. *Social aspects* cover topics such as trust in the LHS and stability of the system [5].

3. Results

3.1. Laboratory Measurements are Important Data in Large LHS

Large integrated managed care systems such as Kaiser Permanente or the Veterans Administration have adopted the notion of a LHS to a large extent. Treatment data are routinely stored in comprehensive electronic health records and clinical data warehouses, analyzed by teams of data analysts and used as a basis for improvement [6]. In Kaiser Permanente, the Laboratory-based Acute Physiology Score (LAPS) integrates 14 laboratory tests into a single variable to predict inpatient mortality. This score has been used for quality improvement efforts and for internal case-control studies [7].

Integrated managed care systems connect many stakeholders to enable holistic patient care. However, the LAPS score and other applications of laboratory data often regard measured values as the true representation of the patient's status. Possible measurement errors are predominantly seen as a data integrity issue [8]. Important questions in laboratory medicine e.g. regarding the effect of measurement accuracy are rarely addressed. In a LHS organized within a single entity evidence pipelines are often bidirectional. Some large LHS share their evidence in the form of traditional publications.

[2] https://www.connectedhealthcities.org/

3.2. External Quality Assessments Schemes Act As a Learning Health System for Medical Laboratories

External quality assessments (EQAs) were introduced in laboratory medicine in the 1960s [9]. In short, EQA organizations send samples containing a known amount of the analyte in question to participating laboratories. These participants measure the analyte and return the result to the EQA organizer. Only if the results deviate less than a certain limit from the known true value the participation is considered successful. In some countries like Germany regular and successful participations are required for medical laboratories to operate. Data from EQA are often reused for other purposes than quality control of individual laboratories thereby forming a learning system of laboratories. Especially for emerging methods, a comparison of different laboratories and their analytical approach identified methods that are not fit for purpose [10]. For more established procedures data from EQAs are used for post-marketing surveillance of analytic devices and reagent lots. It allows stakeholders such as international organizations, governments, scientific societies, manufacturers to take corrective action when EQAs indicate suboptimal measurements [9].

The scope of EQAs regarding medical specialties is limited as they only address medical laboratories and closely related shareholders. Their spatial scope, however, is impressive. For example, Instand e.V. sends samples from Germany to participating laboratories in over 70 countries. EQAs, therefore, benefit from improved logistics. Unidirectional evidence pipelines predominate as EQA organizations send their findings to participants as part of regular, publicly available reports. Feedback is usually collected in an unstructured way only. Evidence generated from the analysis of data spanning several years is reported in form of research publications. Although EQAs usually do not use patient samples raw EQA data is typically not shared or published[3] [11]. EQAs have proved their success in decades of operations and enjoy a high level of trust.

3.3. Emerging Tools Use Increased Connectivity

In recent years a number of tools emerged that reuse aggregated results of patient measurements to detect analytical errors. The “Empower” project is an online platform that allows participating laboratories worldwide to send in the daily median of all patient results of several analytes. The platform analyses these medians and compares them across different analytic devices to detect wrong calibrations or faulty reagent lots [12]. In a similar project, the German Reference Institute for Bioanalytics has introduced the “Preanalytical Benchmark Database” to compare data on hemolytic samples across laboratories and to take corrective action[4].

Some manufacturers of analytical devices further created platforms that make use of wearable devices and increased connectivity. For example, some glucometers for continuous measurements regularly upload data to cloud platforms. These measurements are then presented to patients in a dedicated app or via a special website to enhance their understanding. Manufacturers can additionally reuse this data to detect errors and to improve their products.

The scope of these learning systems is limited to laboratories or even users of a single analytic device, but there are no spatial restrictions. These new platforms were

[3] https://doi.org/10.17605/OSF.IO/F3CD5
[4] http://rfb.bio/cgi/haemolysisLogin

enabled by improved networking technology, wearable devices, and ubiquitous connectivity. In line with the "network effect" the ability of these platforms to generate evidence increases in proportion to the number of users. Because the "Empower" project and "Preanalytical Benchmark Database" are operated by established scientific organizations, trust is high. For some continuous glucose monitors, continuous sharing of measurements and other data with the manufacturer is mandatory, although serious security and privacy risk exist [13].

4. Discussion

Many LHS continuously reuse their existing data from laboratory medicine to generate new evidence. Comprehensive data from large integrated managed care delivery systems could in turn help laboratory medicine to determine necessary analytical performance for optimal patient care [14]. Measurements should be standardized not only within individual organizations but worldwide [15].

Many smaller entities that reuse laboratory data are usually not recognized as LHS because they are narrowly focused on topics that arise in laboratories. Spatially, however, many of these networks cross national borders or even operate worldwide.

Especially EQAs have operated successfully over several decades. To benefit other medical specialties, raw data should be shared more freely. Because no patients are involved, anonymization of participants significantly reduces the potential for data misuse. A standardized encoding of analytes [16] and especially of devices might enhance reusability. The concept of EQAs might be extended to other adjacent areas. Accordingly, an EQA introduced a scheme to examine electronic laboratory reports [17].

Newer projects that make use of increased connectivity and wearable devices showed promising results. But when platforms process patient-level data they need to ensure that patients are able to make an informed decision about whether benefits outweigh the risks. Emerging technologies, in particular Differential Privacy, could be leveraged to enable manufacturers and the scientific community to learn from aggregated data with very little privacy risks [18]. Moreover, scientific communities should consider building similar infrastructures to remain able to evaluate newer devices such as continuous glucometers. Further continuous measurements and related analyses from the "quantified self" movement are likely to gain more importance. These new platforms will also connect experts from laboratory medicine with patients directly.

As noted before, LHS also have a social component [5]. The successful LHS in laboratory medicine usually enjoy a high degree of trust and follow undisputed goals. Initiatives that include specialist from other disciplines have been less successful [19]. Sociological factors might not be appreciated enough when building LHS [4].

5. Conclusion

LHS platforms in laboratory medicine need to be recognized and incorporated into larger systems of systems to generate the best evidence. Further research is required to examine if similar LHS exists in other medical specialties as well. To identify opportunities and obstacles for building and maintaining successful LHS, the potential roles and interests of the individual stakeholders need to be understood also on the fine-grained level of individual departments and functional units of healthcare organizations.

6. Conflict of Interest

None.

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Conducting a Multilingual Study Researching Traumatised Refugees Utilizing a Patient-Reported Outcome System

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Abstract. Approximately 300,000 asylum-seeking children arrived in Europe in 2015. The chance of experiencing a traumatic event is very high for fleeing children. Since the origin of the refugees is widespread, the languages spoken are diverse. Multilingual electronic patient-reported outcome systems (ePROs) can be used to gather medical data in a foreign language and display the results in the health professionals' language, which helps overcoming the language barrier. Utilizing such a system, a two-phase study aiming to screen refugee minors for potential mental health issues has started. Potential eligible participants are examined using questionnaires with good psychometric properties and cross-cultural applicability. To date, 75 minors and 21 of their relatives participated in the study, being German and Arabic the most desired languages for the electronic survey. Developing a system that provides multilingual questionnaires entails several drawbacks like a cumbersome translation process and dealing with writing directions. The proposed translation process and the ePRO can be re-used in similar studies.

Keywords. Patient-reported outcome, multilingualism, mental disorder

1. Introduction

Due to war, persecution, loss of home and security the number of refugees worldwide reached 65.3 million in 2015. Approximately one half of the asylum seekers in 2015 were children and 100,000 children worldwide were fleeing alone or without their families [1].

Nosè et al. demonstrated that experiencing a traumatic event, such as torture and war exposure, was very likely for asylum seekers or refugees: more than one-third suffered from clinically relevant psychological distress and one-fourth were diagnosed with a psychiatric condition like posttraumatic stress disorder (PTSD) [2]. In general, the

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vulnerability of children for mental health disorders increases [3]. Unaccompanied refugee minors are even at higher risk for mental health problems [4].

Migration is a worldwide phenomenon and languages spoken by refugees are diverse. Thus, health professionals often have to treat patients who do not speak a language the health professional understands. In this case, translators have to be used for communication, but they are not always available or requested or not paid by health insurances [5]. Despite challenging circumstances, health professionals are encouraged to use screening tools in order to prevent overlooking mental health problems and ensure early treatment. To overcome the language barrier translated patient-reported outcomes (PROs) can be used, which are defined as "any report of the status of a patient's health condition that comes directly from the patient, without interpretation of the patient's response by a clinician or anyone else" by the Food and Drug Administration of the United States [6]. PROs can be collected using an electronic patient-reported outcome system (ePRO), which avoids typewriting of pen and paper questionnaire results and ensures correctness of the data. At the University of Muenster, an ePRO named "Mobile Patient Survey (MoPat)" was implemented in 2014 and is currently used for routine and research collection of PROs [7].

The aim of this work is to propose an approach on how to support multilingual studies using MoPat and to point out encountered difficulties during the translation process.

2. Method

2.1. Study Design

The study is divided into two segments: short screening and a comprehensive diagnostic process. Screening focusses on posttraumatic stress symptoms and assesses emotional and behavioral symptoms. The screening process includes instruments oriented to proxies and self-reports as shown in Table 1. Children over 11 years receive self-report questionnaires; in case of younger children, their accompanying person(s) fill out external assessment forms.

Table 1. List of all translated questionnaires (*external assessment) and their assignment to the different events (Short screening and comprehensive diagnostic process).

Questionnaire	Screening children	Screening relatives	Diagnostic children	Diagnostic relatives
General Questionnaire	X	X	X	X
Strengths & Difficulties Questionnaire (SDQ)	X	X*		
Stressful life events questionnaire (SLE)	X	X*		
UCLA post-traumatic stress disorder reaction index (UCLA PTBS RI)	X	X*		
Connor-Davidson Resilience Scale (CD-Risc)			X	X
Kidscreen 10			X	X*
WHO Quality of Life (WHOQOL-BREF)				X
Life Events Checklist for DSM-5 (LEC-5)				X
Posttraumatic Checklist for DSM-5 (PCL-5)				X
Patient Health Questionnaire (PHQ-SADS) (PHQ-15, GAD-7, Panic, PHQ-9)				X

If the screening reveals potential mental health issues, children and their accompanying person(s) are invited to get an evaluation at the Department of Child and Adolescent Psychiatry, Psychotherapy and Psychosomatics at the University Medical Center Muenster (UMCM). Refugee minors who are in treatment at the UMCM fill out a more comprehensive set of questionnaires about several traumatic events, quality of life and resilience (see Table 1). Their accompanying person(s) also receive a survey to assess psychopathological symptoms of the child and fill out self-report questionnaires about mental health problems (see Table 1).

Ethics approval for the study was given by the ethics commission of the State Medical Association Westfalen-Lippe, Muenster, Germany (Ethics approval 2016-541-f-S: October 25, 2016). All patients and their legal representative gave informed consent.

2.2. Study Preparation

All study questionnaires were translated forwards and backwards from German into the respective national languages (Arabic, Dari, Farsi and Kurdish) by a certified translator. The initial and resulting version were then reconciled in the next step by consensus. Importing questionnaire meta-data in MoPat is possible using the CDISC Operational Data Model standard (ODM) [8]. Using ODMedit [9], student workers converted the textual questionnaires to machine-readable ODM files, which contained all translations for the questionnaires. After import and adaptation within MoPat, local translators checked every questionnaire to prevent transcription and conversion errors and reported any issue via email to a student worker who corrected the text. In addition, the graphical user interface of MoPat was translated to enable the users to interact with the system in their native language. An in-house research database called Exchange for Trials (x4T) is being used to store the exported survey data [10]. This system enables health professionals to view the responses in any language (e.g. German).

Figure 1. Bundle selection page within MoPat (A) and one example question in Arabic with activated help mode (B) where the participant can click the predefined information buttons to get additional information about the content and the usage of the current page.

3. Results

The translation of the questionnaires represented a cumbersome process. Since the modifications reported by the translators via email led to several inquiries, the process was adapted and the translators visited MoPat's developing lab to modify the texts themselves, supervised by the student worker, directly within MoPat.

The user interface of MoPat is shown in Figure 1. The upper part of the figure (A) displays the bundle and language selection. After selecting one of the bundles, here "Screening Children", one of the available languages, can be chosen to answer the questionnaires. The lower part of Figure 1 (B) shows exemplary a multiple-choice question in the chosen language with activated help mode. The participant can interrupt the survey at any time and resume it. After the participant has finished the survey, the questionnaires are exported to x4T and displayed in German for the health professionals.

To date 96 surveys were carried out using MoPat, divided into 59 screenings of children, 16 screenings of relatives, 16 diagnostic surveys for children and 5 for relatives. The country of origin of the refugees was mostly Afghanistan with 31 out of 75 (41%) and Syria with 17 (22%). The most used language for adults was Arabic with 13 out of 21 encounters (62%), followed by German with 4 (19%). In contrast, children speak mostly German (26 out of 75 - 35%) and less Arabic (20 out of 75 - 27%).

4. Discussion

This work has shown the possibility to survey study participants in foreign languages using an electronic patient-reported outcome system and provide the results in the common language for the medical professionals. Thus, the language barrier for assessing basic medical information and questionnaires-based instruments can be decreased.

To determine mental health problems it is important to use standardized questionnaires. Using non-validated questionnaires may result in pathologization of healthy individuals and/or overlooking of health problems and consequently preventing treatment. Therefore, only internationally established questionnaires for trauma and mental health with proven psychometric quality were selected for this study. Nevertheless, the questionnaires are only valid in German and should be validated for the recently translated languages Dari, Farsi, Sorani and Arabic.

The translation process was one of the most difficult and time-consuming parts for the preparation of the study. Direct communication with the translators should have been established from the beginning of the translation process. An even more efficient process could have been to invite the translation agency to include the translated texts directly in the data collection system. A basic knowledge of the languages or even the alphabet of the languages being translated is highly recommendable for the developing team, as it would help dealing with "simple" mistakes such as incongruence when copy-pasting text from the translated files.

Another drawback was the usage of languages using right-to-left writing like Arabic script. Firstly, the whole application has to use a character encoding like UTF-8, which supports the desired scripts like Arabic or Chinese script. Secondly, displaying web page content with both right-to-left and left-to-right oriented text is very difficult. To display the content correctly, the website could be mirrored at the central shaft of the web page, but this is not applicable due to diverse content on the page. However, best usability could only be achieved if the user interface would be justified culture specific, because

the understanding and interpretation of user interfaces is depending on the cultural background of the specific user [11]. Since the text on buttons and answers is mostly single-lined, MoPat only gives the opportunity to display question texts in right-justified format to be more accurate in displaying questions in right-to-left script. Thirdly, copying the right-to-left script and pasting it into a left-to-right text editor interchanges several characters resulting in wrongly allocated text. The solution adopted was to modify the editor to right-to-left, copy the content and change the editor back to left-to-right.

5. Conclusion

This work has demonstrated the feasibility of conducting a multilingual study utilizing an ePRO. The process of translating and integrating multilingual questionnaires into an electronic system has been optimized and could be reused in similar settings. Once the execution of the study is finished, questionnaire translations will be validated and made openly available if the corresponding licensor agrees.

6. Conflict of Interest

The authors declare that there is no conflict of interest.

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8. Health Science Communication

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Verbal Probabilities: Linear or Logistic? – A Regression Analysis Approach

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Abstract. Verbal probability expressions are quite intuitive and used in a variety of clinical important issues like adverse drug causality appraisal. However, there is insufficient evidence of their numerical meaning and whether they have a linear or logistic relationship with them. We aimed at contributing to answering these questions by means of a comparative regression analysis based on a sample of N=683 participants between 10 and 82 years (mean age 20.33 ± 11.77; median: 18 years) who were asked to numerically rate a given set of sixteen verbal probability phrases on a visual analogue scale. With respect to the explained variance, we found an R^2 for the linear model of 0.574 while R^2 for the logistic model was only 0.392 indicating a superiority of linear model compared to the logistic model. Although we were able to show that ranked verbal phrases are more likely to behave in a linear that in a logistic way other regression options like the double logistic model should be taken into consideration for further research.

Keywords. Verbal probabilities, numerical values, regression analysis

1. Introduction

To provide a patient with the best possible help in a situation which may be utterly unique, unexchangeable and never to recur in the same way, the process of clinical judgement is a core component of every diagnostic process [1]. In particular, prognosis of disease progression and potential attainment of individual therapeutic goals are actual research topics in the field of personalized medicine [2]. They however are limited by the degree of belief and the subjective expectation of uncertainty, which is normally included verbally in statements of physicians like:

"The wound will probably heal within three days…"
or
"A total recovery from your cancer is unlikely…"

Already in 1971, Revenstorf mentioned the use of subjective probabilities statements for the efficiency rating of a therapy using the concept of Markov-Chains [3]. 10 years later, Palm suggested an evidence-framework based on subjective probabilities, which up to day has not been recognized for the use in medical outcome research, although it appeared to be an important step in terms that expert opinions were incorporated into empirical studies [4,5].

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In both cases this might be due to the fact, that there has yet not been much effort in medical research to evaluate verbal probability statements with respect to their numerical meaning. Despite the fact that verbal probability expressions are quite intuitive and used in a variety of clinical important issues like adverse drug causality appraisal, there is insufficient evidence of their numerical meaning. "If for instance the doctor tells the patient, that a cure is possible, she may mean a 5% chance, but it may be interpreted to mean a 70% chance, or vice versa." [6,7].

This problem of translating verbal probability phrases into numerical values however is well known in psychological research. Some of the main findings are:

(1) People prefer to receive numerical values, but many experts, including medical doctors prefer to formulate their probability judgments in words to avoid an unwarranted impression of exactness [6,8].

(2) People have low "variability awareness", they seem not to realize how different the same phrases can be interpreted [6,9].

(3) Probability words are vague, in the sense that the same phrase will be used by different people (and in different contexts) to characterize different probability values. Although most respondents agree that "almost certain" indicates a high probability, and "impossible" refers to probabilities around zero, many intermediate phrases (possible, doubtful, uncertain, not unlikely) can be used to describe very disparate probabilities (for a review, see [10]).

Thus, the question arises, if there are groups of verbal probability phrases which correspond to similar numerical values in particular in the region of zero and one. This as a consequence might lead to a nonlinear, logistic relationship between verbal and numerical probabilities. Another idea would suggest a linear increase when mapping verbal probabilities to numerical values indicating that verbal phrases have a distinct increasing meaning with respect to their corresponding numerical values. This article aims at contributing to answering these questions by means of a comparative regression analysis.

2. Methods

Our approach is based on an online survey of mainly students who were asked to numerically rate a given set of sixteen verbal probability phrases (i.e. "unlikely", "good hope", "certain") on a visual analogue scale (VAS) (Fig. 1) [11].

Something is '*impossible'*

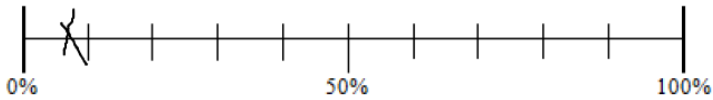

Figure 1: Example of a visual analogue scale

Apart from the numerical ratings from the VAS we also obtained data on gender and age of the participants. For each verbal probability descriptive statistics (min, max, mean, median, standard deviation and interquartile range were calculated). Based on the median values the verbal phrases were ranked. For the logistic model, the percentages p for each rank were transferred into logit values using the logit transformation $y = \log[p/(1 - p)]$. For values of p=0%, and p=100% were set to 0.01% and 99.99% respectively. For both the original and the logit transformed values a linear regression analysis between the ranked verbal probabilities and their numerical values was performed where the linear regression of the logit values corresponds to a logistic regression model of the original values. As a measure of explained variation with respect to the predictive power of the regression models R^2 was calculated. Sensitivity analysis was carried out for the whole sample and for the subsamples splitte by the median age and gender.

3. Results

A total of 683 individuals (n=280 male participants: 41.36%) between 10 and 82 years (mean age 20.33 ± 11.77; median: 18 Years) completed our survey. Table one summarizes the descriptive statistics for each verbal probability term.

Table 1. Verbal probabilities and their numerical correlates in % (IQR: Interquartile Range; Stdev: Standard deviation).

Verbal phrase	Minimum	Maximum	Median [IQR]	Mean ± Stdev
not possible	0	100	7.0 [0.0; 22,5]	15.1 ± 21.4
impossible	0	100	9.0 [0.0; 25.0]	15.7 ± 20.4
low probability	0	100	29.0 [15.0; 45.0]	32.5 ± 21.7
unlikely	0	100	30.0 [20.0; 40.0]	31.0 ± 18.6
uncertain	0	100	30.0 [20.0; 50.0]	35.2 ± 17.9
doubtful	0	100	40.0 [29.5; 50.0]	41.1 ± 19.5
maybe	0	100	50.0 [40.0; 58.0]	48.8 ± 17.9
some hope	0	100	55.0 [30.0; 70.0]	51.3 ± 24.8
possible	0	100	70.0 [50.0; 90.0]	68.9 ± 23.5
good hope	0	100	70.0 [60.0; 80.0]	71.0 ± 17.8
probable	10	100	72.0 [60.0; 85.0]	70.9 ± 17.9
good prospects	10	100	76,5 [70.0; 85.0]	75.8 ± 13.2
relatively certain	10	100	80.0 [70.0; 85.0]	75.8 ± 17.2
very good prospects	0	100	85.0 [75.0; 90.0]	81.5 ± 17.2
most likely	0	100	90.0 [80.0; 95.0]	84.7 ± 17.9
certain	0	100	98.0 [80.0;100.0]	86.6 ± 21.9

In particular the negative probability phrases “low probability”, “unlikely” and “uncertain” had similar orders of magnitude in their mean and even stronger in their median values. On the side of positive connotated verbal phrases a similar effect can be seen in the phrases “possible” “probable” and “good hope” with almost identical mean and median values. The results of the linear and logistic regression analysis are given in Figure 2. As can be seen, the relationship between the ranked verbal probabilities and their numerical values has a better correspondence to the linear approach.

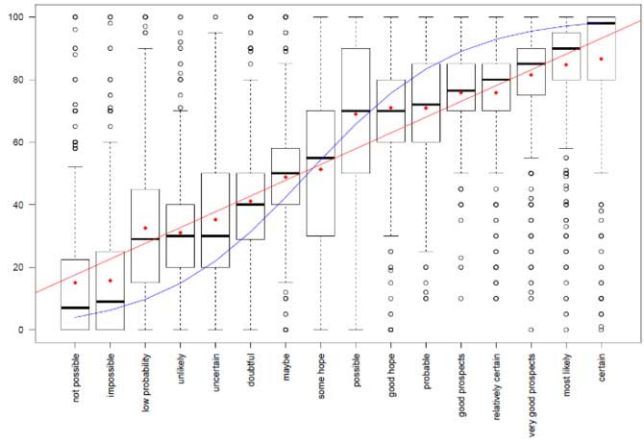

Figure 2. Linear and logistic regression results. Values for verbal probabilities (in %) are given both as boxplots with median (black line) and as mean values (red diamonds).

With respect to the explained variance, we found an R^2 for the linear model of 0.574 while R^2 for the logistic model was only 0.392 indicating a superiority of linear model compared to the logistic model. This also holds, when data is separated with respect to gender. In the first case, R^2 was given as 0.564 resp. 0.376 for the male part of the population and 0.579 resp. 0.404 for the females. Similar results are obtained when splitting the population at the median of 18 years: in the population of younger participants R^2 was given as 0.575 resp. 0.353 while in the older population we found an R^2 of 0.591 resp. 0.449.

4. Discussion

This article provides a short insight on the attribution of numerical values to verbal probability phrases. The values we obtained for the verbal phrases are in good accordance with previous results [12]. However we were surprised about the closeness of verbal probability phrases in regions of p=0.2 and p=0.7. If phrases like “uncertain” or “unlikely” and “possible” and “probable” can be regarded as synonyms this might lead to other results of the regression analysis. As this only explains a certain extend of our results, another explanation is given by the high number of outliers resulting from deliberately false completion of the questionnaire forms in some cases (see Fig.2).

A more person centered explanation is given in [13]. Here we found that persons suffering on a chronic disease tend to underestimate positive verbal phrases like “good prospects” and to overestimate negative phrases like “impossible”. This might be due to their experience that effects of i.e. a new therapeutic approach were not that “certain” as promised. Thus, sensitivity analyses of subsamples should be carried out to control for such confounding factors.

And although we were able to show that ranked verbal phrases are more likely to behave in a linear that in a logistic way other regression options should be taken into consideration. One promising candidate is given by the double logistic curve modeling approach introduced in [14]. This approach might be able to cover the plateau phases more adequate than a simple logistic approach.

5. Conclusion

According to [15] "clinicians should (…) address, as explicitly as possible, the challenge of estimating probabilities for individual patients. Investigations into this topic in particular with physicians and patients thus should be carried out to investigate the impact of verbal probabilities in patient practitioner communication.

6. Conflict of Interest

The authors state that they have no conflict of interests.

7. Acknowledgement

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9. Information Systems and Patient Safety

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German Medical Data Sciences: A Learning Healthcare System
U. Hübner et al. (Eds.)

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Data-Driven Assessment of Potentially Inappropriate Medication in the Elderly

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Abstract. Multimorbid patients taking polypharmacy represent a growing population at high risk for inappropriate prescribing. Various lists for identifying potentially inappropriate medication are spread across scientific journals and difficult to access. To address this ongoing need, a new database named PIMBase is developed which integrates these well-known lists and unifies their rating scales. The analysis of the pharmacovigilance data reveals the benefits of combining the lists. PIMBase is meant to be a web-based system and starting point for the data-driven assessment of polypharmacy to identify inappropriate medication and to improve the quality of prescribing. PIMBase is available at https://pimbase.kalis-amts.de.

Keywords. Drug-Related side effects and adverse reactions, potentially inappropriate medication list, polypharmacy, inappropriate prescribing, medication errors, information systems, decision support systems, computer-assisted decision making

1. Introduction

In aging populations, multimorbidity is increasing with a corresponding increase in polypharmacy, which in turn is the prime risk factor for inappropriate prescribing. The evidence is well-known by several studies that the use of certain groups of medications in elderly and vulnerable patients is associated with falls [1] and an increase in mortality [2]. Furthermore, inappropriate medications can impair cognitive properties [3], reduce the quality of life and cause additional costs for the healthcare system [4].

The major challenges in gerontopharmacology are both over-treatment and under-treatment associated with polypharmacy. In recent years, lists, criteria and classification systems for assessing "potentially inappropriate medication" (PIM) for geriatric patients were developed and published. Besides these PIM lists of medication with a negative risk-benefit balance (i.e. PRISCUS [5], AUSTRIAN PIM [6]), lists with a positive balance (i.e. FORTA [7], EU(7)-PIM [8]) are also becoming the focus of interest. However, those PIM lists are spread across scientific journals and difficult to access for patients or health professionals in the context of treatment. The integration of the various lists into a uniform database and subsequent merging as well as an implementation of a unique rating scale are essential for the qualitative improvement of the drug therapy in elderly and offer opportunities for practical application to identify and reduce inappropriate prescribing.

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2. Methods

The four mentioned PIM lists are published online in different file formats and structures. Thus, all source structures have to be analyzed and transformed into a common schema in order to build up a uniform database of PIMs. Especially the machine processing unfriendly PDF format poses a difficult issue. Additionally, both English and German languages need to be supported. Whereas the FORTA list is available in both languages, the PRISCUS list is provided in German only. The AUSTRIAN PIM and EU(7)-PIM lists are only available in English.

All lists provide the drug name, the expert consensus and selected comments or reasons for each PIM entry. In addition, possible therapy or drug alternatives and dose adjustments are listed in varying detail. Only the EU(7)-PIM list annotates each entry with the respective Anatomical Therapeutic Chemical (ATC) code. The FORTA list groups the entries by indications and the PRISCUS list provides possible contra-indications where applicable.

2.1. Development of the Database "PIMBase"

First, all sources are converted into machine-readable CSV files because all PIM lists can be represented as tables. A simple python script parses the docx file of EU(7)-PIM list and saves only the relevant table as a CSV file. The other lists as PDF files were transferred manually. The ATC classification system is used to annotate all drug names with their respective ATC codes to further unify the different PIM entries for all lists. Additionally, the indications of the FORTA list are annotated with the International Classification of Diseases version 10 (ICD-10) where applicable. The transfer and annotation were validated by several additional people in order to ensure the consistency of the information extracted.

Finally, a python script parses the four generated lists and outputs the uniform MySQL database. Common fields like the drug name and ratings are combined into uniform fields. Additional information such as source references in the EU(7)-PIM list or the drug class in the PRISCUS list is stored in a generic data structure of key-value-pairs where needed.

All PIM entries were rated by expert consensus panels of different research projects. A uniform rating scale was created and all PIM scores transformed into this scale in order to compare all scores efficiently. All lists except the FORTA scale use a 5 points Likert scale for rating. The FORTA list uses four categories for the PIM scores. The missing fifth category in comparison to the other lists is the undecided option. This option is introduced in order to map the FORTA list into the 5 points Likert scale. Since the order of score severities in the FORTA list is descending compared to the ascending severity in the other lists, a simple reversal of the scores results in a uniform rating scale (Table 1).

In future, new or updated PIM lists will be integrated semi-automatically, depending on the form of publication.

Table 1. The uniform 5 points Likert scale for rating potentially inappropriate medication in PIMBase.

1	2	3	4	5
Drug is certainly potentially inadequate for older patients	Drug is potentially inadequate for elderly patients	Undecided	Drug is not potentially inadequate for older patients	Drug is certainly not potentially inadequate for older patients

2.2. Implementation of the Web-based Information System

In order to provide the contents of the PIMBase database in an easy and informative manner to patients and health professionals, a web-based information system was implemented (https://pimbase.kalis-amts.de). The main component is the search tool making it possible to check specific drugs by their name or ATC code for occurrence in the PIM lists. The integrated information outlined above is shown in a user-friendly detail screen and cross-linked to third-party websites (e.g. DrugBank) with further information. Filter parameters allow the selection of all or a subset of PIM lists to be used and the results can be sorted by several criteria. Finally, the results can be printed in a complete overview with all detail information. This print, for example, empowers patients to discuss the results and treatment with their general practitioners. The website uses a REST-API to access the database. This standalone API provides endpoints and an access control management for future developments and interoperability with other systems.

3. Results

A total of 758 different PIM entries were integrated into the PIMBase database. To evaluate the advantage of integrating multiple lists instead of using just a single one, the overlap of ATC codes present in the lists is calculated (Figure 1).

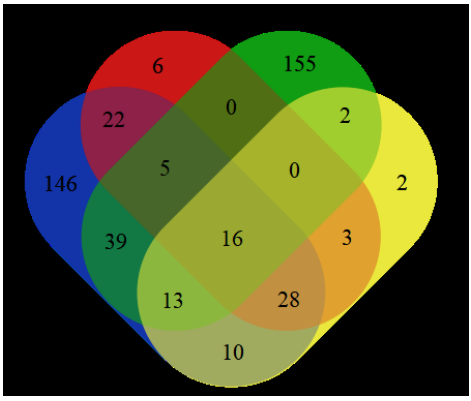

Figure 1. Overlap of different ATC codes covered by the PIM lists.

All lists only overlap in a small fraction of 16 entries. Pairs and triplets of lists show that an overlap is found in all combinations. However, most PIM entries can only be found in a single list, especially EU(7)-PIM and FORTA. This is of course biased by the size

and specificity of the lists. Therefore, using only a subset of the four integrated lists results in a loss of decision-critical information content. Because the lists differ in their focus on different criteria, the combination of information in the PIMBase system allows patients and general practitioners to evaluate different perspectives on the drugs in the context of treatment and improves decision-making.

Evaluation of PIMBase

The systematic analysis of the drug-related problems, i.e. adverse drug reactions (ADRs), is also an important aspect when evaluating the benefits of PIM lists. Therefore, cases of ADRs are analyzed to show the association of polypharmacy and ADRs and the benefits of using the lists integrated in PIMBase.

The data is obtained from the pharmacovigilance databases of the United States (FAERS) [10] and Canada (CVARD) [11]. The databases contain 996,404 cases of patients aged 65 and older with suspected ADRs in the target population.

The analysis of the sample population identifies a high number of cases with patients taking drugs referenced in PIMBase (Table 2). 47.1% of the patient cases in the population take at least one drug that is categorized as certainly or potentially inadequate for the elderly (category 1 or 2), 32.7 % take two or more inadequate drugs.

Table 2. Number of patients taking at least one drug referenced in PIMBase (N (% all cases in population) per category).

1	2	3	4	5	Total	PIMBase lists
Drug is certainly potentially inadequate for older patients	Drug is potentially inadequate for elderly patients	Undecided	Drug is not potentially inadequate for older patients	Drug is certainly not potentially inadequate for older patients		
232,963 (23.4%)	449,861 (45.1%)	20,496 (2.1%)	263,256 (26.4%)	312,345 (31.3%)	604,961 (60.7%)	**All**
41,253 (4.1%)	195,107 (19.6%)	0 (0%)	0 (0%)	0 (0%)	217,439 (21.8%)	**AUSTRIAN**
47,005 (4.7%)	345,035 (34.6%)	17,604 (1.8%)	0 (0%)	0 (0%)	360,418 (36.2%)	**EU(7)-PIM**
216,347 (21.7%)	269,256 (27%)	0 (0%)	263,256 (26.4%)	312,345 (31.3%)	548,129 (55.0 %)	**FORTA**
5,053 (0.5%)	125,152 (12.6%)	2,989 (0.3%)	0 (0%)	0 (0%)	130,941 (13.1%)	**PRISCUS**

The overlap between the ADRs of the PIM lists demonstrates once again the benefit of combining the different lists. The FORTA list seems to be the most comprehensive list in this case. The FORTA list gives a more detailed classification which also takes into account the indication of the drug prescribed. As the EU(7)-PIM list is developed on the PRISCUS list, the majority of identified cases overlap in both lists.

4. Discussion

Pharmacoepidemiologic research has shown that structured interventions like the use of PIM lists help to identify PIMs in elderly patients [9]. The implication for research and daily practice should be the evaluation of interventions like decision support systems that incorporate knowledge sources as PIMBase to improve the assessment of polypharmacy and avoid inappropriate drug usage in multimorbid patients.

5. Conclusion

Based on the standalone PIMBase-API, the database is planned to be integrated as a decision support module in the KALIS system [12] for patient-specific risk assessment of drugs. In future, further third-party integrations of PIMBase into healthcare systems are possible and desirable.

6. Availability and Requirements

PIMBase is available at https://pimbase.kalis-amts.de via a common web-browser.

7. Conflict of Interest

The authors disclose no conflict of interest.

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10. IT Strategy and Leadership

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German Medical Data Sciences: A Learning Healthcare System
U. Hübner et al. (Eds.)

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Requirements for Collaborative Decision Support Systems in Wound Care: No Information Continuity Without Management Continuity

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Abstract. Health IT systems are employed to support continuity of care via information continuity, while management continuity is often neglected. This study aims at investigating issues of management continuity when developing a collaborative decision support system for chronic wounds. Thirty-three experts from a variety of professions and disciplines discussed problems and possible solutions in four workshops. The following topics emerged from the discussion: existing networks involving payers, responsibilities as well as good discharge management. These topics clearly address management continuity and are also relevant for the scenario of inter-professional wound care across different settings.

Keywords. Management continuity, continuity of care, expert workshops, qualitative research

1. Introduction

Chronic Diseases, such as chronic wounds, are demanding for patients and healthcare providers. Due to the chronicity, all actors should be anxious to ensure seamless care. Continuity of care, the underlying concept of seamless care, is composed of information continuity, management continuity and relational continuity [1], whereby management continuity is defined as *"a consistent and coherent approach to the management of a health condition that is responsive to a patient's changing needs"* [1]. This includes discharge management, successful and rapid transfer between services, and the implementation of individualised care plans [2, 3]. When implementing health IT systems, information continuity is the one that is primarily addressed and focussed [4], while management continuity is often neglected. However, organisational and management issues play a paramount role for the success of health IT systems [5].

The aim of this study, therefore, was to investigate the challenges and conditions of management continuity, which healthcare providers must consider when defining

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scenarios for electronically connected care for patients with chronic wounds. This research question was embedded into a project that aims at connecting healthcare providers treating patients with chronic wounds and at supporting them making decisions via a shared health IT platform.

2. Methods

In order to identify the prerequisites of management continuity and to eventually design an IT product according to providers' needs, the user requirements had to be elicited [6]. To this end, four expert workshops were conducted. Three of them were workshops taking place at different locations in Northern Germany during the period June to August 2017. They were either hosted by the University of Applied Sciences Osnabrück or the University Medical Centre Göttingen. A convenience sample of health professionals, who were members of a wound care network in north-western Germany or who were members of a professional society in this field, was invited to participate. During the workshops, the experts were asked about process and action decisions, general frictional losses in the treatment of patients with chronic wounds, potentially problematic decisions of young or unexperienced providers or caring relatives, usage criteria for IT, and information items needed. They were given a process diagram, a case study or mock-ups to stimulate the discussions. Each workshop ranged from 150 to 180 minutes. The results were summarised by two authors (MP, SV) and afterwards discussed and evaluated during a forth, national workshop at the end of August 2017. A selected convenience sample of a wide range of experts including representatives from professionals' associations, members of supra-regional networks and the wound supplies industry was invited.

During the first three workshops, one researcher (SW) visualised the results, which were later documented by photo. Additionally, all workshops were audio recorded and minutes were taken. Audio-records were partly transcribed verbatim. The total results were analysed using inductive content analysis [7] by MP with a focus on challenges and conditions for management continuity.

In total, 25 representatives of health care providing professions participated in the three local workshops: six physicians (dermatology: 2, vascular surgery: 2, casualty surgery, orthopaedics and plastic surgery: 1, internal medicine and diabetology: 1; in-patient care: 3, ambulatory care: 3), four nurses (ambulatory care: 2, in-patient care: 2), eight wound managers (ambulatory care: 4, in-patient care: 4), four orthopaedic mechanics and three other healthcare professionals. In total, 11 participants were male und 14 female. Eight participants attended the national workshop: two representatives from national specialist associations (Initiative Chronic Wounds (ICW), German Society for Vascular Surgery and Vascular Medicine (DGG)), two representatives from supra-regional networks and four from wound supplies manufacturers.

3. Results

The content analysis of the audio taped material yielded four main topics associated with management continuity: 1) working in networks, 2) single point of contact and co-ordination for patients/relatives, 3) discharge management, and 4) payers as part of the network.

Working in networks. Building networks, which the family doctors are part of, emerged as one of the most important points. The experts also expressed, that working in a network shapes the attitude of providers.

"But we are going ways, in which different professions – from in-patient and out-patient care – come together and work in the same manner. You come closer, and that is network-like, and that is good. Today, we must rethink, and we have to bring together the different actors." (Nurse 4)

In order to work properly in a network, participants expressed, that the providers should know their specific part and the services of the other providers. They also suggested an overview of the different services to support their own work.

„In the network, I know exactly, what the colleague can do, what I must tell him, and to make my decision. I have a contact person of whom I know that he can provide the appropriate professional care to the patient. If I do not know which operation someone offers, which know-how he has, I cannot talk to him. Before you link any data, you need to have or find people, who are specialised, have resources, so you can contact them. In my opinion, this is very important for a network, that you know the people." (Physician 5)

Within a network, in which the providers have access to information that is technologically linked, participants stated that findings can be exchanged faster, and "external professionals" can be included. This can lead to less frictional losses.

"[...] If it [the network] is supported electronically, this would be very favourable, there are less frictional losses, as you can access findings more easily." (Physician 5)

"Within this black box, many providers can access the data. Then there is the question, how 'external professionals' can access the data, probably in a restricted way." (Orthopaedic mechanic 1)

Single point of coordination. The participants stated, that patients and relatives need one provider who is in charge and who has an overview about the whole treatment process in order to achieve a coherent treatment. By and large, the family doctor fulfils this role, according to the experts, but they also said that the provider may change during the course of the treatment.

„In my opinion, it is important, that the patient has a person who is in charge, a "director", who says: 'I take care now, I pull the strings.' This can be the family doctor – in most cases it is him – but it can also be another expert, who says, 'Okay, we have the following problem. I try to integrate the others.' Otherwise, you get lost in the system. Then nobody is responsible, and then the information does not come together. A "director" is someone, who takes a matter firmly in hand, or who says at a certain point, 'I handover to another "director"', and passes it on. However, a physician must pull the strings, has to take the responsibility properly." (Physician 3)

Discharge management. During a transfer – if a transfer document is sent –, participants told about wounds partly not documented, about wound-related information not included, or about patients discharged on Wednesday afternoon, Friday, or on weekend without any dressing material or any general practitioner available.

„I've never read in a discharge summary, what kind of wounds a patient has, what diagnostics was done, what the results are, what dressing material was used. That is not common." (Physician 6)

The participants mentioned the German HL7 e-Nursing Summary, a standardised transfer document, to be sent to a nursing home or an ambulatory nursing service.

„There is […] a transfer document from […] the ‚Netzwerk Versorgungskontinuität in der Region Osnabrück'. […] It is the [German HL7] e-Nursing Summary." (Nurse 4)

Additionally, several participants mentioned the soon-to-be legal obligation to implement a proper discharge management, which would be a solution to the problem.

„Beginning with 01.10.2017, there is a clearly defined discharge management. Each hospital that does not adopt it has problems. That starts on 01.10.[2017], and that belongs to the everyday life in a hospital: transfer, discharge management from hospital to the ambulatory setting." (Supra-regional network 1)

Payers as part of the network. The experts, working in a network, identified insurance companies as an important player.

„You are missing an actor, namely the payers. They are whose, who approve prescriptions, or they do not. This is a difficulty, which the public probably has not yet perceived in detail, but which is a huge problem for the patients. (Physician 2)

Referring to the approval process, other participants also supported the importance of the payers as part of the system. In contrast to the past, they perceived the process to last longer, but be more standardised. They complained, that patients suffer, as they must wait longer or the wound worsens. Partly, institutions can compensate this situation temporarily by providing own material, but this is not possible for each patient, they said.

"We always have soft positioning foam mattresses in the house. If someone is at risk, or has a small pressure ulcer …" (Nurse 3) – *"Nice house, but it does this at its own charge. We also do this temporarily. Until a prescription is approved and until it arrives at the patient's home, much time passes, and that will not do. And it is getting worse. The approval processes are getting longer. Many decisions are armchair decisions. It's getting standardised, and nobody is interested any more"* (Nurse 4).

4. Discussion

The results of this study concur with the challenges stated by Haggerty et al. [1] regarding management continuity. As many providers are seeing the patient, a network with coherent standards and clinical pathways can contribute to timely and complementary service delivery. Thus, patients' confusion, which may arise, when different healthcare providers recommend different treatments [8], may be reduced. Such a network may also prevent violation of management continuity, which may arise when patients move from one setting to another [9]. Discharge management was addressed by the German legislation (§ 39 SGB V) and further regulations (Rahmenvertrag Entlassmanagement). However, these regulations do not explicitly oblige institutions to transfer relevant data timely as structured electronic documents based on international standards, like the German HL7 e-Nursing Summary [4] or e-Wound Summary [10]. Another important factor within a network is a co-ordinator for the clinical pathway in order to adapt to changes in the patient's needs and circumstances. The co-ordinators role is perceived to be stronger in a highly integrated network than in an poorly integrated one [11]. One crucial actor in the timely delivery of e.g. aids are the payers approving the prescriptions. Therefore, it is important to integrate them into the networks in order to speed up communication between providers and payers, to shorten the time until an aid or a service reaches the patient.

Management continuity as defined by Haggerty et al [1] may also contribute to decision support, and vice versa. *"A consistent and coherent approach to management*

of a health condition" [1] goes hand in hand with coherent standards and clinical pathways, which should be adapted to guidelines, which also allow flexibility to respond *"to a patient's changing needs"* [1].

5. Conclusion

This study demonstrates the importance of management continuity as part of continuity of care. It also reveals the prerequisites of successful implementation of information continuity via health IT. These requirements embrace health care networks that include the payers and are established prior to the implementation of health IT, single point of co-ordination and finally well-defined discharge management procedures.

6. Conflict of Interest

The authors state that they have no conflict of interests.

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Tensions in Health Care Networks and the Role of IT

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Abstract. Interprofessional health care networks provide the opportunity to improve the care of chronically ill people, making it more effective and efficient. However, interprofessional relations as well as network activities must be mediated, and tensions have to be balanced. IT has the capacity to support the bridging of temporal and spatial distances within these networks. But the introduction of IT may cause unintended changes. Based on a literature study and observations in two networks for patients with chronic wounds, potential tensions and their shift caused by IT were systematically recorded, based on a categorization of Sydow. The findings of this qualitative study enable network managers to raise the awareness of potential tensions. In this way possible consequences of the introduction of IT can be counteracted at an early stage of its occurrence. Furthermore, the study revealed that the introduction of IT results in shifts within the state of tension, and amplifies conflicts in many cases. Ideally, IT may serve as a 'translator' with unifying implications, while fostering the diversity of a network at the same time.

Keywords. Interprofessional relations, medical informatics, organization and administration

1. Introduction

The sectoral separation together with the increasing specialization of several medical providers poses problems with respect to the growing number of chronically ill people. A loss of information at the interfaces between providers results in duplicate examinations, inefficient patient care and rising health care expenditures [1]. The trend towards the emergence of health care networks may provide a solution for counteracting these problems and improving patient care [2].

A network is defined as a group of more than three autonomous organizations or individuals who cooperate on a voluntary basis, and pursue not only individual but also common objectives [3]. In order to express that different professions and disciplines work together, the term 'interprofessional health networks' (IPHN) is used [4]. According to Luhmann IPHN may be classified as complex social systems, where stakeholders with diverging interests, cultures and values come together [5]. With this in mind, tensions cannot be avoided. Hence, network management must attempt to integrate and motivate all actors involved. Moreover, they are supposed to mediate network activities, and balance different interests and tensions within the network [6].

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For efficient networking, the supportive and driving role of information technology (IT), particularly for bridging temporal and spatial distances is emphasized [7]. IT includes any trans-institutional applications such as electronic health records (EHR). The introduction of IT into an existing network is a regulatory intervention that can cause unintended changes and that can create or increase tensions. Sydow points out eight tensions that have to be regulated in networking processes: *diversity-unity, flexibility-specificity, autonomy-dependency, trust-control, cooperation-competition, stability-fragility, formality-informality, economy-dominion* [6].

Relevant publications discuss possible tensions within IPHN and these tensions can be observed in practice. However, they are neither explicitly named nor systematically recorded. Nevertheless, an effective network management and the sustainability of a network require a precise analysis of these tensions. In particular, this is important if IT will be newly introduced to a network.

Accordingly, the aim of this study is to examine whether Sydow's categorization which provides analytical potential and is well-established in management literature, is also suitable for systematically analyzing tensions within IPHN, including shifts of tensions as a consequence of the introduction of IT. To this end the method of qualitative design is applied, and the following research questions are addressed:

- What tensions can be identified within IPHN and how can they be categorized?
- What is the role of IT in this context? How does the implementation of IT affect the state of tensions?

2. Methods

A literature research (EBSCOhost, ScienceDirect) helped to examine the transferability of Sydow's categorization to IPHN in order to implement a systematic and substantive capturing of possible tensions. Furthermore, a substantial reification of each tension including the role of IT was demonstrated. To this end, search terms such as consequences, effects, integrated care, and interprofessional care were combined while browsing only peer-reviewed studies after 2007. Ten articles out of 2899 were selected by two researchers. Articles that focused on patient outcomes, purely on financial benefits, usability and adoption of IT or on intraprofessional relations were excluded. In addition a snowball searching enquiry resulted in seven further articles. Three of the included articles were reviews while the other fourteen were case studies.

In order to confirm, reject or supplement the insights of the literature research, observations with a total duration of seventy hours (e.g. during network meetings, trainings and work shadowing) within two different IPHN for patients with chronic wounds, one of them with a common wound database, were recorded and analyzed. Initially data from the literature research were deductively analyzed based upon the identified categories of states of tension. Subsequently, the results were validated and supplemented with the insights from the observational verification.

3. Results

Both approaches the literature research and the observations in the two networks for patients with chronic wounds revealed evidence as well as individual statements for all

tensions[2]. Table 1 summarizes all relevant insights from the literature *(LIT)* for each tension, the implications of IT *(LIT IT)* and the monitoring results within the two networks *(OV)*.

Table 1. Tensions within IPHN and modifications due to IT.
LIT = literature analysis; *LIT IT* = literature analysis including the role of IT; *OV* = observational verification

Tension	Basis	Findings
Diversity-Unity	*LIT*	Cultural and ´linguistic` differences make it difficult to achieve unity. The harmonization of treatment processes according to uniform standards requires breaking with traditional practices and working methods [8-10].
	LIT IT	If IT can incorporate the diversity and priorities of the actors involved, then the emergence of unity is fostered. However, in reality, IT forces changes in work routines that are often not accepted in many cases [9,11-13].
	OV	In some cases actors use different terminologies and perceive IT as temporary cognitive stress that is unsuitable for their daily routines. IT is recognized as driving force for the creation of unity, but rarely is introduced in day to day work.
Flexibility-Specificity	*LIT*	Flexibility in daily practice is aggravated by specific tasks and activities as well as binding standards. With increasing specificity leaving the network can become more disadvantageous. [10,14-15].
	LIT IT	The application of IT increases the degree of formalization. Rigid structures and restrictions in action alternatives with regard to documentations and information exchange are circumvented by written notes [11,15-16].
	OV	Agreed upon standards are interpreted as recommendations. For nurses IT means regimentation, since it is time-consuming. The benefits of IT are not realized.
Autonomy-Dependency	*LIT*	Dependency occurs when actors collaborate with other medical providers within IPHN and as a result of standardization and shifting of power structures, physicians lose their autonomy while nurses gain autonomy [17–20].
	LIT IT	The treatment quality of an actor depends on data, proficiency and the available technology. Autonomy decreases due to higher degrees of formalization and standardization [13,17,21].
	OV	Physicians are not willing to give up autonomy for the benefit of IPHN. The issue of data quality is also a result of a lack of knowledge and frequent changes in the nursing staff. The selection of partners is often a point of discussion.
Trust-Control	*LIT*	Mistrust exists especially on an intraprofessional basis. It is difficult to create a balance between trust and controlling mechanisms [8,14,19,22].
	LIT IT	Physicians are afraid of transparency, surveillance and increased IT-based control. Trust in the competence of colleagues as well as the quality of data, hard and software are often missing. In contrast nurses tend to appreciate transparency [10,13-14,19,23].
	OV	Mistrust in information and the competency of others results in duplicate examinations. Ownership of data as well as reading and writing permissions are core topics in practice.
Cooperation-Competition	*LIT*	Competition exists despite collaboration within IPHN, especially intraprofessional. Furthermore, interprofessional there is rarely an exchange of experiences [11,18-19,21-22].
	LIT IT	Greater transparency may accelerate competition. IT has the potential to improve coordination and the quality of cooperation but cannot replace face-to-face communication [12,13,19,21,23].
	OV	In practice it is a challenge to find appropriate colleagues. Doubt exists on the goals and interests of potential business partners. Existing power relations and a lack of incentives inhibit effective cooperation.
Stability-Fragility	*LIT*	Increased actor diversity within IPHN results in longer negotiation processes and more fragility [9].
	LIT IT	The application of IT can stabilize a network, while fragility increases due to a variety of parallel systems [9].
	OV	A frequent change in actors introduces more fragility in an IPHN. Initially, regulatory interventions foster fragility.

[2] Only the tension *economic-dominance* (hierarchy) was changed to *hierarchy-democracy* to focus more on the form of leadership and the type of cooperation within the networks and not on the relationship between functional and hierarchical arrangements from the perspective of transaction-cost-theory.

Tension	Basis	Findings
Formality-Informality	*LIT*	Common standards and treatment pathways increase formality and may alter and limit working routines [17].
	LIT *IT*	IT increases the degree of formality. The staff perceives this as over structuring, extra work and information overload. Formal directives are bypassed with additional oral agreements and handwritten notes [13,15-16,21].
	OV	Many actors only want an informal exchange. A formal cooperation contract and directives increases commitment but can also be a deterrent.
Hierarchy-Democracy	*LIT*	Entrenched hierarchical and power structures pose interprofessional barriers. However, within an IPHN competence counts has highest priority. Based on this fact roles and boundaries can be redefined. Democratic structures facilitate a better understanding among actors as well as knowledge accumulation [8,15,19-20,24].
	LIT *IT*	Increased transparency facilitates the visibility of roles and competencies. Moreover, it enables orientation beyond hierarchical structures. A delegation of responsibilities provides nurses with more influence [15,17,23].
	OV	Discussions are more open during intraprofessional meetings. In an interprofessional context hierarchical structures are more obvious, but also observable between different disciplines of the same profession.

4. Discussion

Tensions within IPHN are not always explicit for both actors and network management. Nevertheless, they exist and are perceived and validated in different ways. Based upon a combination of a literature research and observations in two IPHN for patients with chronic wounds, possible tensions were systemized and explained in detail by using Sydow's categorization. A regulatory intervention through the introduction of a common IT system results in shifts in tensions and usually intensifies conflicts**.** But IT may also serve as a 'translator' with unifying implications despite cultural diversity and differences in terminology. However, this is only possible, if IT is integrated in daily routines in a flexible way. Moreover, the prerequisite are regular face-to-face communication and mutual trust in actors and their competencies.

That requires active management of the IPHN involving not only physicians but all health professionals. This can only be accomplished without falling back on hierarchical and power structures that have evolved over time, and without conveying a sense of control. Only in this way a joint knowledge base and standards sustained by all partners can be introduced. However, the observations revealed that these hierarchical and power structures still exist. Negatively perceived dependencies, mistrust in the competency of others as well as mistrust in hard- and software hinder a successful collaboration. Only a few actors are willing to adapt their work routines and to support joint standards.

5. Conclusion

The study revealed that Sydow`s categorization (with one exception) is appropriate for a systematic recording of tensions within IPHN, even if the allocation is not always highly selective. The results and systematic consideration of potential tensions within an IPHN and their transformation due to IT may be especially useful to sensitize network managers or initiators of IPHN and highlights the importance of active network control. These preliminary findings will be contrasted with literature searched via other sources particularly via MEDLINE.

6. Conflict of Interest

There are no competing interests.

7. Acknowledgment

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Determinants of Clinical Information Logistics: Tracing Socio-Organisational Factors and Country Differences from the Perspective of Clinical Directors

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Abstract. The establishment of successful clinical information logistics (CIL) within the care processes is one of the main objectives of strategic health IT management in hospitals. While technical realisations in terms of useful, usable and interoperable IT solutions are essential precursors of CIL, there is limited empirical research on what socio-organisational factors underlie an innovation-friendly culture and how they can affect successful information provision. We applied factor analysis on survey data from 403 clinical directors from Germany, Austria and Switzerland and used the dimensions identified to explain the level of CIL with ordered logistic regression analysis. The intensity of collaboration and exchange with the IT department as well as the degree of executive IT leadership showed to be strongly associated with better CIL while personal views and attitudes of clinical directors were not. Analysing country differences revealed the degree of the exchange with the IT department to be significantly lower in German hospitals. This points at a potential strategic lever for German hospital executives to focus on.

Keywords. Organisational culture, diffusion of innovation, clinical information logistics

1. Introduction

Due to their role as executive managers of the hospital's clinical staff, medical and nursing directors play a crucial role in strategically coordinating care processes – a responsibility that increasingly involves the integration of health information technology (HIT) to generate information at the point of care and to weave it into the care processes [1]. The goal of workflow support through health information technology (HIT) can be tied to the concept of clinical information logistics (CIL), i.e. the principle aiming to provide the right information for the right person, at the right time and in the right quality, making it an important precursor of safe and high quality care [2, 3]. While it is known that adoption rates and IT-quality can substantially differ across organisations [4] and countries [5], we are interested in what generalisable socio-organisational factors are

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associated with higher levels of CIL. This knowledge will help deduce meaningful levers for strategic IT-management.

Establishing successful CIL is a multifaceted challenge. Building on previous results that point at the importance of an organisation's "innovative power" in the digitalisation of hospital care [6], our goal in this study was to look beyond technical realisations and structural hospital characteristics and focus on underlying aspects of an innovation-friendly culture that can drive better CIL. Herein, we did not only look at peculiarities inherent to the organisation itself but also aimed to better understand the role medical and nursing directors play. This included particularly their personal views and attitudes towards IT as they might be able to establish more successful information logistics within clinical workflows.

This research is part of the international initiative "IT Report Healthcare" that aims at measuring the state of digitalisation in secondary care and related topics across German, Austrian and Swiss hospitals in the context of international developments. Thus, we also aimed at gaining insights about differences in CIL and its socio-organisational covariates in Germany compared to Austria and Switzerland.

2. Method

Data analysis was based on a survey that aimed to assess the IT-usage, IT-workflow-support, IT-quality and the socio-organisational environment with regard to innovative HIT from the perspective of nursing and medical directors. Most items were based on existing surveys whereas some scales were newly developed and pretested in two iterations by a total of 14 experts (including executive health professionals, health IT scientists, statisticians, management researchers and one psychologist). The goal was to yield a comprehensive overview on these issues in German, Austrian and Swiss hospitals and therefore targeted medical and nursing directors of 2,421 hospitals. With regard to the socio-organisational environment, we focused on the "innovation capabilities" [7] of the respective respondents and their organisation by issuing a set of statements on related topics and asked for their (dis-)agreement on a 5-point-Likert scale (Tab. 1).

In order to assess socio-organisational determinants of CIL[2] we used a combined approach by applying an ordered logistic regression analysis on factor scores. Factor analysis was applied in order to investigate and extract underlying dimensions of the scale set. We used polychoric correlation coefficients in the correlation matrix together with unweighted least squares (ULS) estimation since all items were measured on ordinal scales (Tab. 1) [8]. Applicability of the correlation matrix was evaluated based on the Kaiser-Meyer-Olkin (KMO) criterion and Bartlett's test of sphericity. Factors were extracted if their eigenvalue exceeded 1, if all components explained at least 50% of the total variance and based on assessing the scree plot.

The factor scores obtained from the final factor solution were then used to predict the level of CIL using ordered logistic regression and controlling for hospital size, ownership (profit vs. non-profit), health system affiliation and the existence of a surgery room. These demographics are known to potentially influence IT adoption levels [9, 10] and therefore might also be influencing successful CIL. Data was tested for

[2] CIL was measured on a single ordinal scale, referring to the entirety of the organisation. Respondents were asked to indicate their (dis-)agreement with the statement "Our IT-system is always able to provide the right information, at the right time, at the right place and in the right quality within the clinical workflows".

multicolliniarity and proportional odds of the independent variables. For comparing country differences of the extracted factors and the perceived level of CIL in German vs. non-German hospitals we additionally performed t-tests on the factor scores and a Mann-Whitney-U-test on the level of CIL.

3. Results

We received completed responses from 403 out of a total of 2,421 hospitals contacted (response rate: 16.6%). 81.4% of responses came from Germany (response rate: 16.8%; n_{GER} = 328), 9.1% from Austria (response rate: 14.2%; n_{AUT} = 37), and 9.5% from Switzerland (response rate: 18.0%; n_{CH} = 38). Data from Austria and Switzerland were pooled and contrasted with those from Germany to obtain more balanced group sizes.

Table 1. Factors and factor loading matrix. Loadings below .25 are left blank (n = 403).

Item	Factor			
	1	2	3	4
Factor "Executive IT leadership"				
"Our executive board actively promotes the initiation of innovative IT projects."	-.89			
"Our hospital shows great agility/flexibility when it comes to implementing new IT solutions."	-.63			
"Our executive board regularly perceives IT as a mere expense factor."	.55			
"Our executive board explicitly demands ideas and suggestions on how to innovate our IT."	-.51			
"I have often received positive feedback for putting forward innovative ideas."	-.37			
Factor "Exchange with IT department"				
"Our IT department gives users a better understanding of the benefits of IT."		.87		
"Our IT is capable to react quickly in face of changing requirements."		.86		
"The CIO incorporates suggestions and works with us to develop appropriate solutions."		.77		
"If there are any questions about IT, I have a personal contact person in the IT department."		.54		
"The IT department does not really seem to be present in our hospital."		-.53		
"We openly communicate/discuss new IT projects in our hospital among all involved staff."		.49		
"In our hospital, IT is considered early on in most medical or nursing innovations."		.42		
Factor "Proactive behaviour"				
"I regularly talk to people from outside our hospital about new IT solutions for our hospital."			.76	
"I regularly take time to think about IT-based optimisations of our hospital operations."			.75	
"I actively call for new IT solutions to improve clinical processes."			.64	
"I regularly seek to discuss strategic IT issues with our CIO"			.55	
Factor "Health IT attitude"[3]				
"IT solutions are often incompatible with the norms and values of nursing and medical care."				.61
"Experience has shown that new IT projects tend to make things more difficult than easier."				.59
"IT undermines medical and nursing autonomy in patient care."				.56

The data set showed to be suitable for factor analysis with a KMO measure of .92 and according to Bartlett's test of sphericity with $p < .001$. We attained a final set of 20 items reflecting 4 factors that explain 62% of the total variance (Tab. 1). Reliability measure ranged from 0.62 to 0.87 indicating solid internal consistency (Tab. 2). The second factor "exchange with IT department" proved to be significantly higher in Austria and Switzerland compared to Germany whereas all other factor scores did not differ across

[3] The factor is inverted. Stronger agreement on these three items technically implies a worse attitude towards HIT since the statements are negatively worded. The same kind of inversion applies to the first factor.

countries (Tab. 2). Furthermore, the Mann-Whitney-U-test result indicated a significantly higher (p=.005) level of successful CIL in the non-German countries ($\bar{x}$ = 3.30 ± 1.10) compared to German hospitals ($\bar{x}$ = 2.77 ± 1.09).

Table 2. Reliability and factor score differences between German and non-German hospitals (n = 403).

Factor	Cronbach's α	$\bar{x}$ German	$\bar{x}$ non-German	SD (German)	SD (non-German)	p-value
(1) Executive IT leadership	.79	.04	-.12	.96	.84	.25
(2) Exchange IT department	.87	-.06	.34	.94	.83	.00
(3) Proactive behavior	.76	.05	.03	.87	.98	.86
(4) Health IT attitude	.62	.01	-.10	.79	.81	.31

Prediction of the level of CIL through ordered logistic regression demonstrate significant influences of "executive IT leadership" and "exchange IT with department". "Proactive behaviour" and "health IT attitude" did not affect CIL significantly. The model controls for potential demographic covariates (hospital size, ownership (profit vs. non-profit), health system affiliation and the existence of a surgery room), none of which were significantly associated with CIL. Overall, it explains about 44% of the variance in CIL (Cox and Snell pseudo R^2). Assumptions about the absence of multicollinearity and proportional odds are met according to VIF measures way below 4 for all predictors and a non-significant result of the test of parallel lines.

Table 3. Ordered logistic regression model (chi^2 = 202.8, df = 8, p<.001) with ** p<.001 for coefficient.

Independent Variables	Ordered Logit Coefficient
Executive IT leadership	-.66**
Exchange with IT department	1.30**
Proactive behaviour	.05
Health IT attitude	-.03

4. Discussion

Four consistent factors could be identified from statements of medical and nursing hospital directors in Austria, Germany and Switzerland: the degree to which the hospital executives promote and lead change towards innovative HIT-solutions, the degree of exchange and collaboration with the IT department, the personal proactive behaviour in IT matters and the personal attitude towards HIT. These factors fit well with comparable structures found in previous works focussing on the CIO's point of view [7]. Support and encouragement from the executive board as well as a collaborative and user-centred approach in IT management have also been pointed at as essential predictors for innovative HIT to prosper [10, 11]. Our study carried on these findings adding the strong association of these two factors with better clinical information logistics (CIL). However, on the personal level of the respondents, behavior and attitudes do not seem to alter CIL.

In accordance with lower adoption rates [5], the perceived level of CIL showed to be lower in German hospitals compared to Austria and Switzerland. When extending the view on factor differences, it is interesting to note that the collaboration with IT departments appears to be closer and more active in the non-German hospitals while the other factors did not differ across countries. This points at a potential deficit in strategic IT management in German hospitals and should be taken up in further research. The studies main limitation stems from modest response rates across countries that might

have caused a non-response bias. Results should therefore be subject to further validation in future research.

5. Conclusion

While shifts in organisational cultural are inherently difficult [12], our study points at the significant influence of an innovation friendly culture in terms of support from the executive board and the exchange of the IT department with the clinicians on successful information logistics in clinical processes. Results should encourage particularly German hospital executives to strategically work towards greater collaboration.

6. Conflict of Interest

The authors declare no competing interests.

7. Acknowledgements

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Experts´ Insights into eHealth-Legislation: Comparing Switzerland and Germany

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Abstract. This prospective longitudinal study aims at better understanding eHealth success factors in different European nations, esp. the role of eHealth-legislation in Switzerland and Germany. Qualitative interviews with 39 matched experts from a large variety of institutions in both nations were conducted. The individual statements in the interviews and the overall satisfaction rating indicate a clear trend for a more optimistic attitude towards the law in Switzerland than in Germany. This result is not surprising given the history of a telematics infrastructure in Germany. Cross-country learning topics for German politicians are the inclusion of the in-patient sector and the focus on one major application. In a next step, interview results from Austria will be included and with that the scope of study findings enriched.

Keywords: Health care reform, eHealth, qualitative research, Switzerland, Germany

1. Introduction

Germany and Switzerland have adopted a new regulatory basis to foster nationwide eHealth in the last years. The Swiss EPDG ("*Elektronisches Patientendossier Gesetz*") is in force since 2017 [1], the German "*E-Health-Gesetz*" passed in December 2015 [2]. Although these initiatives pursue similar political goals (e.g. improving quality of care through better and faster information), the concepts for implementation differ (Table 1). For instance, while the Swiss approach concentrates on one application solely, the electronic patient dossier (*EPD*), and emphasizes the in-patient setting; the German law covers multiple elements, concentrating on the out-patient setting. Health policy plays a decisive role, the wider context contributes to the scale-up, spread and sustainability of new health-IT [3], such as do the involved players [4]. The design of policies in the field of eHealth (e.g. clear goals, with a useful framework, which leaves enough freedom) are considered important success factors for eHealth [5]. Contrasting different nations is a well-known method to reveal facilitators and barriers (see e.g. [5,6]). This prospective longitudinal study aims at better understanding success mechanisms in different European nations. Germany and Switzerland are two nations with different health care- and political systems and with a different history and approach to eHealth. Therefore, the research question at this stage of the investigations was: How is the current eHealth-legislation perceived by the variety of stakeholders in Germany and Switzerland?

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Table 1: Synopsis of the two eHealth-laws.

	***EPDG* [1]**	***E-Health Gesetz* [2]**
Applications	Electronic patient dossier (*EPD*) [in-patient care: implementation period of three resp. five years; voluntary participation of patients and out-patient health care provider (opt-in)]	Medication summary Telemedicine: e.g. x-ray councils Management of patient demographics Emergency data management Electronic patient record
Incentives	National, cantonal subsidies to create the necessary prerequisites for the *EPD*	Subsidy for sending/ receiving medical summaries electronically
Sanctions	Penalty in case of violation (e.g. unauthorized access to the *EPD*)	-1% of physician fee, in case the insurance data is not up to date
Interoperability	Legal obligation to certify, incl. which standards are to be used	Implementing an interoperability register

2. Methods

To address the research question a qualitative study design was chosen as it provides an in-depth understanding of the underlying mechanisms and values, and their complexity [7,8]. The first step was to identify relevant key players. A purposive sampling (using a snowball system) was applied targeting experts from different fields: health care provision (in-patient-, out-patient-care, nursing, telemedicine), industry (IT-provider, pharma industry), health care policy/ polity, and others (academia, data protection, patient organizations). For comparability between the Swiss and the German group a matching process was added. It was carried out with a focus on the professional background of each participant to obtain equal numbers of representatives in terms of specialty in both nations (Table 2). Potential experts were invited via E-Mail.

Table 2: Sample, matching, and data collection.

Sampling	**Switzerland**	**Germany**
Identified and contacted individuals	52	41
Response rate	36,5%	48,8%
Total number of participants	19	20
Matching		
Health care provision	10	10
Industry	4	4
Health care policy/ polity	3	3
Others	2	3
Data collection		
Interview period	12/2016-02/2017	06/2017-08/2017
Average interview duration	40min	38min

Both samples were nearly comparable regarding number of interviewees, background and interview duration. Initially, a preliminary interview guideline was developed: It was discussed with researchers of the field, tested with a selected group of experts of the field, pretested in a pilot, and modified accordingly. The resulting instrument was a structured interview guideline added by a few standardised questions covering: 1) the assessment of the national status quo, 2) country specific forces (potential barriers, facilitators), 3) the current, national eHealth-legislation, and 4) the prospective assessment of the development (expectations). Each expert was interviewed via telephone. All experts were asked to give their permission to record the conversation electronically. The audio material was transcribed. Afterwards, the experts were given the opportunity to review

the content. The collected material was analysed using *MAXQDA®*. A structured qualitative content analysis was carried out, starting with identifying categories openly and inductively, followed by a focused coding concentrating on the topic of eHealth-legislation [7,9]. Complementing the qualitative analysis, the standardized guideline questions were analysed descriptively.

3. Results

How and to what extent the eHealth-legislations were described by the experts indicates Figure 1. It shows the code-co-occurrence model resulting from the analysis for the Swiss and the German subsample, with the absolute number of coded interview segments as "facilitator", "barrier", "positive", "negative". The thickness of the lines between two codes marks how often these codes simultaneously appeared in the interviews. It demonstrates that the Swiss "*EPDG*" was classified as a "facilitator", "positive" more frequently compared to the German "*E-Health-Gesetz*". Though, there is a co-occurrence of "*E-Health-Gesetz*" and "facilitator" or "positive" in the German sample, the most frequent code overlap emerged for "*E-Health-Gesetz*" and "negative".

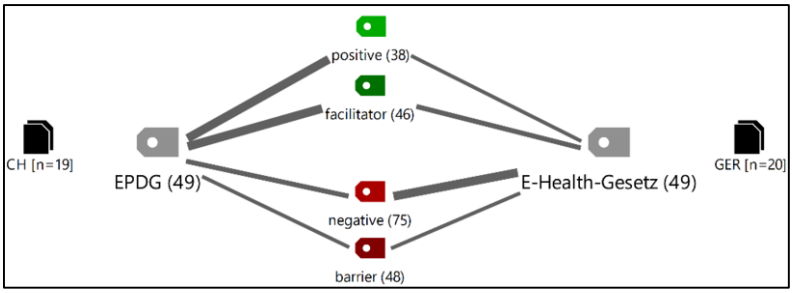

Figure 1: Code-co-occurrence model (*MAXQDA®*), assessment of the *EPDG* resp. *E-Health-Gesetz*. The thickness of the line between two codes indicates the frequency of the co-occurrence.

Many Swiss experts perceived the *EPDG* as a law that aimed at distinct objectives, e.g. efficiency or quality of care. Corresponding comments were made by 12 of the 19 Swiss experts. The perception of the *E-Health-Gesetz* varied and revealed an ambiguous perception among the German experts: 10 of the 20 experts named "political" goals, e.g. good publicity for the ministry of health. Often these perceptions were rather negative, e.g. politicking was implied. Distinct health care related objectives, e.g. quality of care, as named by the Swiss experts appeared only in 5 of the 20 German interviews. In addition, the experts were asked the standardized question: *On a scale from 1 to 10, how satisfied are you with the eHealth-legislation?* (1="not satisfied at all", 10="very satisfied"). The Swiss group showed a higher satisfaction with a median of CH=7 compared to GER=4.25 (Figure 2).

Despite these trends towards a more positive view of the Swiss experts, also multiple, highly differentiated perspectives emerged (Table 3). For instance, the Swiss *EPDG* was acknowledged for its possible harmonising effect (a); but it was also critized for its design focussing on the in-patient sector (b). Also, the German *E-Health-Gesetz* received partly positive appraisals recognising its attemps to promote transparency (c); yet, design flaws were mentioned due to the focus on the out-patient sector (d).

Table 3: Examples from the collected material (translated into English), all excerpts from different participants.

CH	**a)** *That is surely an important part or an important facilitator, looking at the EPDG; that something is developed, which can overcome the cantonal borders a little bit and bring about a certain degree of harmonisation.* **b)** *Regarding the legislative process, as you have seen with the Electronic Patient Dossier, how long it took us. And I think, one of the difficulties is, the way it had passed, there is only the mandatory participation for the hospitals, for the [out-patient] physicians and other health care professionals' participation, it remains voluntary. It protects the autonomy of individuals but inhibits a nationwide implementation.*
GER	**c)** *I believe, creating transparency (and the E-Health-Gesetz is a step forward with the interoperability register), creating transparency, it is surely something that should be demanded and promoted, because it offers all parties access to the market.* **d)** *We notice, the E-Health-Gesetz is concentrating solely on the out-patient sector, I don´t see a single application within the law, that has any connection with the in-patient sector. De facto, the in-patient sector does not take place in the law. And that with an innovation, which intends to break down borders offering opportunities to provide information beyond the borders of care. And we have zero incentives to do so.*

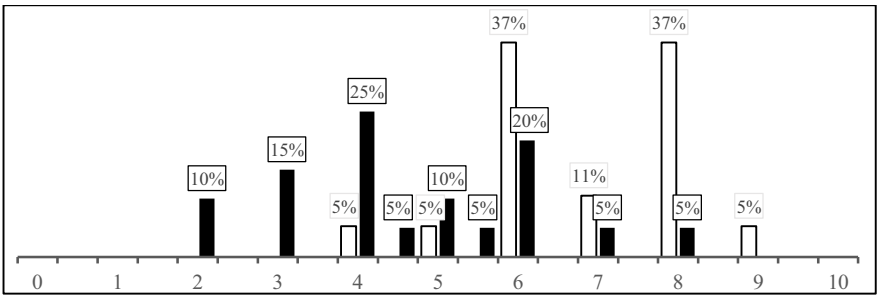

Figure 2: Satisfaction with the eHealth-legislation, 1="not satisfied at all", 10="very satisfied". Relative results, Swiss (n=19; white), German (n=20; black).

4. Discussion

This study compiled statements from two neighbouring nations on their view on eHealth-legislation. The individual statements in the interviews and the overall satisfaction rating indicate a clear trend for a more optimistic attitude towards the law in Switzerland than in Germany. The expert opinions and assessments in both samples are considering idiosyncrasies of their health care system as well as socio-cultural characteristics; therefore, the participants provided reflective, nuanced insights. In Germany, eHealth-legislation has a history dating back to the "Health Insurance Modernisation Act", which passed 2003 [10]. Yet, many expectations towards a health telematics infrastructure were not fulfilled [11]. This can explain the reserved attitude of the German interviewees towards the "new" attempt in terms of the *E-Health-Gesetz*. However, many other nations are facing similar challenges with eHealth and gained mixed experiences [6,12,13]. Therefore, it can be useful to listen to various voices. The combined analysis of the laws and the expert statements offered lessons and opportunities for consideration in German eHealth-legislation: 1) greater inclusion of the in-patient sector (as major IT-player), and 2) concentration on one key application, such as the *EPD* (focused, strategic approach). Due to the qualitative nature of this study, there are some limitations. However, this study explores a range of opinions of a diverse and relevant group of stakeholders. The matching process led to comparable groups, which is a major strength.

This part of the study is limited to Germany and Switzerland. However, interviews conducted in Austria are currently analysed to complete the picture in these neighbouring countries.

5. Conclusions

Although there are major differences between the Swiss and the German health care system, eHealth remains a highly complex topic in both nations. For health policy cross-country learning has a high potential by reflecting on the selected approaches and in pointing out available opportunities. While there is still need for further research, this study formalizes the approach and provides first contrasting findings.

6. Conflict of Interest

There are no competing interests.

7. Acknowledgements

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11. Machine Learning

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Modeling Activity Tracker Data Using Deep Boltzmann Machines

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Abstract. Commercial activity trackers are set to become an essential tool in health research, due to increasing availability in the general population. The corresponding vast amounts of mostly unlabeled data pose a challenge to statistical modeling approaches. To investigate the feasibility of deep learning approaches for unsupervised learning with such data, we examine weekly usage patterns of Fitbit activity trackers with deep Boltzmann machines (DBMs). This method is particularly suitable for modeling complex joint distributions via latent variables. We also chose this specific procedure because it is a generative approach, i.e., artificial samples can be generated to explore the learned structure. We describe how the data can be preprocessed to be compatible with binary DBMs. The results reveal two distinct usage patterns in which one group frequently uses trackers on Mondays and Tuesdays, whereas the other uses trackers during the entire week. This exemplary result shows that DBMs are feasible and can be useful for modeling activity tracker data.

Keywords. Deep Learning, Deep Boltzmann Machines, Activity Trackers

1. Introduction

Wearable devices, more specifically activity trackers, are attracting considerable interest due to their ability to measure physical activity continuously. The identification of patterns in the corresponding vast amounts of data generated in these settings poses a challenge to conventional linear modeling approaches.

Few researchers have addressed the problem of modeling the joint distribution of large quantities of activity tracker data using deep learning techniques. This paper seeks to address this issue by employing a deep Boltzmann machine (DBM) which has been shown to be a promising method in many applications like single-cell genomics, object recognition, and information retrieval [1–3]. Additionally, it provides a generative model, i.e., artificial samples can be generated from a trained model for exploring the structure learned by the approach.

Our analysis aims to develop methods that can detect structures in large unlabeled datasets and broaden current knowledge of weekly activity tracker usage patterns. Furthermore, the results could subsequently provide information on associations of latent activity patterns with health outcomes.

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Due to the complex structure of activity tracker data, the modeling task is often subdivided. Bai et al., for example, employed a two-stage model in which the presence of activity and the activity intensity given any activity at all are modeled separately using linear models [4]. In [5], the authors utilized random forests to model physical activity but were only able to train their model on data from a strictly controlled setting, which is not directly comparable to activity data from the use of activity trackers by the general population. In contrast to the approach considered in our study, [6] use supervised learning methods in a controlled surrounding, i.e., a gold standard is needed in the learning procedure.

2. Methods

The data for this study was obtained from *openhumans.org* where users of activity trackers can donate their activity records for the purpose of scientific studies. We had access to publicly shared Fitbit data from 29 individuals and extracted their daily step counts (median number of steps = 7181). Since our goal was to explore weekly usage patterns data was prepared such that each week represents one sample. Accordingly, step counts were available for 3577 weeks (median number of weeks per individual = 107). Hence, we define X_{ij} to be the recorded number of steps in a given week i at day j, $j = 1$ corresponding to Monday, and so on.

After deleting all weeks in which no data was recorded, we dichotomized the step counts for activity/inactivity of the tracker. This approach was adapted from [4] using an indicator function:

$$\mathbf{1}_A(x_{ij}) := \begin{cases} 1, & x_{ij} > 0 \\ 0, & \text{else} \end{cases} \tag{1}$$

The following subsections give a brief overview of deep Boltzmann machines.

2.1. Deep Boltzmann Machines (DBMs)

In order to identify the usage patterns mentioned above, we employ deep Boltzmann machines. This method has the potential to outperform previous approaches for wearable device data because it can learn a hierarchy of abstract features due to its deep architecture [7]. In addition, DBMs can process vast quantities of unlabeled data which is inevitable in settings where data is obtained from fitness trackers under real-life conditions.

In our approach, we consider a two-layer Boltzmann machine where we denote the visible layer as $\mathbf{v}$ and use $\mathbf{h}^1, \mathbf{h}^2$ for the first and second hidden layer, respectively. Furthermore, we restrict ourselves to a DBM with no within layer connections. The DBM enables modeling of the joint distribution of a large number of Bernoulli variables. In this context, these variables represent whether an individual made use of an activity tracker or not at a specific day. Broadly speaking, DBMs consist of stacked sets of visible and hidden nodes in which each layer captures complex, higher-order correlations between the activities of hidden features in the layer below [2].

Following the definitions and outline of [2] we define the energy of the state $\{\mathbf{v}, \mathbf{h}^1, \mathbf{h}^2\}$ as:

$$E(\mathbf{v}, \mathbf{h}^1, \mathbf{h}^2; \theta) = -\mathbf{v}^T \mathbf{W}^1 \mathbf{h}^1 - \mathbf{h}^{1T} \mathbf{W}^2 \mathbf{h}^2 \tag{2}$$

where $\theta = \{\mathbf{W}^1, \mathbf{W}^2\}$ are the model parameters, representing the symmetric interactions between layers. In energy-based models, low energy corresponds to high probabilities whereas high energy represents a low probability.

Next, we define the probability of the visible vector $\mathbf{v}$:

$$p(\mathbf{v}; \theta) = \frac{1}{Z(\theta)} \sum_{\mathbf{h}^1, \mathbf{h}^2} \exp\left(-E(\mathbf{v}, \mathbf{h}^1, \mathbf{h}^2; \theta)\right) \tag{3}$$

Where $Z(\theta)$ denotes the intractable normalizing constant, i.e., the partition function. Furthermore, the conditional distributions over the visible and the two sets of hidden units are given by logistic functions σ:

$$p\left(h_j^1 = 1 \middle| \mathbf{v}, \mathbf{h}^2\right) = \sigma\left(\sum_i W_{ij}^1 v_i + \sum_m W_{jm}^2 h_j^1\right) \tag{4}$$

$$p(h_m^2 = 1 | \mathbf{h}^1) = \sigma\left(\sum_j W_{im}^2 h_i^1\right) \tag{5}$$

$$p(v_i = 1 | \mathbf{h}^1) = \sigma\left(\sum_j W_{ij}^1 h_j\right) \tag{6}$$

2.2. Training Procedure

To carry out stochastic gradient ascent on the log-likelihood we make use of the following parameter update rule for the simplified case with one hidden layer $\mathbf{h}$:

$$\Delta \mathbf{W} = \nu\left(\mathbb{E}_{P_{data}}[\mathbf{v}\mathbf{h}^T] - \mathbb{E}_{P_{model}}[\mathbf{v}\mathbf{h}^T]\right) \tag{7}$$

Here, $\mathbb{E}_{P_{data}}[\cdot]$ is referred to as the data-dependent expectation while we denote $\mathbb{E}_{P_{model}}[\cdot]$ as the data-independent expectation. Also, the learning rate ν determines the influence each individual training sample has on the updates of $\mathbf{W}$. To train the model's expectations we used stochastic approximation procedures which are outlined in [2].
The data-dependent expectation was approximated using variational learning, where we can characterize the true posterior distribution by a fully factorized distribution [8]. Besides, the data-independent expectation was approximated using Gibbs sampling [8]. By stacking multiple restricted Boltzmann machines (RBMs), where only two layers are considered simultaneously, the resulting DBM can learn internal representations which enable us to identify complex statistical structures within the hidden layers [8]. To this end, we adopted the greedy layerwise pre-training which is detailed in [2]. In this framework [2] introduce modifications to the first and the last RBM of the stack so that the parameters $\theta = \{\mathbf{W}^1, \mathbf{W}^2\}$ are initialized to reasonable values. On this basis, the parameters can be improved during the approximate likelihood learning of the entire DBM [7]. In our analyses, we set the number of visible nodes $\mathbf{v}$ and the number of nodes in the first hidden layer $\mathbf{h}^1$ to seven in order to represent each weekday. We use one node in the terminal hidden layer $\mathbf{h}^2$ since we aimed to detect two groups of usage patterns. Hence, an active node in the terminal hidden layer represents one group while an inactive node represents another pattern. Furthermore, we set the learning rate ν to 0.007 during

pre-training and increased it to 0.008 for the training of the entire DBM. The number of epochs was held constant at 40 for pre-training as well as for training the entire DBM. Data analysis was performed using Julia Version 0.6.2 and the Julia package BoltzmannMachines.jl (*https://github.com/binderh/BoltzmannMachines.jl.git*).

3. Results

After having obtained estimates of the parameters θ, we used the DBM to generate new observations for the visible layer to explore the learned structure. Specifically, we used the DBM to compute the deterministic potential for the activation of the hidden nodes $\mathbf{h}^1$ given that the nodes in the terminal layer $\mathbf{h}^2$ were active/inactive. We then propagated the deterministic potential through the network to obtain the visible potential. Subsequently, we generated 10,000 uniformly distributed random numbers between 0 and 1 and assigned the value 1 if the visible potential was higher than the random number. Next, we used the generated data to plot the learned patterns in a heat map displayed in Figure 1. From the graph, we can distinguish two clear usage patterns. The upper pattern denoted as "on" shows that there is the tendency to use activity trackers at the beginning of the week on Monday and Tuesday and slightly increased usage on the weekend, while the lower pattern indicates a high usage throughout the whole week. A robustness check of the results was done by dropping each case individually and calculating the visible potential again, resulting in similar patterns.

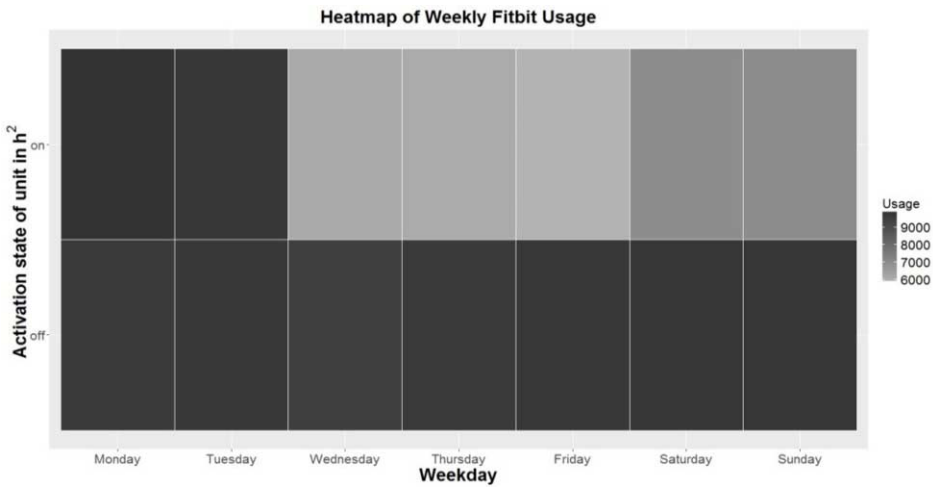

Figure 1. Heatmap of weekly activity tracker usage generated by the DBM. The upper pattern "on" shows a group that frequently uses activity trackers on Mondays and Tuesdays. The lower row indicates a group with regular usage throughout the whole week.

4. Discussion

One limitation is our sample size of 29 individuals. Being able to acquire more data and validate the model is an essential next step. Besides, integrating other measurements like heart rate and sleeping behavior could improve our model substantially. Based on

this, we could adapt the structure of our DBM to find even more usage patterns or model activity intensity. Another downside regarding our methodology is the dichotomization of step counts which gives away valuable information in the data. This can be addressed by using different cutoffs to incorporate the number of steps per day, but other, direct modeling approaches may be more useful.
Nevertheless, we believe our work could be the basis for future studies which take the continuous measurements from activity trackers into account. To further our research, we plan to use the partitioning approach presented in [7]. Doing this, we could potentially combine a binary RBM, to model the presence/absence of tracker usage, with a Gaussian RBM, for learning the activity intensity patterns and for modeling the temporal structure over a longer time period.

5. Conclusion

We have presented a deep learning approach, more precisely a deep Boltzmann machine, to model the complex joint distribution of activity tracker data and showed that it can be used successfully to extract meaningful activity tracker usage patterns. Most importantly, we were able to reveal two distinct weekly usage patterns in which one group mostly uses trackers on Mondays and Tuesdays, whereas the other uses trackers during the entire week. Understanding these usage patterns, which might e.g. correspond to individuals with planned activity vs. individuals with continuous self-optimization, could be useful for subsequent analyses.

6. Conflict of Interest

The authors state that they have no conflict of interests.

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Finding Needles in the Haystack: Identifying Patients with Rare Subtype of Multiple Myeloma Supported by a Data Warehouse and Information Extraction

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Abstract. Finding patient cases with extremely rare pathologies is a laborious task. To decrease time spent on manually searching through thousands of discharge letters and reports, a data warehouse with a fast fulltext search index was queried. Our use case is to find "macrofocal myeloma", i.e. Multiple Myeloma patients with few large lesions. We guessed the number of those patients in the University Hospital Würzburg at about 20. Most criteria were available in the data warehouse in an unstructured form requiring information extraction. 8 patient cases were found by searching for different spellings of "macrofocal myeloma" in discharge letters directly. With an indirect search combining several criteria, we found additional 23 candidate patient cases, from which 10 were classified by a domain expert as correct. The most difficult criteria were determining the degree of bone marrow infiltration. We achieved an F1 score of 93.2 % for this task. The number of patient cases to be screened manually for this disease decreased from about 25000 to 23.

Keywords. Macrofocal myeloma, information extraction, data warehouse, NLP

1. Introduction

Data warehouse systems such as i2b2 [1] or PaDaWaN [2] are often used in university hospitals for patient recruitment for medical studies. While it is straightforward to query structured data like age and sex of a patient, laboratory values, ICD10 and OPS codes, unstructured text data from discharge letters or examination reports benefit from a preprocessing step with information extraction (IE). Such unstructured parameters are often needed for studies that deal with rare diseases like macrofocal myeloma.

Multiple myeloma presents often with a patchy disease pattern. New unpublished data suggest that patients with few large lesions may have a different prognosis. We call this rare subtype macrofocal myeloma. There are some cases published [3]. The criteria

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to find patient cases with this subtype are: Bone marrow infiltration should be less than 10% and the number of bone lesions should be between 1 and 10. Both parameters must be extracted from various textual documents, e.g. in histopathological examination findings. From there, the values are usually transferred to the diagnosis section of discharge letters. The bone lesions or extramedullary lesions of a patient are visualized by different radiological procedures, which is why data can be found in the corresponding radiological findings.

In the context of macrofocal myeloma, this work focuses primarily on the extraction of the fraction of bone marrow infiltration from diagnostic sections of medical reports. The process of extraction and an associated evaluation are described. In addition, the problems of extraction of imaging data from radiological findings are considered. The overall data warehouse queries are described and the results discussed.

2. Methods

To identify patient cases that meet several criteria at the University Hospital Würzburg, the best way is to use Web-PaDaWaN [2], a web application that makes it possible to specify queries easily for the underlying data warehouse. The data warehouse is built on basis of MSSQL and indexed by a Solr[2] server for fast response times.

Since macrofocal myeloma is a very rare subtype, not all required data is contained in the data warehouse in a structured and retrievable form. For example, the percentage of bone marrow infiltration is a numeric value, but it is only available as text in unstructured data sources like discharge letters and several investigation reports. So before a query in the data warehouse can be answered, the missing data must be mined from unstructured text by IE techniques. So the whole process is this: Identify the relevant attributes. If these are not already present in the data warehouse, perform an information extraction and integrate the results in the data warehouse to make them searchable. Make a request in Web-PaDaWaN.

2.1. Information Extraction

To extract facts from unstructured text data, an approach similar to [4] was used, but instead of building terminologies for a whole medical domain, the focus was on the extraction of a single parameter. ATHEN [5] was used to build and fine-tune a minimalistic terminology for the extraction of the percentage of bone marrow infiltration (BMInf), containing only three attributes: “BMInf exact”, “BMInf <” and “BMInf >”. All three attributes describe the same value, but since it is numeric, a distinction between exact values and values that are referred to as “BMInf < 10%” or “BMInf > 5%” must be made to keep this information available in the data warehouse after the extraction process. A simple extraction of the found numbers without respect to this classes leads to wrong query results, because “BMInf < 10%” and “BMInf = 10%” can make a big difference from a medical point of view.

The used IE framework supports a special kind of regular expression (regex) as synonyms of attributes. By writing a regex in curly braces, the regex gets split at whitespaces and all possible permutations of the resulting parts get calculated. The resulting permutations are joined by wildcards (incl. leading and trailing ones) and added

[2] http://lucene.apache.org/solr/

as further synonyms. For example, the regex *{a b c}* results in the synonyms: *.*a.*b.*c.* ; .*a.*c.*b.* ; .*b.*a.*c.* ; ...* where a, b and c can be regular expressions themselves with the restriction that they must not contain whitespaces any longer. To detect all occurrences of bone marrow infiltration in the text, the regex *{(km|knochenmarks?|plasmazell.*) (befall|infiltration)}*[3] was used, because it covers all previously seen variations of this attribute.

Since the extraction algorithm works on segments as described in [5], we can use a segment that contains a match for BMInf as context in which to look for the corresponding numerical value. Sometimes there are authors of documents who describe a range including the real value rather than only writing the exact value. Therefore, we extract all found numbers and take the minimum. This is implemented by post processing candidates with a subset of the Python programming language. Our Python script not only detects numbers, but also verbal descriptions of the value, which get translated into corresponding numerical values for the BMInf classes "<"' and ">". Exact values do not have verbal descriptions. For example, the German phrase "geringe Knochenmarksinfiltration" is mapped to the attribute "BMInf <" with a value of 10. These mappings were defined by a medical domain expert.

Further, the presence or absence of bone lesions was extracted from radiology reports. In the first place the concrete number of such lesions should be extracted, but due to lack of exact documentation in the text documents[4], the task was restricted to extract the Boolean attribute "bone lesion present". Since we're looking for patient cases with at least one lesion, the Boolean decision is sufficient. To determine whether a segment contains a manifestation of lesion, an information extraction with a synonym list was done with respect to negated segments. A segment is negated if it contains a negation keyword like "keine" or "nicht" and has to be ignored in this case. All extracted facts were imported into the data warehouse and indexed by the Solr server.

Our framework is based on the architecture for unstructured information management UIMA [9], similar to cTAKES [6], the most frequently used system for clinical IE [7]. Therefore, it can be easily integrated in cTAKES and is transferable to many other clinical data warehouse systems. Another rule-based approach using UIMA dealing with breast cancer information is shown in [10]. Another popular IE architecture is GATE [8], on which i2b2's NLP cell is based on.

2.2. Data Warehouse Query with PaDaWaN

In order to make the best automatic preselection of patient cases, the following criteria were formulated as PaDaWaN requests, which are refered to as Query 1 and Query 2.

Query 1 (direct search):

- discharge letter contains match for regex *ma[ck]ro(f|ph)o[ck]al.**

Query 2 (additional indirect search; all three criteria must apply):

- ICD10 codes C90.x or D47.2 present (in structured data)
- "bone lesions present" (in radiology reports)
- "BMInf <" <= 10 or "BMInf exact" < 10 (in diagnosis section)

[3] The regular expression contains the German translation of bone marrow infiltration and synonyms, since this work describes the extraction from German documents.

[4] The documents contain verbal descriptions very often like „diffuses osteolytisches Bild". The exact calculation of the number of osteolytic lesions is therefore very hard, if possible at all.

3. Results

Query 1 returned 8 patient cases, and all of them were actually correct. For Query 2, the first criteria referred to structured data. The second and third criteria required information extraction and were evaluated separately.

The extraction of the percentage of BMInf was evaluated on a set of diagnosis sections containing C90 (Multiple Myeloma) from discharge letters. There were 4257 documents in total, from which 100 were randomly sampled each for training and evaluation. The training set was used to find errors in the regular expressions used and possibly to discover unknown spellings. A single run on the evaluation set was controlled manually to get the following results. Since all errors were eliminated that occurred while evaluating the training set, an F1 score of 100% was achieved on the training data, while for the evaluation set, an F1 score of 93.2% was achieved (see Table 1). Wrong values were never extracted (precision 100%), but some values were not found (recall 87.2%). Overall, the evaluation set contained 39 entities, of which 34 were found. In addition, there were 4 occurrences with a verbal description of the value, of which 3 were correctly translated into the number value.

Table 1. Results of the extraction of BMInf (TP = True Positive; FP = False Positive; FN = False Negative).

	TP	FP	FN	Precision	Recall	F1
Test set	26	0	0	100%	100%	100%
Evaluation set	34	0	5	100%	87.2%	93.2%

The evaluation of the second criterion in Query 2 "bone lesion present" and "bone lesion absent" in radiology reports showed an accuracy value of 95% (TP: 13, TN: 82, FP: 4, FN: 1, documents: 100, results manually controlled). Since TN (true negative) is relevant here, the accuracy measure was used instead of the F1 score.

Using the extracted attributes evaluated above, Query 2 delivered the following results: A list of 23 patient cases was generated. The list was controlled manually by a physician. Ten of the cases actually described a macrofocal myeloma. With Query 1 and Query 2 together 18 positive cases were found.

4. Discussion

Patients with multiple myeloma and few large lesions without significant bone marrow infiltration are rare. Patients with this subtype are hard to detect in thousands of patients from a retrospective perspective. Dimopoulos et al. [3] reported a favorable prognosis in patients with macrofocal myeloma as primary diagnosis compared to regular myeloma patients. Many different criteria are used in the diagnosis of multiple myeloma. The use of a data warehouse facilitates such queries.

Since the extraction of the percentage of BMInf contributes significantly to the overall result, the errors in the evaluation set were examined in more detail. The five incorrectly extracted values can be broken down as follows: Once a synonym was actually missing, once the Python post-processing selected the wrong number. Twice, the instance could not be unequivocally attributed to bone marrow infiltration in the text, as this assignment could only be deduced from the context. In addition, the literal value "diffuse KM-Infiltration" could not be mapped to a number. Except for the assignment

of the word "diffus" to a number, all errors can be corrected by further fine-tuning. Overall, the extraction of the required value works satisfactorily.

The errors that occurred in the classification of radiology documents are as follows: Negation detection failed in four false positives, from which two were due to an unknown negation expression and two due to wrong segmentation (the negation word was outside the segment). One document was classified as false negative due to a missing synonym.

While it is easily possible to correct the extraction knowledge to find these particular false positives, it is an open question how to generalize the query so the recall is near 100% with a similar F1 value (i.e. a slightly less precision) to find really all needles in the haystack.

5. Conclusion

This work shows that a data warehouse can not only be used for large studies, but can also meet very specific queries, if effort is put into the information extraction, provided the frameworks as in [4]. In the future, further studies with special parameters, which can only be determined by information extraction, will be carried out in order to further improve the existing frameworks and workflows.

6. Conflict of Interest

The authors state that they have no conflict of interest.

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De-Identification of German Medical Admission Notes

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Abstract. Medical texts are a vast resource for medical and computational research. In contrast to newswire or wikipedia texts medical texts need to be de-identified before making them accessible to a wider NLP research community. We created a prototype for German medical text de-identification and named entity recognition using a three-step approach. First, we used well known rule-based models based on regular expressions and gazetteers, second we used a spelling variant detector based on Levenshtein distance, exploiting the fact that the medical texts contain semi-structured headers including sensible personal data, and third we trained a named entity recognition model on out of domain data to add statistical capabilities to our prototype. Using a baseline based on regular expressions and gazetteers we could improve F2-score from 78% to 85% for de-identification. Our prototype is a first step for further research on German medical text de-identification and could show that using spelling variant detection and out of domain trained statistical models can improve de-identification performance significantly.

Keywords. De-identification, anonymization, medical admission notes, named entity recognition, personal health information

1. Introduction

The task of de-identification of German medical texts is a precondition for any further research on these resources [1]. The main obstacle for research on German medical texts is the lack of shared medical corpora [1]. If medical texts are available, privacy issues make serious research a hard task. Medical texts can only be analyzed without legal restrictions if tokens containing personal information, e.g. personal health information (PHI) [2], had been deleted by de-identification [2].

The objective of this project was to build a de-identification tool based on named entity recognition (NER) and spelling variant detection, regular expressions and

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[2] In this paper we use PHI specified in the Health Insurance Portability and Accountability Act. A set of eighteen named entities need to be de-identified to comply with this regulation. For further details see: https://en.wikipedia.org/wiki/Health_Insurance_Portability_and_Accountability_Act.

gazetteers. Our medical data consisted of medical admission notes as binary MS-DOC format. The data included a time period of 2004-2016. The whole un-annotated data set consisted of 180,000 notes including 132 million tokens in total. The notes contained texts from the cardiology domain and have the following basic structure: i) Header containing addressee, sender, patients name and address; ii) Salutation; iii) Diagnosis, anamnesis, etc. (as free texts, tables, images); iv) Summary.

Previous de-identification approaches mostly targeted English medical texts and were based on rules, gazetteers and/or machine learning models [3,5,6,7,8,9,12]. Most recently Yuwono *et al.* used spelling variant detection for de-identification of unstructured medical texts, based on structured PHI entities in a database [4]. Stubbs *et al.* 2015 extensively reviewed existing de-identification systems [10].

Due to the total lack of shared German medical data the use of supervised learning NLP tools is restricted as they need training data to work properly [1]. There are several commercial research activities on German medical texts, but currently no publications are available.[3]

2. Methods

2.1. Definitions

To analyze medical data the consent of the patient is the legal basis generally preferred by law. Due to the vast amount of data, this is typically not possible to fulfill [2]. If there is no individual consent, medical texts can only be analyzed without legal restrictions if the personal data is fully removed [2].

The task of removing personal data is described with several different terminologies.[4] In this task we use the term de-identification, which is the most general term. De-identification describes the process of removing the association between a set of identifying data and the data subject.[5]

2.2. Three-Step Approach

To benefit from well-established de-identification methods and to gain experience on statistical methods we are using a three-step approach for our task.

Spelling Variant Detection: The header mostly contains the name and address of the patient, the name and address of the recipients of the note and the contact information of the clinic. To parse the medical text for spelling variants we are using the approach of Yuwono *et al.*, who used the minimum edit distance measure to identify PHI tokens [4].

Named Entity Recognition: After investigating NER tools for German we identified the German Stanford NER as the best performing NER tool for the recognition of person (PER) and location (LOC) entities on out-of-domain data [11]. We trained the best model on a combination of three popular NER data sets.[6]

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[3] Commercial vendors: Averbis and Statice: https://averbis.com, https://www.statice.io/.

[4] De-Identification, anonymization and pseudonymization are often used in this research area. All these terms are not consistently defined in different privacy laws. In summary anonymized data can never be re-identified, while pseudonymized data can only be re-identified with the use of external information.

[5] https://en.wikipedia.org/wiki/De-identification (ISO/TS 25237:2008).

[6] For further info see: https://github.com/MaviccPRP/ger_ner_evals.

Regular Expressions and Gazetteers: For the rule-based approach we combined plain gazetteers for first names, surnames and German cities and towns with gazetteers containing German street names with regular expressions.

In addition to the NE classes used by the NER model with RegexNER we are trying to recognize the following named entity classes: SALUTE (Herr; Frau); EMAIL (ex@xyz.com); PHONE (0123/3456); URI (www.xyz.de); DATE (01.02.1999); PLZ (12345); TITLE (Dr., Prof.).

2.3. Implementation

We implemented the methods using the Stanford Core NLP pipeline in a Java program including the preprocessing and de-identification steps. After preprocessing, parsing the header and annotating the medical text with a rule-based NER and a statistical NER model the algorithm starts the de-identification step. It replaces all spelling variants of tokens found in the header with its NE (e.g. <PER>, <LOC, etc…) label or <O> if it is not recognized as a NE by the NER models. If the token is not a spelling variant but annotated as a NE it will be replaced by its NE label, too. If the token is neither a spelling variant nor labeled with a NE class, the algorithm checks if the token contains location suffixes, pre-defined in a list. If the token contains a suffix it will eventually be de-identified and replaced by <LOC>. If the token is neither a NE, a spelling variant nor contains a suffix it will be directly written to the output.[7]

3. Results

Our baseline is the purely rule-based approach with Stanford RegexNER. We had an annotated test set of 15 medical admission notes with in total 14,134 tokens and 680 annotated named entities. They have been annotated with 10 pre-defined named entity tags. Table 1 shows the results for the binary de-identification task (PHI recognized vs. PHI not recognized). Table 2 shows the results for the NER task, the baseline, and for the full featured model.

Table 1. Evaluation of the binary PHI recognition task. Comparing all three models.

Model	Precision in %	Recall in %	F2-Score in %
Baseline	80	78	78
Baseline + Spelling Variant	79	85	83
Full Featured	71	89	85

Table 2. Evaluation of the NER task (B: Baseline, F: Full-Featured).

NE	Precision B/F	Recall B/F	F2-Score B/F
DATE	98/98	93/93	94/94
LOC	**93/66**	**51/56**	**56/57**
PER	**26/33**	**48/87**	**41/66**
PHONE	100/100	58/58	73/73
PLZ	88/88	100/100	97/97
SALUTE	100/100	98/98	98/98
TITLE	100/100	95/95	97/97

[7] To keep semantic value, we replace the PHI token using their NE class. The replacement mechanism can be adapted to specific needs, E.g. dates can be replaced my its year part, to track courses of time.

4. Discussion

Evaluating our de-identification task, ignoring the different named entity classes, table 1 shows that the spelling variant detector improved our baseline F2-score by 5%, while almost keeping precision. Our NER model decreases precision considerably, while improving recall to a lesser extent. There were 605 correct classified PHI tokens but still we got 251 non-PHI tokens de-identified and 75 PHI-tokens not de-identified.

Evaluation of the NER task showed a high F2-score for purely rule-based classes like PLZ, SALUTE, TITLE and DATE. The PER class improved F2-score by 25% whereas the LOC class made almost no improvement. A closer look at precision and recall in table 2 showed a huge improvement of recall for PER class by almost 40%, while precision could be improved by 6%. The LOC class lost 27% in precision, gaining 5% in recall. Due to ambiguities of entries in LOC and PER gazetteers (e.g. Schöneberg, Rostock, etc.) precision of LOC class was suffering.

To get a better understanding of the false negatives, we analyzed them more precisely. From 165 PER tokens there were ten false negatives, of which eight were one letter abbreviations of a first name and two not de-identificated surnames. From 104 LOC tokens, there were 35 false negatives, majority of which (29) belonged to the term 'Chest Pain Unit'.

The F2-score of the PHONE class was the lowest from the rule-based recognized entities. While precision was 100% recall was 58%. From 26 phone number tokens nine tokens were not recognized. All of them represented the last token of a phone number, not matched by the regular expression.

The precision of the PER class is comparably low, as we got several false positives. We got 165 PER entities in the gold standard but the full featured model classified 294 PER tokens. 68 false positives were 'Die', because this token was listed in the gazetteer 'German and international first names'. Further 32 false positives were 'Sehr' (classified as PER by the statistical model). The same goes for the tokens 'Pain' and 'Kollegin'.

To keep the focus of this paper on the technical aspects of de-identification, we used the PHI definition from the Health Insurance Portability and Accountability Act (HIPAA) which specifies more precisely which tokens need to be removed. While comparing "data concerning health" in European General Data Protection Regulation (GDPR) and PHI in HIPAA both terms are very similar.[8] However, PHI classes chosen in this paper do not necessarily comply with the GDPR.[9] Any de-identification task needs to be closely related to specific legal aspects and the intended use of the de-identified data.

5. Conclusion

Though there is a lack in German training data we could achieve satisfactory results for our de-identification and our NER task. While traditional rule-based approaches worked well for numerical and well-structured named entities like DATE, PLZ, SALUTE and TITLE, they achieve worse results for irregular entities like PHONE, LOC and PER. As

[8] For a more detailed comparison between GDPR and HIPAA see: https://iapp.org/news/a/gdpr-match-up-the-health-insurance-portability-and-accountability-act/.

[9] Our approach can be adapted to different legal conditions, as it is based on generic methods. In addition there are situations where full de-identification is not necessary [13].

there was no German corpus available for training a model on the PHONE class, we could use out-of-domain corpora to improve recall for the PER and LOC class. Still there are problems recognizing the LOC and ORG classes. The ORG class had to be excluded from our task, as medical texts contain several abbreviations and acronyms in capital letters, which confuse the out-of-domain trained model and led to very low F2-scores.

Besides the improvement of PHI and multiclass NER recognition scores the NER step added semantic information to our de-identified data, as additional PHI tokens could be replaced by semantically valuable NE labels.

Because of the lack of published research work on de-identification of German medical texts this project understands itself as a proof of concept. Further research needs to be done to improve recall while keeping precision high. As Uzuner *et al.* integrated a precision improvement step using support vector machines this could be done for our medical text set, too [7]. In addition Wellner *et al.* showed that already small annotated training sets can increase recall significantly [5]. Therefore we established an annotation project based on WebAnno[10] to encourage authorized medical researchers to annotate training and evaluation data. An encouraging research for the use of smaller training sets was done by Scheuerwegs who used 100 Dutch medical records for training and Liu *et al.* who trained a bidirectional LSTM model on a training set of 600 annotated mental health records. [8, 12]

6. Conflict of Interest

None.

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Prediction of Readmissions in the German DRG System Based on §21 Datasets

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Abstract. Hospital readmissions receive increasing interest, since they are burdensome for patients and costly for healthcare providers. For the calculation of reimbursement fees, in Germany there is the German-Diagnosis Related Groups (G-DRG) system. For every hospital stay, data are collected as a so-called "case", as the basis for the subsequent reimbursement calculations ("§21 dataset"). Merging rules lead to a loss of information in §21 datasets. We applied machine learning to §21 datasets and evaluated the influence of case merging for the resulting accuracy of readmission risk prediction. Data from 478,966 cases were analysed by applying a random forest. Many cases with readmissions within 30 days had been merged and thus their prediction required additional data. Using 10-fold cross validation, the prediction for readmissions within 31 – 60 days showed no notable difference in the area under the ROC curves comparing unedited §21 datasets with §21 datasets with restored original cases. The achieved AUC values of 0.69 lie in a similar range as the values of comparable state-of-the-art models. We conclude that dealing with merged cases, i.e. adding data, is required for 30-day-readmission prediction, whereas un-merging brings no improvement for the readmission prediction of period beyond 30 days.

Keywords. Patient readmission, decision support techniques, diagnosis-related groups, hospital information systems, machine learning

1. Introduction

Readmissions, shortly after hospital discharge, are a common problem in most healthcare systems. Beside unpleasant circumstances for the patients and their families, high numbers of unplanned readmissions also come along with high costs for the healthcare system [1]. In USA, UK, Canada and New South Wales (Australia), the rate of 30-day unplanned readmissions lies between 7% and 16%, whereof many could probably be avoided [2].

In a systematic review from 2016, Zhou et al. [1] give an overview of models for 30-day readmission prediction that were published in 2011 – 2015. The models had an area under the receiver operating characteristic curve (AUROC) from 0.55 to 0.88. Many of them were risk score models, which had been derived from the results of machine learning models and allowed an estimation of the readmission risk by hand [1].

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In a recent approach from 2018, Maali et al. [2] applied gradient boosted tree algorithms to data from New South Wales and created a risk score model, which had an AUROC of 0.71 for 30-day unplanned readmissions. They concluded, that their model was comparable to recent risk score models, which showed AUROCs from 0.68 to 0.75. Although Electronic Health Record (EHR) data were considered, most of the mentioned models relied on further data sources.

The German-Diagnosis Related Groups (G-DRG) system, used for structured reimbursement of comparable treatments, is supervised by the institute for reimbursement in hospitals (InEK) [3]. G-DRG codes are generated based on diagnoses (ICD-10-GM) and procedures (OPS codes) by a grouper software. Based on these data, the resulting reimbursement fees for each hospital stay are determined. By law, every year at the end of March, each German hospital has to deliver a dataset, which contains the calculated G-DRG reimbursement fees together with the underlying data, designated §21 datasets in the following. The contents of this dataset are standardized [3-5]. During G-DRG grouping, subsequent cases are merged in the following situations, leading to reduced reimbursement fees [6]:

- §2-1 FPV: If a patient is readmitted within the upper length-of-stay threshold within the same basis DRG (initial three characters of the DRG code).
- §2-2 FPV: If a patient is readmitted within 30 days within the same major diagnostic category (initial character of the DRG code), if the initial case fee was from the medical or other partition and the subsequent case fee is from the surgical partition.
- §2-3 FPV: If a patient is readmitted within the upper length-of-stay threshold because of hospital-caused complications.
- §3 FPV comprises additional rules for case merges with involved hospital transfers.

For merged cases in §21 datasets, the length of stay and the assignment of procedures, diagnoses, etc. to the separate cases gets lost. The aim of the present paper was to evaluate the influence of restoring merged original cases in §21 datasets for the accuracy of readmission predictions.

2. Methods

Our §21 datasets comprised 478,966 cases recorded in the time from 2013-2017 at five independent hospitals in Germany, including one university hospital. For each hospital, five files were provided (see Table 1). Data protection was assured through contracts between the participating hospitals and Agfa HealthCare GmbH on the one hand side, and on the other hand side by a corresponding contract between Agfa HealthCare GmbH and AIT, which received de-identified data in the framework of their collaboration in the funded HIS-PREMO project (see acknowledgment).

For the analysis of 31-60-day readmissions, §21 datasets and §21 datasets with restored original cases were used. To restore the merged original cases, we received additional information. However, the additional information did not contain all the fields of the §21 dataset. In table 1, the fields of the csv files were grouped according to their availability for the underlying original cases. Fields like the “year of birth” remained unaffected and could be copied for all separate original cases. Features, which varied for each separate original case, like the “date of admission” could not be restored and had to be extracted from the hospital information systems (HIS) as additional information. We

rejected some fields of the §21 dataset for our analyses, as no additional information was given to restore them.

We used our Predictive Analytics Toolset for Healthcare (PATH) for data pre-processing, feature engineering, modelling and evaluation. PATH is a Matlab (The MathWorks, Nattick, US) based toolset, which has already been applied to various predictive modelling scenarios (see [7, 8]). PATH allowed aggregating the data to one row for each admission by various grouping and linking functions.

Time-based features were calculated with the given dates (e.g. length-of-stay = date of discharge minus date of admission). Some features were extracted from particular digits of encoded fields. Hospital specific encodings of the additional files were converted. Codes, which comprised information that would be unavailable in a prospective situation were not considered. Charlson's comorbidity score [9] was calculated from ICD codes. ICD and OPS codes were disassembled to obtain the underlying classification hierarchy. Only inpatient stays were considered.

For the machine learning task, the "fitensemble" Matlab function was used to apply a bagged ensemble of 25 regression trees. [10]

Table 1. Overview of the fields of the various csv files, which were available for separate original cases. Keys, targets and constants are crossed out. For the separation of case merges "unaffected" features could be copied, "HIS required" features needed additional data from the HIS and for "rejected" features, no additional data from the HIS was available. ENTGELTE.csv files were available too, but not considered.

Name	Content	unaffected	HIS required	rejected (no HIS data)
FALL	cases	hospital (ID), ~~fee category~~, ~~pseudonymised case ID~~, insurance (ID), year of birth, month of birth (if age < 1 year), sex, ~~merged case (yes/no)~~, ~~reason of case merge~~, age in years at admission, ~~pseudonymised patient ID~~	date of admission, reason of admission, date of discharge, reason of discharge, age in days at admission (if age < 1 year), hours of mechanical ventilation	type of admission, admission weight (if age < 1 year), number of intermittent dialyses, starting date of preadmission treatment, number of preadmission treatment days, end date of post-discharge treatment, number of post-discharge treatment days, transfer hospital (ID)
ICD	diagnoses		type of diagnosis, ICD version, ICD code, localisation, secondary code, localisation for secondary code	
OPS	procedures		OPS version, OPS code, localisation, OPS date	
FAB	hospital departments		hospital department, hospital department admission date, hospital department discharge date	

3. Results

The analysis of the time to readmission for §21 datasets and §21 datasets with restored original cases showed a significant number of merged cases (13%) within the first 30 days (see Figure 1). For readmissions within 31 – 60 days, very few original cases were merged and thus, this time span was chosen as the basis for our further analysis.

For 31-60-day readmissions, we compared the prediction accuracy of a model based on §21 datasets and a model based on §21 datasets with restored original cases. Both models were based on "unaffected" and "HIS required" features only (see Table 1). For

both analyses, the AUROC was 0.69 (see Figure 2) and at the chosen cut-off, the sensitivity was 0.64 and the specificity was 0.64.

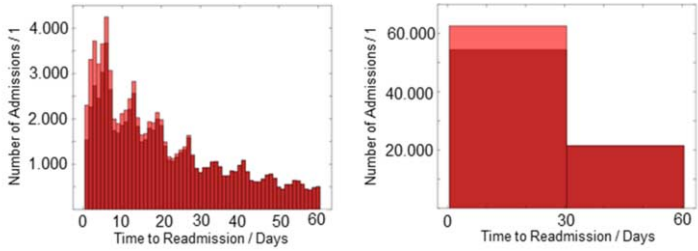


Figure 1. Comparison of days to readmissions after discharge for §21 datasets (dark red) and §21 datasets with restored original cases (light red). *Left*: bin size of 1 day, *Right*: bin size of 30 days, only very few case merges lie within 31 – 60 days to readmission.

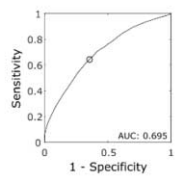
4. Discussion

Our predictive models have been derived from retrospective training datasets. In the future, they could be applied in a prospective way to unseen data to predict future outcomes. For the learning phase, case merging needed to be reversed to re-construct all separate hospital stays and, thus, obtain a comprehensive training set from §21 datasets. In clinical routine use, predictions would rely on information, collected into a data warehouse in near real-time from the hospital information system and other related source systems.

Merged cases did not contain all details of the underlying cases (admission and discharge dates; assignment of diagnoses, procedures, etc. to separate cases). We found that most of the case merges of §21 datasets were done within 30 days after discharge, although they can occur within up to 144 days. Within 31-60 days after discharge, only very few cases were merged and thus, this time window was chosen for the comparison of the prediction accuracy between models based on §21 datasets and based on §21 datasets with restored original cases.

a)

	Truth (-)	Truth (+)
Prediction (-)	221,196	6,818
Prediction (+)	122,905	12,291



b)

	Truth (-)	Truth (+)
Prediction (-)	224,968	6,870
Prediction (+)	126,531	12,251

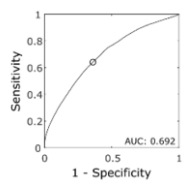


Figure 2. Results of the 31-60 day readmission prediction analyses: a) for §21 datasets, b) for §21 datasets with restored original cases, *left:* Confusion Matrix, (+) … readmission within 31-60 days, (-) … no readmission within 31-60 days, *right:* ROC curve with marked cut-off (circle). Both analyses resulted in a Sensitivity = 0.64, a Specificity = 0.64 and an AUROC = 0.69.

Our analysis showed, that for short-term hospital readmissions within 30 days, the restoration of original cases is required, whereas no improvement of the prediction accuracy could be observed for 31-60-day readmissions. Our resulting AUROC of 0.69 was comparable to recent state-of-the-art readmission prediction models.

Once our model has been validated, it will be a valuable tool, which could be used in real-time at the point of discharge to decide, whether a patient is ready to leave the hospital whether additional treatment/monitoring may be necessary or whether additional measures (rehab, etc.) are indicated.

5. Conclusion

Case merges in §21 datasets limit their utility for research in general. Our approach shows that this is the case in particular for 30-day-readmission predictions, whereas they do not significantly impact the readmission prediction within 31 – 60 days.

6. Conflict of Interest

To the authors' knowledge, no conflicts of interest were given.

7. Acknowledgement

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Understanding the Characteristics of Frequent Users of Emergency Departments: What Role Do Medical Conditions Play?

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Abstract. Frequent users of emergency departments (ED) pose a significant challenge to hospital emergency services. Despite a wealth of studies in this field, it is hardly understood, what medical conditions lead to frequent attendance. We examine (1) what ambulatory care sensitive conditions (ACSC) are linked to frequent use, (2) how frequent users can be clustered into subgroups with respect to their diagnoses, acuity and admittance, and (3) whether frequent use is related to higher acuity or admission rate. We identified several ACSC that highly increase the risk for heavy ED use, extracted four major diagnose subgroups and found no significant effect neither for acuity nor admission rate. Our study indicates that especially patients in need of (nursing) care form subgroups of frequent users, which implies that quality of care services might be crucial for tackling frequent use. Hospitals are advised to regularly analyze their ED data in the EHR to better align resources.

Keywords. Emergency hospital service, patients, ambulatory care, medical informatics

1. Introduction

Patient volume in hospital emergency departments (EDs) has increased steadily worldwide. Over the past decades the average annual growth rate in Germany was 4.9 % [1] with some German hospitals even reporting an increase of up to 9 % in 2015 [2]. This surge in demand is the major cause for ED overcrowding, which negatively impacts patient safety and mortality rates [3]. In light of this, considerable attention has been drawn to the group of patients who make use of ED services excessively.

Frequent users (FU) are conservatively defined as presenting at an ED for three or more times per year [4]. While FU account for 12 – 18 % of ED visits overall, their proportion to all ED patients amounts to merely 3.5 – 7.7 % [5]. The common belief, that frequent ED users present with light complaints and are unnecessarily clogging EDs, was not supported by most research. Although it was found, that FU are triaged more often with a lower urgency level than non-FU [6] and overestimate their need for emergency

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treatment [3], FU are often sick patients with chronic conditions and exhibit higher admission rates [4]. Yet FU do form a heterogeneous group [5]. Even though numerous studies have focused on FU related diagnoses, chronic conditions and socio-demographic features of FU, no studies so far have identified distinct subgroups of FU. An interesting finding, however, is that patients do not show patterns of frequent ED use over an extended period of time [4]. This suggests, that certain circumstances lead to frequent ED use that could be related to temporary issues of health or care. Especially, ambulatory care sensitive conditions (ACSC) [1] might pose a prominent risk factor, when ambulatory care fails to provide appropriate treatment or is out of reach. This study aims to further elucidate some of the constellations that elicit frequent use. We first examine how ACSC affect the risk of patients for exhibiting frequent use. Secondly, we determine distinct diagnoses subgroups of FU, regarding their morbidity, degree of acuity and admission rate. Finally, we readdress the issue, as to whether FU has an effect on complaint acuity or admission rate.

2. Methods

We performed a retrospective longitudinal study including ED patients of an urban German teaching hospital (35,000 cases per year). Data were extracted from the hospital data warehouse (MS SQL Server 2016). Subjects were included only if they were at least 18 years old, triaged upon arrival and received at least one diagnosis. The resulting 23,364 patients corresponded to 29,520 ED visits. Patients were defined as FU, when they had three or more visits within the study period (January 1 until December 31, 2017). Complaint acuity was measured by triage (Manchester Triage System [7]). With most ICD-10 diagnosis codes occurring only once or rarely in our data set, we used the Clinical Classifications Software (CSS) [8] to collapse these into 260 meaningful categories. This led to a roughly uniform distribution of conditions. We used the complete set of 40 ACSC from Sundmacher et al. [9], since they comprise relevant conditions in an ED context [1].

For simple FU vs. non-FU group comparisons regarding demographic variables we performed a t-test (age) and chi square test (gender). In order to test, whether FU are associated with higher acuity and admission rates we performed multiple linear regression analyses for triage time and admission as criterion respectively with FU as predictor and adjusting for gender and age. Relative risk of frequent use for ACSC was determined by estimating FU prevalence per ACSC compared to prevalence among patients without ACSC. This was done (1) regarding main diagnoses only and (2) for all diagnoses, since ACSC might accompany the emergency condition in a systematic way.

For discovery of FU subgroups, we used nonnegative matrix factorization (NMF) as a means for clustering patients according to their CCS features, triage and admission. This procedure is well suited for identifying latent features in very sparse data [10], as is the case for the patients' diagnoses. To determine an optimal number of clusters, we performed 50 randomly initialized factorization runs for ranks in range 2 to 15. For each factorization, we calculated cophenetic correlation coefficients and residual of sum squares, by which optimal rank can be selected [11]. Cluster robustness was verified by computing consensus matrices. The resulting factors where then labeled semantically according to their associated diagnoses (CCS features).

3. Results

Frequent users made up 5.5 % (n = 1,292) of the study sample, accounting for about 15.5 % (n = 4, 583) of visits. Mean age in years for FU (64.1, SD = 21.0) and non-FU (54.0, SD = 21.8) differed significantly ($p < .001$), whereas gender ratio did not (53.1 % vs 52.7 % male; $p = .81$). Analyzing the risk of becoming a frequent user when having at least one of the ambulatory care sensitive conditions showed that a main diagnosis of heart failure increased the risk for frequent ED use by at least 75 %. Pressure ulcers, diseases of the urinary system as well as gastritis and duodenitis increased this risk at least by 25 % (Fig. 1 top). When including secondary diagnoses, risk for FU to be associated with certain ACSC was two to three times as high for patients with alcoholic liver disease, and at least 75 % higher for heart failure and malnutrition, bronchitis and COPD. ACSC with the least estimated risk for FU in both diagnose sets were soft tissue disorders and dental diseases. In accordance with Brunet [11], we defined k = 4 as optimal number of clusters for identifying disease subgroups of FU. Consensus matrices also showed a good cluster robustness for this k value. The four clusters with their respective CCS diagnosis codes, top associated ICD-10 codes and ACSC are shown in Table 1. Finally, regarding the potential influence of FU on admission rate and acuity, the regression models showed no significant effect of FU (yes/no) as variable of interest with respect to admission rate ($p = .81$), and a negligible effect of FU on acuity as measured by triage time (+1.2 minutes; $p < .05$).

Table 1. Diagnose clusters for frequent ED users.

Cluster	Summary	CCS codes	TOP ICD10	TOP ACSC (main diagnoses)
I	Nutritional deficiencies, infections, hypertension, gastritis & duodenitis; triage: 30 min	259, 3, 55, 159, 98 155, 52	Z74, B96.2, E87, N39.0, I10	Diseases of urinary system, intestinal infectious diseases
II	Device or implant complication, injuries, wounds, fractures; triage: 30 – 120 min; no admission	237, 239, 235, 205	T83, S30.0, S01, M54	Mental and behavioral disorders due to use of alcohol or opioids, back pain
III	Immobility, dementia, incontinence; triage: unspecific	95, 163, 68, 254	R26, F03, R39, Z50	Diseases of urinary system, metabolic disorders
IV	Heart failure, chronic renal failure, cardiac dysrhythmias; triage: 10 min	108, 259, 158, 106	I50.1, Z92.1, N18, I48	Heart failure, ischemic heart diseases, bronchitis & COPD

4. Discussion

This study contributes to a better understanding of what kind of patients typically present repeatedly at the ED of an urban teaching hospital. Analysis of FU in conjunction with ACSC implies, that certain conditions, such as alcoholic liver disease and heart failure, pose a strong risk for frequent ED visits. The results are in line with several other studies that have reported association to selected diagnoses or generally ICD-codes [4]. Our findings clarify the relation to ACSC by estimating relative risk for FU. Yet it remains to be determined whether these higher risks stem from individual circumstances or insufficient ambulatory care, especially for these specific conditions. However, many of the higher risk ACSC apparently have complex factors in common. Our approach specifically addresses the question of what diagnoses in FU typically go together and how they are related to ACSC. As could be shown, the results from FU cluster analysis outline four of these diagnose compounds. While both clusters I and III obviously relate to care dependent patients (as indicated by ICD-10 codes Z74 and R26), they comprise

Figure 1. Relative risk of ED frequent use for main diagnoses (above) and all diagnoses (below), that qualify as ACSC, ordered by lower bound of confidence interval (95 %). Values express factors for risk change compared to non-FU (e.g. 1.5 means 50 % higher). Only conditions with at least 20 cases in overall data are shown.

two clearly different groups of conditions. Interestingly, these clusters also relate to conditions, which are typical patients with nursing needs [12]. Cluster II is related to lighter ED cases, due to lower triage and ambulatory treatment and includes less distinct conditions. Cluster IV comprises patients with chronic heart and renal conditions that also go together with diabetes mellitus and have already been identified as typical for FU [13]. Clusters, in which ICD-codes and ACSC diverge, point to underlying causal mechanisms with varying manifestation (e.g. alcoholism, nursing care needs).

The clusters presented here can serve as a starting point for characterizing FU subgroups in a more detailed fashion. Our results do not show any notable relation between FU and admission rates or acuity as compared to non-FU. Age however was highly significant and associated with both higher acuity and higher admission rates in our regression models. This might explain opposite findings that did not adjust for age (cf. [13]). This study has several limitations. Risk analysis for ACSC indicates that crucial factors are actually underlying causes. Alcoholism not being treated certainly leads to a number of conditions that end up in an ED sooner or later, no matter whether these qualify as ACSC or not. Ultimately, the extracted FU subgroups from factorization are too few in number to be isolated phenomena in the sense of patient phenotypes (cf. [14]). Therefore, subsequent clustering and phenotyping of FU groups needs to be pursued.

5. Conclusion

This is the first study to examine risk of frequent ED use for ACSC and to extract diagnose subgroups from frequent users. It contributes to a better understanding of common patterns and characteristics of patients, who excessively visit ED. Apparently, the most prominent FU subgroups are care dependent patients with typical ACSC, which

points to quality of ambulatory care as a crucial factor that might explain frequent use. The study further demonstrates that secondary use of EHR data is highly desirable for hospitals. We recommend hospitals to regularly analyze their ED patients based on EHR data in this manner to better align resources and competencies in emergency departments.

6. Conflict of Interest

The authors declare no conflict of interest.

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12. Networks and Systems Medicine

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German Medical Data Sciences: A Learning Healthcare System
U. Hübner et al. (Eds.)

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Visualization and Analysis of miRNAs Implicated in Amyotrophic Lateral Sclerosis Within Gene Regulatory Pathways

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Abstract. MicroRNAs (miRNAs), approximately 22 nucleotides long, post-transcriptionally active gene expression regulators, play active roles in modulating cellular processes. Gene regulation and miRNA regulation are intertwined and the main aim of this study is to facilitate the analysis of miRNAs within gene regulatory pathways. VANESA enables the reconstruction of biological pathways and supports visualization and simulation. To support integrative miRNA and gene pathway analyses, a custom database of experimentally proven miRNAs, integrating data from miRBase, TarBase and miRTarBase, was added to DAWIS-M.D., which is the main data source for VANESA. Analysis of human KEGG pathways within DAWIS-M.D. showed that 661 miRNAs (~1/3 recorded human miRNAs) lead to 65,474 interactions. *hsa-miR-335-5p* targets most genes in our system (2,544); while the most targeted gene (with 71 miRNAs) is *NUFIP2* (Nuclear Fragile X Mental Retardation Protein Interacting Protein 2). Amyotrophic Lateral Sclerosis (ALS), a complex neurodegenerative disease, was chosen as a proof of concept model. Using our system, it was possible to reduce the initially several hundred genes and miRNAs associated with ALS to eight genes, 19 miRNAs and 31 interactions. This highlights the effectiveness of the implemented system to distill important information from otherwise hard to access, highly convoluted and vast regulatory networks.

Keywords. microRNAs, amyotrophic lateral sclerosis, metabolic networks and pathways, gene expression regulation, KEGG, miRBase, miRTarBase

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1. Introduction

Amyotrophic Lateral Sclerosis (ALS) is the third most common neurodegenerative disease following Alzheimer's and Parkinson's disease. *ATXN2*, a major risk factor for ALS [1], has been shown to be important in miRNA function in flies. Furthermore, *TDP-43* and *FUS* take part in the processing of both coding and non-coding RNA and are suggested to influence miRNA biogenesis [2, 3]. Such findings suggest a significant role for miRNAs in the onset and progression of ALS and motivates further investigation.

Mature miRNAs (~22 nt) are derived from distinct hairpin structures (pre-miRNAs) present in primary RNA transcripts and are post-transcriptional regulators of gene expression. They function via binding to complementary sequences on their target messenger RNAs (mRNAs). MiRNAs can target many mRNAs with multiple target sites. Conversely, mRNAs can be targeted by several miRNAs leading to mRNA degradation or blocking of translation. Currently, miRBase (miRBase 21) is the primary database for miRNA sequences and annotations. TarBase and miRTarBase contain miRNA target information and interactions derived from experimental and computational techniques.

KEGG and similar databases (e.g.: Reactome) contain large bodies of information on gene regulatory pathways. The roles of miRNAs as important regulators of cellular processes, has led to an increasing number of studies utilizing biological pathway analysis. Most such studies lack miRNA visualization methods within large biological networks and focus on predicted miRNAs. DIANA miRPath v.3.0 [4] is an online tool which combines miRNAs with KEGG pathways but lacks the necessary tools for network editing and simulation. Another tool is miTALOS [5] which uses miRNA target prediction, KEGG and NCI PID data [6].

Network reconstruction is important for investigating the impact of miRNAs on biological networks. For example, miRGen [7], integrates various biological databases and includes miRNAs. Such networks should be analyzed in combination with other resources and VANESA [8] (https://agbi.techfak.uni-bielefeld.de/vanesa/), a systems biology software, provides a suitable platform for this, as it draws data from many databases within DAWIS-M.D. [9] and allows for construction, merging, visualization, and analysis of biological networks. It also provides free access to our custom miRNA database and was employed in this study to integrate multiple resources yet avoid huge interaction networks. The KEGG ALS pathway has 575 interactions including miRNAs (Figure 1A). Merging it with a miRNA enriched GeneMANIA [10] pathway (Figure 1B) resulted in 694 interactions (Figure 1D); but the intersection of the two networks, reduced the complexity to only 31 (Figure 1C). Thereby, we were able to show that this approach was able to help distill information from multiple larger regulatory networks.

2. Materials and Methods

In-house scripts were used to collect data from miRBase 21 (http://www.mirbase.org/ftp.shtml) containing both sequence and annotation information for the experimentally proven human miRNAs. TarBase V6.0 (http://diana.cslab.ece.ntua.gr/tarbase/tarbase_download.php) was used for obtaining information regarding the target sequences of experimentally proven human miRNAs. MiRTarBase 6.1 (http://mirtarbase.mbc.nctu.edu.tw/php/download.php) was downloaded and the data was modeled and imported into our database and DAWIS-M.D. MySQL Workbench 5.2 CE was used for creating our miRNA database.

3. Results

The addition of our combined miRNA-target database to DAWIS-M.D. enables the analysis and visualization of miRNAs within biological pathways. MiRNAs can be added to gene pathways if they are co-expressed with genes (e.g.: within an intron). MiRNAs that come from intergenic regions can also be added if they target a gene in the pathway. This extends the applications of VANESA and allows for a variety of different analyses on networks. Advantages of VANESA include techniques such as Petri net modeling and graph analysis as well as direct access to a variety of databases available within DAWIS-M.D. This allows drill down analysis of the function of miRNAs or families of miRNAs from a parent network to other related or targeted networks.

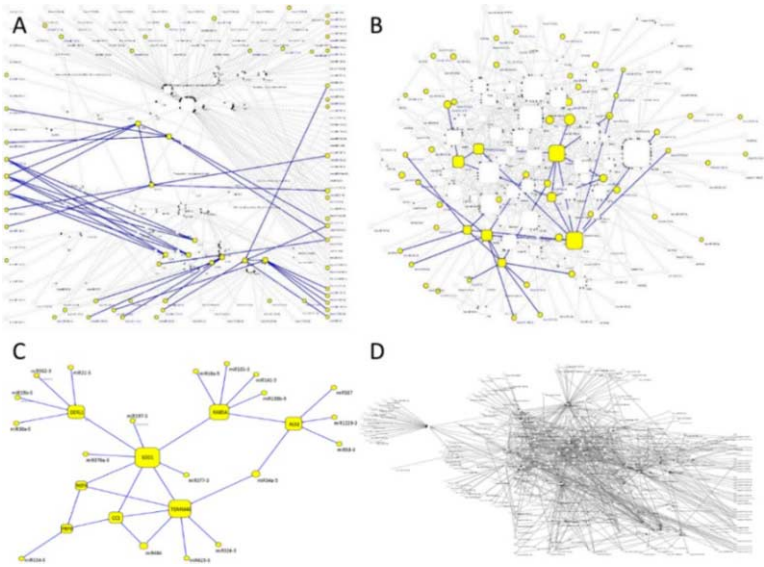

Figure 1. ALS KEGG pathway enriched with miRNAs (Panel A). Square nodes represent genes and circular nodes represent miRNAs. The colored edges in Panel B highlight the overlaps between the ALS network from KEGG and GeneMANIA (Panel B). Intersection of the two miRNA enriched ALS pathways from KEGG and GeneMANIA (Panel C). Merged KEGG and GeneMania ALS pathways enriched with miRNAs (Panel D).

2,588 miRNAs were extracted from miRBase along with 10,966 targets from TarBase and 34,777 targets from miRTarBase and integrated into our database, publicly accessible through VANESA. 3,022 genes and 346 miRNAs were found to be common in both TarBase and miRTarBase. In total, the 2,588 mature miRNA extracted from miRBase have 45,344 mRNA targets, targeting 12,347 distinct genes.

For human KEGG pathways, 99.15% showed miRNA mediated regulation. Within the 234 miRNA regulated KEGG pathways 65,473 miRNA gene interactions were found. Such high numbers show how versatile miRNAs are in regulating events within the cell. Some miRNAs such as *has-miR-335-5p* have higher potential activity by targeting 2,544 genes uniquely whereas the median for all miRNAs is 100 targets, but this may change due to pending miRNA target validations. This pattern is also seen at the gene level where some genes such as *Nfat5* (13) and *NUFIP2* (71) are targeted by multiple miRNAs.

When enriching a network built from the 30 known ALS implicated genes and *ATXN2*, the main risk factor for ALS, our system identified 119 miRNA interactions. Furthermore, two genes namely *ERBB4* and *CHMP2B* contained three and one miRNA

in their transcription unit, respectively. The most targeted ALS associated gene is *BLC2* (targeted by 47 miRNAs), while *DAO* is the only gene not having a known associated miRNA. *VCP*, *LMNB2*, *FUS*, *TARDBP*, *SOD1*, *LMNB1* and *TAF15* present the ALS associated genes within our network which are highly targeted by miRNAs (Figure 1B).

The ALS KEGG pathway was enriched with miRNAs using VANESA requiring that miRNAs are either co-expressed with any of the genes in the pathway or are directly targeting genes in the pathway. The enriched network consisted of 204 nodes, with 39 genes, 142 miRNAs and 242 interactions (Figure 1A). Regulatory networks tend to grow quickly with increasing number of genes involved and including miRNAs further adds to the already complex gene regulatory networks. The KEGG pathway for ALS is focused on the *SOD1* related pathogenesis mechanism. An alternative ALS-related GeneMANIA network was constructed, imported, and enriched with miRNAs in VANESA. All ALS associated genes along with *ATNX2*, as the main risk factor for ALS, were collected from literature and the resulting 31 genes were processed with GeneMANIA to obtain further interactions. The resulting network of 50 genes was downloaded and enriched with miRNAs under the same constraints as for KEGG resulting in a total of 260 nodes, consisting of genes and miRNAs. 575 interactions were found between 50 genes and 210 miRNAs (Figure 1B). The enriched KEGG and GeneMANIA networks were merged in VANESA resulting in a large complex interaction graph. This merging resulted in a total of 293 nodes (77 genes and 199 miRNAs) with 694 interactions (Figure 1D). Since such a complex network convolutes understanding and complicates experimental analysis, the intersection of the two networks was established in VANESA. The intersection graph contains only the shared nodes and edges, significantly reducing the complexity of the network to 27 nodes (8 genes and 19 miRNAs) with 31 interactions (Figure 1C). Out of the 8 genes, 4 (*ALS2*, *NEFH*, *PRPH*, and *SOD1;* collected from ALSoD) have previously been implicated in ALS. Among the 19 miRNAs, 2 (*hsa-miR-21-5p* [11] and *hsa-miR-124-3p* [12]; with 1351 targets) were previously shown to have a role in ALS.

4. Discussion

Currently, large amounts of biological data are available and the pool is growing at an increasing pace. This database and tool inflation has led to data fragmentation and thus data integration has gained increasing importance. DAWIS-M.D. facilitates data integration including resources such as KEGG and BRENDA as well as our custom miRNA database. Using VANESA which can query DAWIS-M.D., we were able to observe the significant impact of miRNAs on biological pathways, underlining the great importance of miRNAs for pathway analysis.

We demonstrate the ability of our system using ALS as an example. Previous studies have identified differentially expressed miRNAs both in disease models and patient samples such as *hsa-miR-143-3p*, a *TDP-43* binding miRNA, in the CSF and serum of sporadic ALS patients [13]. Here, we show that this miRNA along with *hsa-miR-365a-3p*, *has-miR-29a-3p* and *hsa-miR-21-5p* target *BCL2* in ALS KEGG (Figure 1A).

Intersecting miRNA enriched GeneMANIA and KEGG networks reduced the network size making it feasible for experimental validation efforts (Figure 1C). Among the 8 genes, 4 have been implicated in ALS before (*SOD1*, *ALS2*, *PRPH* and *NEFH*). Out of the 19 miRNAs, 2 (*miR-21-5p* and *miR-124-3p*) have been associated with ALS previously. These findings demonstrate how our system can help to elucidate some of

the possible miRNA-mediated pathology mechanisms in complex diseases such as ALS, but it is clear that experimental validation must be performed for confirmation.

5. Conclusion

Our findings demonstrate how our system and approach can help to further elucidate some of the possible miRNA-mediated pathological mechanisms in complex diseases such as ALS. The stability of miRNAs and their potential for use as biomarkers make these molecules strong candidates for further more in-depth studies especially in respect to the cross-talk among regulatory networks.

6. Conflicts of interest

The authors declare no conflicts of interest.

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13. Nursing and Interprofessional Applications and Teaching

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German Medical Data Sciences: A Learning Healthcare System
U. Hübner et al. (Eds.)

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Smart Glasses in Nursing – Situation Change and Further Usages Exemplified on a Wound Care Application

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Abstract. Smart Glasses are a promising technology that can be leveraged to improve flexible service processes. Especially in the field of nursing where practitioners are facing complex tasks and challenges. Introducing such pervasive computing devices in service processes may have both positive and negative consequences. This leads us to the following research questions: How does the usage of the Smart Glass applications change the caring situation? Which ideas for future usage of Smart Glasses do nurses have? To answer these questions we followed a design science research approach to design a prototype for support of wound care management in nursing. We evaluated the prototype in a real life situation. Five nurses used the application in a real world setting to perform a wound documentation. Afterwards, we conducted semi-structured interviews with the nurses. The intent of the interviews was not only to get information on the current prototype, but to generate knowledge about dimensions of changing the caring situation which should be considered further. The nurses gave the application an overall positive evaluation. They stated that they would expect an improvement of the quality of the wound documentation when using the device. In addition, they mentioned a change in the communication structure with the patient would be necessary. Furthermore, opinions regarding further use cases did differ.

Keywords. Nursing, information technology, evaluation studies as topic

1. Introduction

Ongoing technological improvement of Smart Glasses offers promising new use cases and directions for the digitalization of flexible service processes. Furthermore, life expectancy in Germany and most other industrial nations is rising. Several challenges emerge from this that will have to be dealt with in the near future. These include political and societal changes in order to incorporate care for elderly people [7]. Another approach would be to support the daily work of nurses by supporting their working tasks through technology.

Usage of IT in nursing is already rising [6]. Thus, more and more digital data exists, which makes it easier to implement Smart Glasses. As one artifact of pervasive

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computing, introducing Smart Glasses may have positive as well as negative effects [3]. Prototypes for usage of Smart Glasses in nursing already exist [1; 4; 5; 9]. Scientific evaluations of current prototypes have primarily focused on prototype specific aspects. An evaluation of changings of the caring situation and of possible future use cases is missing. This leads us to the following research questions:

- How does the usage of Smart Glass applications change the situation of care giving?
- Which ideas for future usage of Smart Glasses in nursing do nurses have?

To answer these questions we designed a Smart Glass wound care application for nursing. Nurses could use the application and get first ideas about general usage of Smart Glasses in nursing. Afterwards we questioned the nurses.

2. Methods

In order to design a Smart Glass application to support nurses in their daily work, our research follows the design science research method (DSRM) [10]. We applied this methodology as follows: (1) We identified the problem in three workshops with nursing directors, nurses, IT specialists, a computer scientist and a nursing scientist. Resulting from these workshops was a list of possible use cases which were prioritized by feasibility and usefulness. One argument to implement a wound care application is that it is an independent process, which allows for implementation of a system without integration into existing software solutions.
(2) In subsequent workshops with wound managers and by conducting a literature search, we derived requirements for the artifact. One of the main requirements is the possibility to use the application hands-free. This guarantees a hygienic and time saving process. We implemented the application on the Vuzix M100.

(3-5) In a next step we developed an artifact iteratively. During the evaluation of the prototype it became clear that it is not effective to use the Smart Glass for all steps of wound documentation since typing letters is very cumbersome with Smart Glasses and the display size of the Vuzix M100 is very small. Because only the process of adding initial data of the wound to the system requires typing, but the majority of documentation time arises from daily documentations afterwards, we decided to divide the documentation into “initial documentation” and “continuous documentation”. Whereas the initial documentation should be completed on a tablet or any other device suited for convenient textual input, we decided to implement the continuous documentation on the Smart Glass. The application is described in the results section.

The artifact was evaluated in several stages of its evolution via workshops and qualitative interviews. The final evaluations of the application were performed with five nurses in two different nursing homes. To evaluate the application, the nurses received a brief introduction into the application. In a next step the nurses conducted a wound care documentation in a real life setting using the application. Data was not recorded. Afterwards, we conducted semi-structured interviews with the nurses and asked them about their experiences and further thoughts about Smart Glasses in nursing.

Whereas two interviews were conducted by a single interviewer (Nursing scientist), three were conducted by two interviewers together. For the evaluation of the interviews we conducted a qualitative content analysis. This analysis is useful to get systematized

and generalized results [8]. (6) As a last step of the design science research method we intend to communicate our results in future workshops.

3. Results

The elements of the continuous documentation which should be conducted on a smart glass are the following: affected tissue structure, length, width, depth, wound condition, wound exsudate, wound margin, additional symptoms, and wound assessment. The Smart Glass application used speech recognition to enable hands-free documentation. The item to be described was displayed with the relevant options. The options were assigned to numbers which the nurses used for the input. In a last step a picture of the wound could be taken by the command "take a picture".

Next, we describe the results of the final evaluation. Following our research questions we defined two codes deductively. These are "Change of the situation" and "Future usage". Furthermore, we defined the subcodes to these codes inductively. This leads to the following list of codes and subcodes.

Table 1: List of codes and subcodes.

Code	Subcode
Change of the situation	Proposed time savings
	Improved focus on the wound
	Contemporary documentation
	Disadvantage if device is broken
	Advantages of wound pictures
	Difficulty to communicate with the resident while treating the wound
	Enhanced communication between colleagues
	Patient can be informed before taking wound treatment
	Lost attention for the surroundings
	Communication can happen after taking care of the wound
Future usages	Training courses for nurses necessary
	Usage of Smart Glasses is a matter of habituation
	Expected acceptance of residents and nurses
	Rejection due to lack of understanding expected
	Acceptance is expected to be divided
	Usage of Smart Glasses during the whole working day is seen as infeasible
	Usage of Smart Glasses during the whole working day is seen as possible
	Smart Glasses with a reminder function could be useful
	Display of additional information could be useful (medication, therapy recommendations)
	Communication with physicians through Smart Glasses could be useful

The overall impression of the application was positive for most of the nurses. Whereas it became clear that using the application would change the process of the wound care management, nurses expected time savings due to the usage of the application. In addition, they mentioned a stronger focus on the wound during wound treatment. This aspect has several side effects. On the one hand, it enhances the quality of the documentation. On the other hand, it makes it difficult to talk to the resident while caring for the wound. However, some nurses came up with a solution for talking to the patient before or after caring for the wound. One example is displayed in the following quote which we translated to English:

„I think if you inform the resident in advance and he understands it, then you can make clear to that person that it's about a better healing of the wound, because you were able to do a much better documentation of the wound."

Future usage of Smart Glasses in nursing is seen as possible. Nurses mentioned training courses for the application as one important element of a successful usage of the application. There was no consistent common expectation regarding the acceptance of the application. While some of the questioned nurses expected a high level of acceptance, others expected divided opinions or even rejection of the application. We questioned the nurses about acceptance of the wound care application as well as about the possibility of wearing Smart Glasses during the complete working day.

„I can imagine the acceptance to be very high, if the colleagues are trained such that it can run smoothly. Than I could imagine that the colleagues would have fun using it and would really use it."

This quote illustrates that nurses find facilitated training sessions with the system important to improve technology adoption outcomes. Finally, opinions on the possibility of wearing Smart Glasses during the whole working day were mixed. It was seen unproblematic by some and infeasible by others.

4. Discussion

This exploratory research presents one possible use case for Smart Glasses in nursing and provides first answers to the questions:

- How does the usage of Smart Glass applications change the situation of care giving?
- Which ideas for future usage of Smart Glasses in nursing do nurses have?

Several aspects for possible changes of the caring situation, which should be considered in future developments, are illustrated. The main aspect mentioned is the missing possibility to talk to the patient during wound treatment due to the usage of speech recognition. Communication and appreciative behavior are important factors when providing care [2]. This seems to be clear for the questioned nurses as they reflected upon solutions to talk to the patient before or after treating the wound.

Nurses' feedback, that using Smart Glasses may improve the quality of the documentation focuses mainly on a timely documentation and on taking a picture. Taking wound pictures and conducting a timely documentation would be possible without using Smart Glasses. This shows that possible improvements due to the usage of Smart Glasses always depend on the actual implemented process.

The questioned nurses had some ideas for future use cases of Smart Glasses in nursing. Some could imagine wearing Smart Glasses the whole working day whereas this was not considered an option by others. The nurses mentioned the relationship to the patient as the most relevant factor. Data security or privacy was not mentioned. In contrast, these aspects are described as relevant in the literature [3]. Due to the small sample size we cannot claim completeness for the aspects mentioned relevant for developing applications of Smart Glasses in nursing.

Whereas we could get first results about possible changes of the situation and ideas for future usage, answers may be limited due to the example of one specific technology. It would be rewarding to repeat this research with further prototypes.

5. Conclusion

Our results show that using Smart Glasses for wound care management in nursing may change the situation in several ways. Nurses expect benefits regarding the accuracy of the wound documentation but also found communication with the patient more challenging. Nurses mentioned the possibility to talk to the patient before or after treating the wound. Regarding potential use cases for Smart Glasses in other nursing processes, differing opinions exist. Aspects to consider for future research are privacy and acceptance as well as the effects on the relationship between the residents and their nurse.

6. Conflict of Interest

The authors declare that they have no competing interests.

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Building a Community of Inquiry Within an Online-Based Health Informatics Program: Instructional Design and Lessons Learned

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Abstract. The rapid medical and technological progress and the resulting need for life-long learning lead to the proliferation of online-based programs in health informatics for health care and health informatics professionals. Effective learning in these online-based programs demands close interaction and cooperation between students and instructors. In such cooperative online-based programs, special emphasis is thus put on fostering a community of inquiry between all participants. The online-based master program on Health Information Management at the University UMIT is firmly based on the constructivist theory of situated learning in such a community of inquiry. Online activities in this master program are designed to promote social presence, cognitive presence and teacher presence as preconditions for successful learning. To verify whether these three forms of presences are visible and whether learning was successful, the first four six-week online-based modules with overall 60 participants were analyzed in depth, combining data from community of inquiry surveys, student evaluations, workload assessments and final assessment results. Results indicate that it was possible to build a community of inquiry (with scores of 4.2 and higher) and to foster effective learning. We conclude with recommendations on how to build a community of inquiry in online-based cooperative learning and also discuss some constraints.

Keywords. Learning, education, distance, cooperative behavior, health informatics

1. Introduction

The rapid medical and technological progress and the resulting need for life-long learning of health care and health informatics professionals leads to the proliferation of online-based programs in health informatics. Online learning offers many advantages, such as self-directed and self-paced learning “anywhere” and “anytime”. However, online learning is associated with challenges such as high dropout rates [1]. Collaborative approaches to online-based learning are based on the assumption that learning within a community of inquiry fosters effective and successful learning and reduces dropout rates.

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2. State of the art

From a constructivist point of view, learning is a constructive and social process that works best in interaction with other persons [2]. Therefore, interaction and cooperation between students and instructors is often considered a key element for successful learning in online settings [3]. From this perspective, online courses need to support interaction and collaboration, so that students can "gradually construct systems of shared meanings" [4].

Studies show advantages of cooperative learning, such as more engaged learning, increased motivation and attention, more active processing of information, improvement of meta-cognitive and social skills and overall better knowledge acquisition and retention [5].

The Community of Inquiry (CoI) is a model that describes critical success factors for sustainable online-based cooperative learning [6]. This model postulates three success factors: Cognitive presence is the extent to which the students are able to construct and confirm meaning through sustained communication and reflection. Social presence is the ability of participants to identify with the group, communicate purposeful in a trusting environment, and develop inter-personal relationships. Teaching presence is defined as the design, facilitation, and direction of cognitive and social processes for the purpose of realizing personally meaningful and educational worthwhile learning outcomes [7-9].

3. Concept: Establishing a Community of Inquiry in online-based learning

The online-based 5-semester master program on Health Information Management (www.umit.at/him) at UMIT is firmly built on a constructivist perspective and applies a cooperative instructional strategy. Based on the Community of Inquiry model, online activities are especially designed to promote social presence, cognitive presence and teacher presence as preconditions for successful learning.

In particular, elements from social constructivism are used by emphasizing student activation through weekly learning activities that always comprise communication and interaction. Elements from situated learning are also applied which involve authentic learning activities and collaborative problem solving in a community of practice [10]. In addition, self-regulation is fostered by asking students to define personal learning goals, by demanding reflections on the individual learning progress, and by organizing regular peer feedback during the course [11]. The overall design is also inspired by andragogy [12], with a focus on activating preliminary knowledge, on interaction in interdisciplinary teams, and on applying new competencies in the own professional context.

4. Implementation: Master programme in Health Information Management

The master program "Health Information Management" is offered by the University UMIT in cooperation with the University of Innsbruck. It started in fall 2017. In the meantime, four online-based modules have been successfully conducted that are presented in this paper.

All modules followed the same structure. Each module contained meta-information, including information on learning objectives, content, instructor, and literature. Then, each module consisted of a set of weekly learning activities. Each week, participants had to complete two to six learning activities. The structure of the learning activities was based on the concept of E-tivities [13]: Each learning activity comprised a description of the intended learning objectives, the tasks to be performed by the participants, and the expected interaction (e.g., discussion in a forum). The learning activities were not meant to test competencies, but to allow the students - alone or in interaction with the others - to accomplish the intended learning objectives. Examples for learning activities: Reading and presenting literature on a given question, preparing and presenting a case study, writing a project plan, developing a tutorial, contrasting different approaches, criticizing a given approach, conducting a statistical analysis, or developing a data model. For each learning activity, the necessary materials (presentations, papers, chapters, or web sites) were provided by the instructor. All four modules had a different instructor. Moodle was used as learning management system.

The entire communication in the modules was asynchronous. Students could work on the learning activities and post or reply to messages at any time they wanted. The instructor was present each day, helped in case of questions and problems, monitored discussions, provided tailored input and summaries, and gave feedback to the student submissions [11]. Assessments comprised oral or written presentation of learning outcome, reflections of personal learning progress and engagement in the course.

In all presented modules, student workload was collected through daily self-documentation. The validated community of inquiry survey [6] was applied at the end of each module after careful translation to German [14, 15]. Table 1 shows results from the four modules. Module 4 was a pilot module without formal assessment.

Table 1. Evaluation results of four online-based modules.

	Module 1: Project management	**Module 2: Process Management**	**Module 3: Applied Informatics**	**Module 4: eHealth (pilot)**
Number of participants	13	13	13	21
Number of learning activities, (% required)	29 (80%)	21 (75%)	20 (60%)	30 (100%)
Number of student's posts	1.572	1.453	1.072	1.568
Final assessment results (0-100, with <60 is a fail)	91 ± 7 (n=13)	91 ± 9 (n=13)	91 ± 6 (n=13)	13 (out of 21) passed (only pass/fail)
Student evaluation (mean): 1 = very good, 5 = very bad	"good – very good" (1.5) (n=11)	"good – very good" (1.5) (n=12)	"very good" (1.2) (n=11)	"very good" (1.2) (n=14)
Student workload (hrs/wk) (Expected workload: 10 – 15 hrs/week)	15 ± 6 (n=13)	14 ± 6 (n=10)	13 ± 4 (n=10)	14 ± 2 (n=15)
Community of Inquiry Survey: Teaching /social/ cognitive presence (mean) 5 = high, 1 = low	4.5/4.5/4.5 (n=11)	3.9/4.3/4.4 (n=10)	4.6/4.5/4.6 (n=12)	4.5/4.2/4.2 (n=16)

5. Lessons learned

The presented results indicate that it was possible to build a community of inquiry and to foster effective learning. This is visible in the Community of Inquiry survey results, the high number of posts as well as in the positive student evaluations. Also, final assessment results showed good learning outcome for most of the students.

Based on the experiences with these first online-based modules, we developed a guideline for instructors within our master program. For this guideline, we especially focused on how to develop social presence, cognitive presence and teacher presence as precondition for successful learning. We derived the recommendations from free text within student evaluation, from interviews with all instructors, and from available evidence from the literature. Table 2 summarizes some of these recommendations.

Table 2. Recommendations to foster social presence, cognitive presence and teacher presence.

Social presence	Cognitive presence	Teacher presence
Invite to face-to-face networking days before start of program	Provide clear course structure	Introduces him-/herself
Develop student-defined netiquette	Activate previous knowledge	Shows daily presence
Build an atmosphere of mutual trust	Invite personal learning goals	Helps in case of questions
Invite respectful peer-feedback	Offer formative assessments	Introduces new topics
Invite regular group work and group discussions	Invite individual reflection on learning progress	Provides additional materials
Build learning activities that demand interaction and cooperation	Invite peer-feedback for joined learning	Provides summaries of discussions
Provide rewards for effective peer-support	Offer authentic learning activities	Fosters good discussions
	Offer various degrees of difficulties of activities	Praises good results
	Provide competence-oriented assessments	Supports struggling students
		Gives feedback

6. Conclusion

Online-based programs allow learning any-place and any-time. This supports life-long learning of health care and health informatics professionals in a rapidly changing professional environment. The potential disadvantages of online-based learning are isolated learning, no deep learning, no applicable knowledge, and high drop-out rates [16, 17]. We addressed these challenges by developing an instructional strategy that is firmly based on modern instructional theories such as social constructivism, situated learning, and community of inquiry. To make sure that all modules follow the presented and successful instructional approach, all instructors get an individual training and coaching. To support this, we have developed a guideline for instructors. In addition, we are currently developing indicators to monitor activity and interaction of students within cooperative online-based learning environments [18].

First experiences from our online-based modules show that students are satisfied with such an activating and cooperative way of learning, even when the workload is high. In the student evaluations, students especially stress the benefits of cooperative working based on authentic learning activities. As a limitation, the presented way of teaching and learning is quite time-intensive for both students and instructors. We responded to this challenge by reducing the number of E-tivities that students have to complete from 100% (in the pilot course) to 60-80%. This helps to reduce workload,

but also allows the students to select those E-tivities where they feel they profit the most, in turn fostering self-regulation and increasing student motivation. To assure that learning objectives are still reached, a certain number of core E-tivities in each module are defined as mandatory and cannot be dropped by the students.

Results of assignments confirm that students seem to acquire the intended competencies in a way comparable to face-to-face teaching. Compared to face-to-face-teaching, students seem to especially benefit from the E-tivities that demand to acquire and then immediately apply competencies as well as from the discussions among each other, which are much more intense than in a comparable classroom setting.

7. Conflict of interest

All authors are instructors within the program Health Information Management.

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What Are Inter-Professional eHealth Competencies?

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Abstract. As health IT supports processes along the entire patient trajectory and involves different types of professional groups, eHealth is inter-professional by nature. The aim of this study, therefore, is to investigate which competencies are at the intersection of the individual groups of health professionals. 718 international experts provided relevance ratings of eHealth competencies for different professional roles in an online survey. Communication and leadership proved to be important competencies across all professions, not only for executives. None or very little differences between professions were found between physicians and nurses, between IT experts at different levels and between IT experts and executives. However, there were a number of competencies rated differently when contrasting direct patient care specialists with executives. These findings should encourage organisations issuing educational recommendations to specify areas of shared competencies more extensively.

Keywords. eHealth, medical and health informatics, competencies, inter-professional education

1. Introduction

eHealth competency development of health care professionals has been recognized as a major driving force of health IT adoption, which resulted in several influential educational recommendations [1,2]. Competencies comprise a combination of cognitive, motivational, moral and social skills that are directed to meet requirements, solve tasks and problems or achieve goals through the necessary knowledge and the corresponding actions [3]. As health IT supports processes along the entire patient trajectory it involves different types of professional groups that are interacting to achieve the patient outcomes targeted. Thus, in principle, eHealth is inter-professional by nature and must also be reflected in education and training, e.g. of physicians, nurses and other professionals working in direct patient care. Furthermore, shared terminologies and competencies were found to be highly valuable between executives (decision makers) and IT specialists [4,5] or clinical experts in direct patient care. Although inter-professional eHealth education has been postulated as desirable, the empirical base of what competencies should be

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common across different professions is small. Furthermore, little is known about clusters of health professionals with similar eHealth competency profiles and those with different ones. The aim of this study, therefore, is to investigate which competencies lie at the intersection of the individual groups of health professionals. As different educational recommendations deliberately address a global audience, we were interested in pursuing the goal of this study on an international level. The following research questions guided this study: (a) Are there common relevant eHealth competencies across different health professions? (b) Do the relevance ratings of eHealth competencies differ between selected professional groups?

2. Methods

In order to answer these research questions, a quantitative survey to capture the current state of needs about eHealth competencies was performed within the Horizon 2020 project *EU*US eHealth Work* that investigated transatlantic cooperation in eHealth workforce development. This survey also contained questions about other fields in eHealth, which are not part of this study. The section on eHealth competencies embraced questions on assessing their relevance for twelve different professional roles on a scale from 0 to 100. The list of competencies was compiled drawing on a list of 24 items that had been previously developed based on international [1,2] and national recommendations [6], validated and used in two other similar international surveys [7]. This list was augmented by competencies incorporated in the HITCOMP tool [8], a database of competencies and roles. Both sets of competencies were evaluated for overlaps and eventually, 33 eHealth competencies[2] emerged from this mapping process. The survey participants could give relevance ratings for several professional roles they felt competent for. The survey was internally and externally pre-tested and was finally made available online from the middle of February to the end of June 2017. The link was sent either to individuals with known eMail addresses, e.g. HIMSS TIGER (Technology Informatics Guiding Education Reform) or to organisations that are representing eHealth professionals, e.g. at the international level *International Medical Informatics Association* or at the national level *UK Royal College of Nurse*s and *German Association for Medical Informatics, Biometry and Epidemiology*. The target organisations were chosen according to their mission to represent health professionals with direct patient care, health executives, health IT specialists and leaders and experts from science & education, thus aiming to garner views from a wide range of professionals. Due to this deployment approach the sample addressed could not be exactly specified by number.

3. Results

892 participants responded to at least one section of the questionnaire, out of which 718 experts provided answers for the section on eHealth competencies (Table 1). As these experts could rate the relevance for several professional roles, 1,571 replies resulted, which were used for the descriptive top ten rank analysis.[3]

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[2] Overview of all 33 competencies https://netcase.hs-osnabrueck.de/index.php/s/4wfTXA8tYnQtOpT

[3] Overview of competence roles https://netcase.hs-osnabrueck.de/index.php/s/QR4LzuFXvEp1xvP

Table 1: Participants' demographics.

Professional background			
Academic or non-academic teaching and / or education [n=101]			Health data management [n=65]
Physician [n=25]			Clinical CEOs [n=20]
Nursing [n=106]			Technical CEOs [n=24]
Pharmacist [n=5]			Clinical CIOs [n=18]
Other health care professions (e.g. physiotherapy) [n=22]			Technical CIOs [n=16]
Other health care institution worker or staff [n=51]			Science & research [n=18]
Engineering or IT specialist [n=129]			Other [n=118]
Gender [n=711]	**Age [n=713]**	**Countries by continent [n=718]**	
Female 70.6 %	Mean 50 years	Africa (2)	Europe (28)
Male 28.6 %	SD 10.99	Asia (10)	North America (2)
Other 0.8%		Australia (1)	Central and South America (8)

In a first step, we looked at which eHealth competencies were represented in the top ten among all professional roles. These were the competencies leadership and communication. In a second step, the professional roles were clustered into the four groups: direct patient care (physicians, nurses, pharmacists, other health care professions), executives (technical and clinical CEOs / CIOs), IT (engineering / IT specialist) and science & education. Health data management was excluded in this analysis. Table 2 shows the common competencies that were found in the top ten among the different professional roles in each cluster.

Table 2: Common top ten eHealth competencies for professional roles (number of replies in brackets).

Direct patient care (401)	Executives (132)	IT (172)	Science / education (274)
Leadership	Leadership	Leadership	Leadership
Communication	Communication	Communication	Communication
Ethics in health IT		Ethics in health IT	Ethics in health IT
Documentation		Documentation	Learning techniques
Teaching, training and education in health care		Data protection & security	Teaching, training and education in health care
Quality and safety management		Health care processes and IT integration	Principles of health informatics
Info. / knowledge management in patient care		Interoperability and integration	Info. / knowledge management in patient care
		Process management	Data protection & security
		Project management	
		Info. / communication tech. (applications)	

Professions within direct patient care shared seven and science & education shared eight competencies out of the top ten, professional roles belonging to the cluster IT shared all competencies (cluster consists of a homogeneous group) while executives only shared two. Again, leadership and communication were found to be a horizontal competency across all professional clusters (green), followed by ethics in health IT shared by three clusters (blue). Competencies that occurred twice are marked in orange and those only once in grey.

In order to test these descriptive findings, we carried out binary logistic regressions for selected pairwise comparisons (Table 3). In particular, we wanted to know, whether physicians and nurses were different regarding the relevance ratings of competencies, whether IT differs from executives, whether IT engineers differ from IT leaders and finally, whether professions engaged in direct patient care differ from those working as executives. To this end, the relevance rating of an expert was limited to his / her

professional background. Thus, the relevance rating for the professional role e.g. "physician" was only counted if the rater's professional background was "physician" in order to increase the authenticity of the individual answers. This, however, reduced the number of ratings per professional role.

Table 3: Binary logistic regression with professions as criterion and competencies as predictors.

Physician [n=18] Nursing [n=82][1,2]	IT [n=140] Executives [n=77][1,3]	Sig. p≤0.05	OR	IT [n=140] CIOs [n=37][1,3]	Sig. p≤0.05	OR
No significant differences	Medical technology	.035	0.977	Change and stakeholder management	.018	1.083
	Teaching, training, edu. in health care	.035	1.032			
	Financial management	.034	1.035			

Executives [n=77][1] vs. Direct patient care [n=118][3]	Sig. p≤.05	OR
Teaching, training and education in health care	.000	0.844
Medical technology	.000	0.863
System lifecycle management	.001	0.896
Information and knowledge management in patient care	.025	0.925
Legal issues in health IT	.007	0.927
Documentation	.036	0.943
Public health informatics	.036	0.951
eHealth, mHealth, telematics and telehealth	.045	1.039
Clinical decision support by IT	.030	1.053
Interoperability and integration	.028	1.062
IT risk management	.029	1.079
Process management	.028	1.080
Strategic management	.026	1.088
Ethics in health IT	.005	1.103
Leadership	.009	1.119

[1] Reference category; [2] Calculated with combined mean value over the roles physician, nursing, other health care professions (direct patient care); [3] Calculated with the combined mean value over all roles.

Table 3 reveals that there were no differences in rating the relevance of all 33 competencies (for direct patient care) between physicians and nurses. The averaged ratings for each of the 33 competencies shows that IT experts and CIOs differed only regarding change and stakeholder management, which received a higher odds ratio (OR) value in favour of CIOs. IT experts also differed from executives - in this case regarding three competencies. The largest group of competencies with different relevance rating occurred when comparing executives and professionals with direct patient care responsibilities. A total of fifteen competencies showed significant OR values either marking a larger chance for executives, e.g. leadership, ethics in health IT or a larger chance for clinicians, e.g. information and knowledge management in patient care.

4. Discussion

This study provides a first overview of what inter-professional eHealth education could mean as investigated from a global perspective, which provides evidence that physicians and nurses can be educated together. It seems that there are several differences regarding the relevance of competencies between executives and direct patient care specialists. Leadership and communication geared to eHealth adoption and use are competencies that exist across the entire professional spectrum. While leadership and communication competencies [2,7] can be found explicitly or implicitly in recommendations, their

paramount importance and link between the professions has not been pointed out that clearly before. Some findings are surprising, e.g. ethics in health IT as well as data protection and security not appearing as common competencies for executives. However, they are included if the top 15 are analysed.

This study has some limitations due to small number of experts when testing the findings for significance, in subgroups. Caution therefore needs to be exercised. Still, these results point to the fact that there is a considerable number of competencies at the intersection of health professions which underpin the role of eHealth as a catalyst and focal point of inter-professional cooperation.

5. Conclusion

These findings should encourage organisations issuing educational recommendations to specify areas of shared competencies more extensively and explicitly.

6. Conflict of Interest

There is no conflict of interest.

7. Acknowledgment

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14. Research Data Management and Reproducible Research

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German Medical Data Sciences: A Learning Healthcare System
U. Hübner et al. (Eds.)

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OSSE Goes FAIR – Implementation of the FAIR Data Principles for an Open-Source Registry for Rare Diseases

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Abstract. The Open Source Registry for Rare Diseases (OSSE) provides a concept and a software for the management of registries for patients with rare diseases. A disease is defined as rare if less than 5 out of 10,000 people are affected. Up to date, approximately 6,000 rare diseases are catalogued. Networking and data exchange for research purposes remains challenging due to the paucity of interoperability and due to the fact that small data stocks are stored locally. The so called "Findable, Accessible, Interoperable, Reusable" (FAIR) Data Principles have been developed to improve research in the field of rare diseases. Subsequently, the OSSE architecture was adapted to implement the FAIR Data Principles. Therefore, the so-called FAIR Data Point was integrated into OSSE to provide a description of metadata in a FAIR manner. OSSE relies on the existing metadata repository (MDR), which is used in to define data elements in the system. This is an important step towards unified documentation across multiple registries. The integration and use of new procedures to improve interoperability plays an important role in the context of registries for rare diseases.

Keywords. OSSE, FAIR, interoperability, rare diseases, registries

1. Introduction

Approximately 30 million people with rare diseases live in the European Union (EU) [1]. According to the World Health Organization (WHO), a disease is rare if it affects less than 5 out of 10,000 people. Although, there are about 6,000 different recognized rare diseases [2]. Large-scaled collection of registry data has evolved to an important tool in

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health services research. Especially in the field of research on rare events or diseases, registries or comparable real-world data sources have substantial advantages [3,4]. Registries are generally considered as a form of medical documentation. In this context, health relevant data, e.g. information on a certain disease, are collected. In addition to providing support in clinical research, they serve quality assurance and quality improvement and the description of epidemiological connections [4].

1.1. Background & Motivation: OSSE – Open-Source Registry for Rare Diseases

OSSE was developed as part of the National Action Plan for People with Rare Diseases, promoted by the German Federal Ministry of Health in the years of 2013 and 2015 [5].

OSSE provides technical and legal concepts and a software-system as a registry toolbox to collect longitudinal and medical data. Data elements are defined and stored within the metadata repository (MDR) following the International Organization for Standardization (ISO) 11179. For each data element, its data type, value range, and unit of measure are defined [6].

However, OSSE does not provide a limited interface to communicate data with registries of other software solutions. An open approach, with standardized vocabulary and ontologies would be desirable in order to facilitate a simple connection. The FAIR Data Principles have been developed for this purpose, which is explained in the next chapter. This paper describes the first steps towards the architecture extension and implementation of the FAIR Data Principles in OSSE. The focus is to build a first prototype.

1.2 The FAIR Data Principles

Interoperability of registries plays a crucial role in increasing data communication in the field of rare diseases. At present, OSSE offers the possibility to integrate other registry solutions into a registry landscape through the OSSE bridgehead, but the access is limited to the participating registries. In order to see which registry includes what kind of metadata, is currently only available via the MDR. However, only owner of an OSSE registry gain access to the MDR, which prevents other research institutes from retrieving the metadata. The FAIR Data Principles are taking this aspect into account. An important aspect of FAIR is to provide existing and new records in a interoperable format, which is legible by computers and humans. Through the semantic annotation of data and metadata, computer systems can automatically combine different data sources, resulting in greater knowledge discovery. The main reason to adopt FAIR in OSSE is to provide a standardized interface to communicate with registries of other software solutions and registries. The criteria of FAIR are examined in more detail below [7].

(1) Findable: By describing metadata, people and computers can interact with the data to search for specific records. (2) Accessible: Data is stored long-term, with defined license and access conditions, both at the level of metadata as well as the level of the data. (3) Interoperable: Data sets can be combined with other data sets. (4) Reusable: Data can be used for further research using computational methods [7].

1.3 Expose Datasets FAIR: The FAIR Data Point

The FAIR Data Point (FDP) is a software component that allows data owners to store data in a FAIR manner and, on the other hand, retrieve data records via metadata. In

addition, retrieving the data is only possible if permitted by the license conditions. FDP are used in some domains, but in this paper we focus solely on patient registries. The FDP has the following objectives: Publish datasets in a FAIR compliant way, allow to retrieve data usage information of the FDP, provide information on the available records, allow data consumers to access the data and make an interaction via GUI and API available. The API (Application Programming Interface) is used by software systems to integrate data directly into a system while the GUI (Graphic User Interface) is designed to provide users with a simple interface to access metadata [8]. In summary, the FDP has two purposes: First, it can be used as a stand-alone web application, where data owners give access to their datasets in a FAIR manner and second, it can be integrated in a larger data interoperability system, like a FAIR port, where many FAIR Data Points of different systems are available. The Dutch Tech Centre of Life Sciences (DTL) is working on a "DTL Data FAIRport" to make FAIR Data Points available to connect different systems in life sciences [8].

2. Methods

To implement the FDP, the existing OSSE architecture needs to be adapted. OSSE already has a component that is ideally suited to connect a FAIR Data Point. This includes the MDR, which provides the data of the registry as metadata. Specifically, the FDP is implemented as a new, standalone component with connections to the electronic data capture of OSSE (OSSE.EDC) and the MDR, as shown in Figure 1. The OSSE.EDC is responsible for managing and storing the data in the registry and thus contains the medical data, while the MDR provides information about the data, stored in the registry. Both components are needed to get access to the registry data and provide metadata information for the FDP. The MDR provides a Representational State Transfer (REST) interface to retrieve the metadata of the registry and it can be used for communication between systems [9]. Also, the OSSE.EDC retrieves the registry data of the OSSE.Store component, which is responsible for the database access. This component also provides a REST interface.

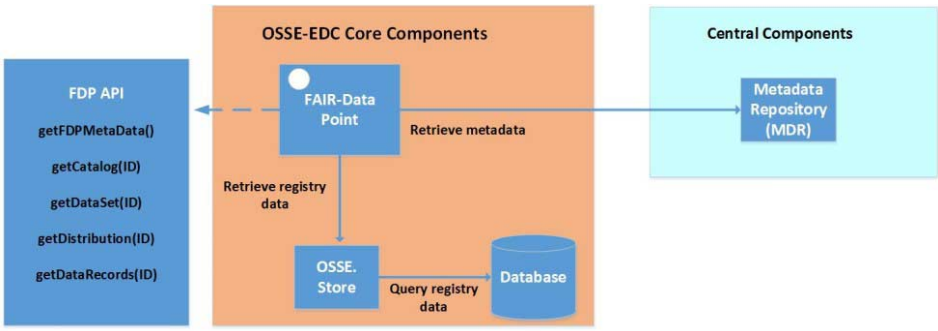

Figure 1. Extended OSSE-Architecture.

3. Results

The main goal of the FDP is to provide meta information about the data in the registry. Which means that only the descriptions of the medical data elements are available, no patient-identifying and medical data is provided. For metadata description the FAIR Data Point API (FDP-API) is the most important interface for sharing this information with other registry systems (shown in Figure 1). The FDP-API contains the Metadata Provider API. This component includes a public service for retrieving the data, which includes several layers to retrieve data. Each layer defines a set of (minimally required) metadata and recommends ontologies for that metadata. The FDP layer describes the FDP as a basic data information source (Data Repository). For example, information about the title, description, a unique ID and information about the data owner of the FDP can be obtained. The description is based on R3Data, "the Metadata Scheme for the Description of Research Repositories" [10]. Each layer points to the respective lower layer. For example, the FDP layer contains a reference to one or more catalogs. A catalog is a collection of metadata about datasets.

The Catalog, Dataset and Distribution layers are based on the Data Catalog Vocabulary (DCAT) published by the W3C in 2014. The main goal of DCAT is to improve the data catalogues interoperability and make applications easy consume metadata from multiple catalogues [11]. The catalog layer contains metadata from the catalog of the FDP. Again, a unique ID, title and description is provided. There are also references to the dataset metadata. The dataset layer gives information about each of the offered datasets. The distribution layer gives information about how the dataset is distributed. Information about the access URL, download URL, format and media type is given here. The bottom layer, data record layer, describes a collection of data, which is available for access or download in one or more formats. In OSSE, each of the described layers can be accessed via a RESTful API. To access the available catalogs, datasets, distributions and data records, only the identification of the respective element is needed. The individual layers were implemented as functions in the OSSE FDP REST API. For example, the *getFDPMetadata* function returns appropriate information about the FDP, so for every layer a function to get metadata out of the layer is available.

A corresponding RDF document is provided, which is created with the Apache Jena framework, which allows the writing and reading of RDF [12]. The concrete metadata uses the FDP according to the function *getUserRootElements* of the MDR-REST API, which returns all metadata for the specific register. In addition, OSSE offers a configuration web page for the FDP to set individual parameters. This includes, for example, an indication of the website or the name of the institution that publishes the FDP.

4. Discussion

The increasing number of FAIR infrastructures is evident. Various collaborations and research institutions as the European Reference Network for Rare Multisystemic Vascular Diseases (VASCern) are currently promoting the utilization of FAIR Data Principles. It is planned to implement a decentralized registry infrastructure with OSSE registries and the FAIR approach in the L-ACMAG (Longitudinal Study Registry of Aortic, Myocardial, Arterial and Genetics in aortic diseases) [13]. L-ACMAG aims to implement a data privacy compliant research collaboration among fifteen German

Marfan Reference Centres to improve the diagnosis and treatment of patients with rare genetic vascular diseases. To meet the changing requirements in the field of digital health care, the European Commission proposed a comprehensive reform of data protection rules in the EU. The novel regulation will apply from 25 May 2018. Time will show if registries can remain FAIR and data privacy compliant at the same time.

5. Conclusion

In the field of rare diseases, registries have evolved to an important research tool. The FAIR Data Principles have been developed to improve data exchange and networking of registries. However, semantic interoperability is not considered in FAIR. Therefore, the OSSE-MDR offers a possibility to uniformly describe and define data elements. The MDR can thus be used as an additional tool for creating semantic interoperability in a FAIR infrastructure.

6. Conflicts of Interest

The authors declare that there is no conflict of interest.

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15. Sequencing and High Dimensional Data

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Using RNA-Seq Data for the Detection of a Panel of Clinically Relevant Mutations

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Abstract. Somatic single nucleotide variants (SNVs) are genomic events with increasing implications in cancer treatment. The clinical standard for SNVs detection is whole genome/exome sequencing (WGS/WES) in matched tumor-normal samples. Yet, this is a very costly approach both economically and biologically and very often only tumor samples are sequenced. On the other hand, RNA sequencing (RNA-Seq) is the most popular technology to study gene expression, and has also the potential for a cost-effective identification of SNVs as an alternative to tumor-only WES. Here we present a method for the identification of SNVs in tumor-only RNA-Seq data putting a special focus on a small panel of clinically relevant SNVs. For evaluation purposes, we analyzed matched tumor-normal WES and tumor-only RNA-Seq data from 14 cancer patients. We compared SNVs detected in i) RNA-Seq by our method, ii) WES tumor-only by Mutect2 and iii) WES matched tumor-normal by Mutect2. We did a detailed evaluation for a reduced panel of clinically relevant SNVs and reliably identified in RNA-Seq data a subset of mutations for which we had pathological annotation. Hence, RNA-Seq rises as a cost-effective option to detect in parallel gene expression as well as a small panel of clinically relevant SNVs in research.

Keywords: RNA-Seq, SNVs, Mutect2, variant calling, GATK

1. Introduction

Somatic single nucleotide variants (SNVs) are genomic events known to drive cancer. Whole genome and exome sequencing (WGS, WES) in matched tumor-normal samples are the clinical standard for detecting somatic SNVs. There are many tools for identifying SNVs on WGS or WES data, thoroughly compared in different contexts [1-3]. According to these studies, two tools outperform the rest: Mutect [4] and Varscan2 [5]. The first performs better at identifying SNVs with low allele frequencies, whereas the latter detects the highest number of SNVs and outperforms any tool at positions with high coverage. On the other hand, RNA sequencing (RNA-Seq) has become the most popular technology -after replacing microarrays- to study gene expression. Unlike microarrays, RNA-Seq can easily be used to detect alternative splicing, RNA editing, fusion genes, other RNA species, and, potentially, SNVs. Calling somatic SNVs in RNA-Seq data has been done in some studies by applying tools specific for WES/WGS data [6-8]. Besides

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obvious false negatives produced in regions with low or no expression, these studies reported false positive SNV calls in RNA-Seq data mainly due to: PCR cycle bias, strand bias, RNA editing and difficulty to align the transcriptome to the reference genome due to splicing. Sheng and colleagues tried to address some of these issues both on DNA and RNA-Seq [9]. An added problem is the fact that clinical samples are usually limited to tumor-only profiling. Detection of somatic SNVs in WES tumor-only samples is challenging and has been addressed with machine learning approaches [10] or the use of whitelists and blacklists as in Mutect [4]. However, the same has not yet been attempted for RNA-Seq data. All in all, its cheaper cost compared to WES/WGS together with all its possible applications makes RNA-Seq a technology with high interest for clinical use (e.g. parallel detection of SNVs and functional activation of genes). It seems worthwhile developing a method to call SNVs in RNA-Seq data optimized for a panel of well-known SNVs.In this study we present a method to call SNVs in RNA-Seq tumor-only samples. We assess its performance putting special focus on optimizing the method for a panel of known SNVs with high clinical interest. We compare our method's performance on a matched dataset comprising RNA-Seq and WES data. We chose Mutect2 to detect SNVs in WES data.We compare the SNVs detected in RNA-Seq data by our method to tumor-only and tumor-normal results by Mutect2.

2. Materials and Methods

2.1. Databases

The panel of known SNVs with high clinical interest was based on the Clinical Interpretation of Variants in Cancer (CIViC, version from 01/06/2017) [11] and Cancer Genome Interpreter (CGI, last updated 02/08/2017) [12]. In both cases, we filtered for SNVs predictive of drug response. Genomic coordinates were transformed from hg19 to hg38 built using the rtracklayer R package. Both databases were merged by aggregating duplicate entries. The panel of actionable variants contains information on 442 variants in 92 genes.

2.2. Collection of Patient Samples

Tissue samples were collected by the surgery departments of the University Medical Center Göttingen. The collected tissues are from seven metastatic brain and seven metastatic liver tumours with origin from either colorectal or breast cancer. Fresh frozen tissue samples for WES and RNA-Seq were separated. From these 14 patients we collected EDTA-blood for WES as well. EDTA-blood samples served as control samples to differentiate between germline and somatic mutations. In total 42 samples were sequenced, 3 samples per patient. The study is approved by the Ethics Committee of the University Medical Centre Göttingen, application number 21/3/11 and 14/10/05.

2.3. Data Preprocessing and Analysis

WES and RNA-Seq reads were quality assessed using fastqc. All WES brain samples showed high duplication levels and a drop in quality at the end of the reads due to high levels of contamination with nextera adapters. Hence, TrimGalore-0.4.3 was applied to all brain samples. Between 11% and 38.4% of base pairs were trimmed. WES paired-end reads were aligned against the reference genome of Homo sapiens version GRCh38 with bowtie2 (version 2.2.3). Samtools was used to create bam files, and Picard (version

2.0.1) to mark duplicates. Then, GATK (v3.8.0) best practices were followed to perform read realignment (IndelRealigner) and base recalibration (BaseRecalibrator). RNA-Seq single-end reads were aligned against the reference genome of Homo sapiens Ensembl Version GRCh38.91 with the splice-aware aligner STAR (v2.5.2b). Picard (version 2.0.1) was used to remove duplicates. We used Mutect2 (GATK v3.8.0, beta version) to detect somatic variants in WES data using matched tumor-normal samples (referred to as "clinical standard" in the text) as well as only tumor samples ("tumor-only" mode). Cosmic (version 83, Coding and Non Coding vcf files) and dbSNP (version 138) were provided as input to Mutect2 to adjust the threshold for evidence of a variant in the normal sample. To confirm germline mutations detected by tumor-only samples, GATK Haplotypecaller (v3.8.0) with dbSNP (version 138) was used.

3. Results

3.1. Detection of SNVs in RNA-Seq Data

We implemented a variation of pileuping nucleotide bases at each position in the transcriptome, using mpileup from samtools. In case of a transcriptome-wide analysis, mpileup of all base positions in the RNA-Seq data is performed. Afterwards, the distribution of bases in each position is annotated with supplemental database information (CIVIC, ClinVAR, Cosmic). In case the -panelmode flag is selected, mpilup is called only for the 442 curated SNVs. A minimum of 3 reads or 10% of the reads supporting the alternative variant are used as default thresholds. The output comprises the distribution of bases at each position, the decision whether the position contains an SNV, and clinical annotations from CIViC and CGI.

3.2. Comparison of WES and RNA-Seq Data in Detecting SNVs

We generated WES matched tumor-normal and RNA-Seq tumor-only data from 14 cancer patients. We applied a standard pipeline to detect somatic SNVs in WES matched tumor-normal samples, referred to as clinical standard. We also applied a tumor-only mode to WES tumor samples, referred to as tumor-only. Finally, we applied our method to detect SNVs in RNA-Seq tumor samples. SNVs detection was focused on 442 cancer-specific variants with clinical interest. As shown in Figure 1A we found 109 SNVs in all samples by the three methods: 10 in the gold standard, 104 in tumor-only and 73 in the RNA-Seq (average of 0.7, 7.4 and 5.2 SNVs/sample, respectively). The 10 SNVs detected by the clinical standard were also detected in WES tumor-only and RNA-Seq. We found a higher overlap between WES tumor-only and RNA-Seq (68 SNVs) than between the two methods on WES (10 SNVs). This finding is explained by the fact that our panel includes germline mutations and polymorphisms; the clinical standard is optimized to reliably detect mutations only present in the tumor sample by filtering out any mutation present in the normal sample. Accordingly, the clinical standard only detected SNVs known to be somatic. Nonetheless, all somatic SNVs were also detected by tumor-only and RNA-Seq (Figure 1B).

The variants uniquely detected by WES tumor-only (36 SNVs) could be explained in the majority of the cases due to low expression in the RNA-Seq data. The only exception presenting high expression was the *MGMT* promoter SNP rs16906252. Yet, this SNP is known to be associated with low MGMT expression, leading to allele specific expression [13]. On the other hand, only 5 SNVs were exclusively detected by RNA-

Seq. Two of them - *TPMT* Y240C and *TPMT* A154T - are known to be an haplotype of the TPMT enzyme (TPMT*3A) [14] and were indeed found in the same patient (BM4). These haplotype was not confirmed by WES tumor-only due to high duplication levels, which did not pass Mutect2 filters. The other three mutations (*ETS2* mutation in patient BM1, *NQO1* in patient BM7 and *XRCC1* mutation in patient LM3) were not found in WES tumor-only also due to the same reason. As a matter of fact, these 5 germline polymorphisms detected exclusively in RNA-Seq data could be confirmed by a germline SNV caller (Haplotypecaller) in normal samples. Last but not least, 7 out of the 8 pathologically validated mutations in *BRAF*, *KRAS*, *NRAS* and *PIK3CA* were consistently detected by the three methods (Figure 1). *PIK3CA* E545K mutation in patient LM5 was not detected by any method.

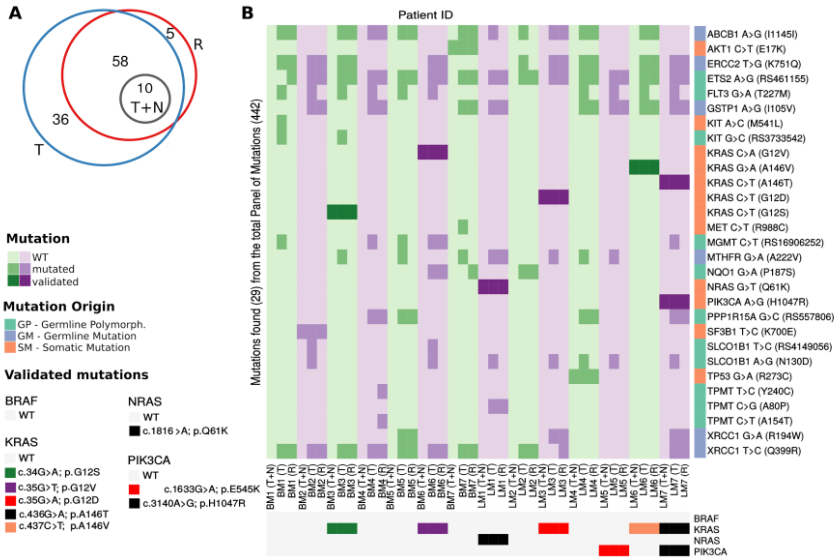

Figure 1: A) Venn diagram depicting the number of SNVs identified by each method across all samples (T+N: Mutect2, Tumor+Normal samples, T: Mutect2, Tumor samples, R: Wileup RNA-Seq). B) Heatmap visualization of 29 unique SNVs which were found by at least one of the methods in any of the 14 patients. Wild Type (WT) mutations are shown light green and purple, mutations found by the methods are in green and purple, mutations agreeing with the pathological annotation (validated) are marked in dark green and purple. The details of the pathological mutations are described in the annotation bars at the bottom of the figure. The origin of the mutation is annotated in the bar at the right sight of the heatmap.

4. Discussion

We showed a high overlap between RNA-Seq and tumor-only WES. Previous studies reported high numbers of false positives in RNA-Seq data, however, by using a whitelist of well-defined SNVs we avoid this problem. In this setting, detecting SNVs in RNA-Seq data is a comparable approach to WES tumor-only; yet, in RNA-Seq it is regarded as an extra analysis which can be quickly performed (average of 11-15 min/sample) for no extra cost. More important, RNA-Seq appears to be a reliable approach for detecting the selected panel of clinically relevant SNVs, as confirmed by the pathologically validated data (for somatic variants) and by the analysis of normal WES samples (for germline variants). Of course, the user has to accept false negatives in non expressed

genes, but that is inherent to RNA-Seq data. For future implementations, it would be important to consider RNA editing processes as well as including indels in the analysis.

5. Conclusion

We showed that RNA-Seq is a cost-effective option to detect a curated list of SNV in parallel to gene expression. Although WES will remain to be the clinical standard, the method presented here can become an alternative when WES is not available.

6. Conflict of Interest

The authors declare no conflict of interest.

7. Acknowledgements

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16. Standards, Classifications & Terminologies

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German Medical Data Sciences: A Learning Healthcare System
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XML Skeleton Definitions for Human Posture Assessments

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Abstract. In this paper, we show how the XML dialect SKAML (Skeletal Assessment Markup Language) can be used to use data from one or more Motion Capture systems to perform human posture assessments with multiple assessment methods. We show an implementation example using an inertial measuring suit and both OWAS and REBA assessment methods. SKAML makes it possible to implement classifiers for a Motion Capture system once and adapt the classifier *by-configuration* to various ergonomics assessment methods. We anticipate our work as help for researchers and developers that implement new assessment methods or motion capture systems.

Keywords. Skeleton markup, motion capture, posture assessment, posture classification, musculoskeletal disorders, SKAML, OWAS, REBA

1. Introduction

One of the primary causes of unfitness for work and early retirement for workers in physically demanding occupations are musculoskeletal disorders (MSD) [1]. Appropriate preventive measures may delay or even prevent the emergence of MSDs if the individual or workplace-related risk factors for MSDs are identified early [2].

A task that was -- and often still is -- performed using manual pen-and-paper assessment methods such as OWAS [3], REBA, EAWS, and others. Despite the practical usability of manual assessments, they have drawbacks and limitations: observer bias, limited view of human observers, non-repeatability, no automatic documentation and post-processing. As a result Motion Capture (MoCap) systems are increasingly used for ergonomics assessments, i.e., the determination of risk factors for occupational diseases, e.g., work-related musculoskeletal disorders (WMSD) [4]. Researchers that must implement a method for ergonomics assessment in software (to use it in conjunction with MoCap systems) define skeleton model as well as the rules of the assessment method in code, which is costly and error-prone. However, if the MoCap system is to be replaced by another system, e.g., by an inertial measurement system with less or more sensors, this provides another skeleton and probably does not provide spatial joint coordinates but relative limb orientations. For example, an optical MoCap system such as Kinect provides a predefined abstract Skeleton structure and updates the spatial joint coordinates accordingly. A computerized assessment method must be specifically adapted to this pre-defined skeleton to work properly. Here the description of an abstract skeleton structure that can be annotated to support multiple MoCap systems, as well as multiple assessment methods, is needed.

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The Skeletal Assessment Markup Language (SKAML) [5] used here has been developed to fill a need for a method to combine the input of different Motion Capture systems with the various available methods for assessing postures and motions.

2. State of the Art

There are various markup languages for several aspects of human behavior (e.g., HBML) to virtual avatars (e.g., HumanML), but to our knowledge, no markup language for the description of abstract skeleton representations in combination with motion capture and ergonomic assessments exists.

However, with the Unified Robot Description Format (URDF) [6], there is an XML markup language that allows robots to be modeled in their (possibly humanoid) structure as well as their movement capabilities. URDF starts at a relatively low level and allows, for example, the modeling of robot geometry from geometric primitives. The link with ergonomic assessment methods was apparently not considered.

The Motion Capture Markup Language (MCML) [7] aims for the integration of heterogenous MoCap file formats. The MCML is similar to SKAML with its approach to improve the usage of multiple MoCap systems with its different file formats. The language models the data close to the actual MoCap formats whereas SKAML tries to model the human skeleton on a more abstract level leaving details of the MoCap system to the implementor of a digitized assessment method.
Nevertheless, the authors of MCML recognize the importance of open and easy-to-implement formats and propose an XML-based markup language, too. MCML lacks the conjunction with assessment methods.

3. Concept

The Skeletal Assessment Markup Language proposed here fits into the gap of XML-based description languages between descriptions of human motion on an abstract level and descriptions of the human biomechanics. SKAML describes the skeletal system of a human body from a full-body motion capture observer perspective. It is not possible nor intended to describe details of the human biomechanics with SKAML. Implementors may refer to the XSD-specification[2].

3.1. General

On the technical side, SKAML allows the definition of simplified skeleton structures mainly consisting of joint and bone definitions. Root of an SKAML-document is the `<body>` tag, which has the direct child `<skeleton>`. A skeleton consists of a set of joints and bones. The `<joints>` tag contains one or more `<joint>` tags. Each joint may have one or more children so that the topology of joints (kinematic tree) can be described, e.g., a joint *hip* has two child joints *back_mid* and *femur_left* each having children as well:

[2] See http://www.skaml.org/SKAML.xsd

```
<joint id="hip">
    <joint id="back_mid"><joint id="neck"/></joint>
    <joint id="femur_left"> ... </joint>
</joint>
..
<bone id="arm_right" from="ellbow_right" to="hand_right"
length="0.279">
```

Bones of the skeleton can be defined using the bones tag followed by one or more bone tags. A bone is defined as a rigid connection between two joints.

3.2. Sensor Input

In a motion capture scenario using an inertial sensor suit, IMUs (Inertial Measurement Units) are placed on the limbs, in most cases around half the distance between two joints on the limb. One can model this using an abstract sensor definition and attach it to a bone definition, e.g.

```
<bone id="upper_spine" from="neck" to="back_mid">
    <sensor type="IMU" id="5" transformation="1 0 .. 0 1" />
</bone>
..
<joint id="ellbow_left">
    <sensor type="Kinect" id="ELLBOW_LEFT" />
</joint>
```

The transformation attribute is a 4x4 matrix that describes the translation and orientation of the sensor on the bone originating at the start joint if applicable.
It is also possible to attach a `sensor`-tag to a joint, which is useful if the motion capture system provides joint coordinates. This is the case for most optical MoCap systems, e.g., Kinect.The sensor type is not enforced by SKAML but application-specific.

3.3. Assessments

SKAML allows for the definition of various skeleton structures. Screening methods for posture or motion assessments consider only specific joint or bone combinations, in most cases different to the structure defined in SKAML. To address this, an assessment tag was introduced. The tag can be placed as child to a joint or bone, e.g.

```
<joint id="joint42">
    <assessment type="OWAS" joint="FOOT_RIGHT"/>
    <assessment type="REBA" joint="FootR"/>
</joint>
```

The attributes type and joint of the assessment tag refer to user-defined strings, i.e., identifying strings specific to the used assessment method and software.
With the assessment tags, the model skeleton on one side and the assessment method on the other side are loosely coupled.

4. Implementation

In this section we show how to use SKAML to implement a classifier for OWAS [3] assessment method using motion data coming from a Kinect sensor as well as IMU-based motion capture suit. The software for the classifiers is only implemented once, and the adaption for both Motion Capture systems is done using a SKAML document.

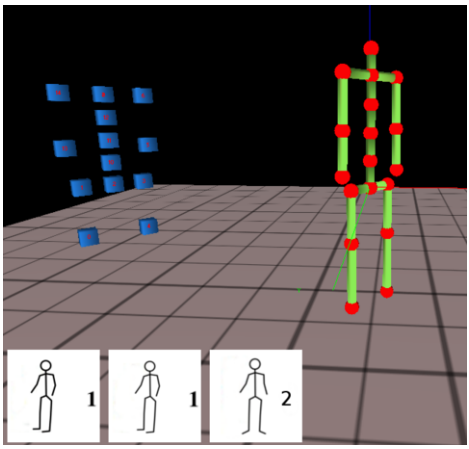

Figure 1: Skeleton visualization (center) based on SKAML description utilizing IMU-sensor data (blue cubes on the left) classified by an OWAS classifier (OWAS posture with code 112).

The software shown in Figure 1 was created for the SIRKA sensor suit [8]. The application can load recorded motion capture data from a file, both Kinect and SIRKA sensor suit. We created a SKAML configuration for each MoCap system for easier handling, although it would be possible to combine both configurations in one file. Once data and configuration are loaded, the classifier code can use some generic load functions to access the skeleton (C++ in our implementation):

```
Skeleton* skel = SKAMLLoader->loadFromFile(..);
..
// getGenericBone expects an assessment id (here: OWAS)
// and a bone id specific to this assessment method
Bone* bone_upperarm_l = skel->getGenericBone("OWAS", "ARM_LEFT");
..
bone_upperarm_l = skel->getGenericBone("REBA", "upper_arm_left");
..
// access bone properties
bone_upperarm_l->getOrientation();
bone_upperarm_l->getLength();
```

We have modeled classes for the SKAML entities Skeleton, Joint, and Bone, which provide methods for accessing both markup properties (e.g., length) and sensor-based properties (e.g., orientation matrix/quaternion for given timestamp). The assessment method classes are using the generic Bone and Joint objects as shown above, so one can change the underlying MoCap system without changing one line of code in the classifier.

5. Lessons Learned

An XML description language was introduced that allows the definition of abstract body skeletons that can be annotated for various MoCap systems and various ergonomic assessment methods.

We have shown that the description language can be easily integrated into software applications. SKAML makes it possible to increase the maintainability, reusability, and adaptability of the software. This reduces the development effort for new technologies and methods. Also, the formalization of the description avoids errors and enables automatic consistency checks using the XSD grammar. Additionally, SKAML may be used in conjunction with formally defined digitized assessment methods, e.g., formulated using Arden Syntax [9].

In subsequent additions, we plan to expand SKAML by modeling motion constraints, and we will provide an Open Source library (libskaml) as a convenience for developers and researchers.

6. Conflict of Interest

No potential conflict of interest was reported by the authors.

7. Acknowledgements

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17. Surveillance, Screening and Registries

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German Medical Data Sciences: A Learning Healthcare System
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User Evaluation Indicates High Quality of the Surveillance Outbreak Response Management and Analysis System (SORMAS) After Field Deployment in Nigeria in 2015 and 2018

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Abstract. During the West African Ebola virus disease outbreak in 2014-15, health agencies had severe challenges with case notification and contact tracing. To overcome these, we developed the Surveillance, Outbreak Response Management and Analysis System (SORMAS). The objective of this study was to measure perceived quality of SORMAS and its change over time. We ran a 4-week-pilot and 8-week-implementation of SORMAS among hospital informants in Kano state, Nigeria in 2015 and 2018 respectively. We carried out surveys after the pilot and implementation asking about usefulness and acceptability. We calculated the proportions of users per answer together with their 95% confidence intervals (CI) and compared whether the 2015 response distributions differed from those from 2018. Total of 31 and 74 hospital informants participated in the survey in 2015 and 2018, respectively. In 2018, 94% (CI: 89-100%) of users indicated that the tool was useful, 92% (CI: 86-98%) would recommend SORMAS to colleagues and 18% (CI: 10-28%) had login difficulties. In 2015, the proportions were 74% (CI: 59-90%), 90% (CI: 80-100%), and 87% (CI: 75-99%) respectively. Results indicate high usefulness and acceptability of SORMAS. We recommend mHealth tools to be

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evaluated to allow repeated measurements and comparisons between different versions and users.

Keywords. mHealth, eHealth, systematic evaluation, disease surveillance, outbreak response, open source, Africa, infectious disease, medical and health informatics

1. Introduction

During the West Africa Ebola virus disease (EVD) outbreak in 2014-15, it became very clear that response teams must be equipped with technologies enabling real-time digitalized reporting and response management in order to improve efficiency in outbreak containment [1]. This experience led to the development of the Surveillance, Outbreak Response Management and Analysis System (SORMAS). SORMAS is a mobile and web application that provides a platform to detect outbreaks, manage tasks, validate cases, coordinate with laboratories and perform contact tracing. SORMAS uses a bi-directional information exchange and feedback to the different users within the Integrated Disease Surveillance and Response System (IDSR), which defines the disease notification procedures, established in Africa. We piloted the first version of SORMAS in 2015 in the states of Kano and Oyo, in Nigeria and conducted a user evaluation on its usefulness and acceptability. In 2016 and 2017, we reprogrammed the system in a new platform and included additional functionalities. In December 2017, we initiated the deployment of the system with the new version (SORMAS 2017) in Kano State and conducted a new user evaluation in order to measure whether the usefulness and acceptability of the system had improved.

2. Methods

For SORMAS 2015, we used the In-Memory Database technology in the cloud-based SAP HANA platform [2] [3]. We programmed SORMAS 2017 based entirely on open source platforms using a PostgreSQL Relational Database Management System and VAADIN framework [4][5]. The hospital informants operated SORMAS with smartphones in 2015 and with tablets in 2017. In 2015 version, the login procedure was a 3-tier authentication protocol, whereas in the 2017 version user login required a one-time encrypted password stored on the device and a 4-digit pin to authenticate the app anytime it was in use. We conducted six design-thinking workshops with a total of 65 epidemiologists, clinicians, and laboratory experts from Germany and 8 African Countries. The team developed the disease specific process models for Ebola Virus Disease (EVD), Cholera, Measles and Highly Pathogenic Avian Influenza (HPAI) in SORMAS 2015 and additionally for Lassa fever, Monkeypox (MPX), Dengue fever, Yellow fever, Cerebrospinal Meningitis (CSM), and Plague in SORMAS 2017 [6, 7, 2, 8]. The tool contains user specific interfaces, including that for hospital informants who are in charge of reporting notifiable disease occurring in the health facilities, and for laboratory officers in charge of processing laboratory diagnosis. In 2015, we selected eight Local Government Areas (LGA) at random from both Oyo and Kano States to participate in the pilot [9]. Within these LGAs, the state epidemiologist selected 31 private and public health facilities based on previous reporting activities resulting in 31 participating hospital informants. In 2017, SORMAS was deployed in all private and

public health facilities of two LGAs [10]. After a two-day training, we conducted a five-week pilot in June 2015 [10]. For SORMAS 2017, all participants attended a two-day basic and a one-day refresher training. We implemented a post-pilot questionnaire in July 2015 after the hospital informants had used SORMAS 2015 for 5 weeks and post-implementation questionnaire in February 2018 two months after deployment of SORMAS 2017. We asked the users whether they would want to use the tool for their routine surveillance work (usefulness: yes/no), if they would also recommend the tool to their colleagues (acceptability: yes/no) and whether the users had experienced problems in the login (log-in difficulty yes/no). The 2015 questionnaire was paper based [11], while the 2018 questionnaire was administered electronically with SurveyCTO [12]. Questionnaires from both years contained questions on 'age', 'sex', and 'type of facility and rural/urban settlement types'). We performed a comparison of the respondents' characteristics between 2015 and 2018 using chi-squared test and calculated response proportions and their 95% confidence intervals (CI) using STATA version 14 [13]. We secured written consent from study participants and survey completion was anonymized. The study proposal was submitted to the ethics committee at the Hannover Medical School and cleared by the national agency in charge in Nigeria (NCDC).

3. Results

In 2015, 31 hospital informants (100%) that participated in the pilot submitted their responses. In 2017, 80 hospital informants were recruited and trained, out of which 74 (93 %) participated in the post implementation survey. The mean age in 2015 was 40.0 years (standard deviation= 8.97) and 35.7 years in 2018 (standard deviation= 8.27). See Table 1 for respondents' distribution for age, sex, facility type and settlement. Table 2 demonstrates statistically significant improvements for usefulness, and ease of login and no significant differences with respect to acceptability, comparing 2015 and 2018.

Table 1: Comparison of survey respondents' characteristics between 2015 and 2018.

Variable	2015 (N=31) n (%)	2018 (N=74) n (%)	Chi2	p-value
Age (years)				
18-34	8 (26)	33 (45)	3.2	0.072
35-54	23 (74)	41 (55)		
Sex				
Female	8 (26)	12 (16)	1.3	0.254
Male	23 (74)	62 (84)		
Facility Type				
Public	20 (65)	50 (68)	0.1	0.762
Private	11 (35)	24 (32)		
Settlement				
Urban	16 (52)	54 (73)	4.5	0.034
Rural	15 (48)	20 (27)		

Table 2: Comparison of survey respondents' assessment on usefulness, acceptance and ease of login between 2015 and 2018 surveys.

Variable	2015 (N=31)		2018 (N=74)		Chi2	p-value
	n (%)	95% CI	n (%)	95% CI		
Usefulness: would want to use SORMAS in surveillance work						
Yes	23(74.2)	(58.8 - 89.6)	70(94.6)	(89.4- 99.7)	8.98	0.003
No	8(25.8)	(10.4 – 41.2)	4(5.4)	(0.3 – 10.6)		
Acceptability: would recommend SORMAS for their colleagues						
Yes	28(90.3)	(79.9 - 100)	68(91.9)	(85.7 - 98.1)	0.07	0.793
No	3(9.7)	(0 - 20.1)	6(8.1)	(1.9 - 14.3)		
Ease of login: encounter problems with SORMAS login						
Yes	27(87.1)	(75.3 – 98.9)	14(18.9)	(10.0 – 27.8)	42.67	<0.001
No	4(12.9)	(1.1 – 24.7)	60(81.1)	(72.2 - 90.0)		

4. Discussion

Overall, the user responses revealed significant improvement for usefulness and ease of login in the 2017 version of SORMAS compared to the 2015 version. In addition, in both surveys, a large majority of the respondents would recommend the tool to their colleagues. In 2015, the complex login process led to frequent typing errors and eventually caused the devices to be blocked. The changes in the login procedure alongside with the large screens of the mobile tablets enabled users to easily navigate through the app and most likely explain the improved user login experience of 2018 version. Even though we are now finding an improvement from the user perspective, demands on data security are increasing and may force developers to implement login procedures, which in the given setting lack practicability. In order to address this challenge, an enhancement of multi-factor authenticated key exchange protocols could be a solution [14]. The increase in usefulness in the 2018 survey is remarkable.

The following differences are likely to account for this improvement. SORMAS 2018 covers 10 disease instead of four in the 2015 version and includes the laboratories as additional users in the network. We believe that the major difference in usefulness rating results from multiple improvements of user interface design and user training and the fact that increased smartphone/tablet use in the country may also have contributed. However, this comparison also contains several limitations: The study populations are not identical, even if their demographic characteristics between the 2015 and the 2018 did not differ. Furthermore, improvements in the survey tool also limit comparability.

5. Conclusion

Hospital informants have repeatedly rated SORMAS very high on usefulness, acceptability, and reported clear improvement with respect to login procedures from the 2015 to the 2018 version. We will take the opportunity of continued SORMAS implementation in 2018 to analyze other data quality dimensions, such as completeness or timeliness.

6. Conflict of Interest

There is no conflict of interest.

7. Acknowledgment

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